

RAMIFICATION AND INTRINSIC PRIMARY BOUNDARIES IN MULTIPLICATION FAILURE

QIAN QI

ABSTRACT. We show that a nonreduced multiplication-failure scheme can intrinsically recover a smooth genus-one curve or a K3 surface although its reduced divisor is rational and independent of the quadratic multiplication relations. For a nonempty open class of cube-zero algebras of Hilbert function $(1, 3, 3)$ or $(1, 4, 6)$, the embedded support on the entire projection-corank-at-most-one open is a vector bundle over the ramification variety of a finite quadratic projective morphism, times a projective space. Its full failure ideal is the product of the determinant-divisor ideal and the embedded-support ideal. We determine an irredundant primary decomposition of every ordinary power, its exact nilradical index, and the canonical embedded torsion and its annihilator. These identify the embedded support without choosing a primary representative; birational cancellation then recovers the ramification curve or K3 surface from the abstract failure scheme. The associated families and all specified ideal-power thickenings are flat over the admissible relation space. We also give universal determinantal primary flags in arbitrary embedding dimension, a complete classification for Hilbert function $(1, 2, 2)$ including the extreme-corank locus, and sharp codimension-controlled stabilization and relative conductor comparisons with global section multiplication under explicit regularity hypotheses.

1. RAMIFICATION RECOVERED FROM MULTIPLICATION FAILURE

The reduced support of a multiplication-failure scheme need not determine its embedded geometry. We show that a fixed rational reduced divisor can underlie failure schemes from which one recovers a smooth genus-one curve or a K3 surface. The recovering invariant is intrinsic to the nonreduced scheme: it is the annihilator of its nilradical, followed by birational cancellation. Thus neither an ambient framing nor a choice of embedded primary component is needed for the recovery statement.

Here is the construction. Let $e \in \{3, 4\}$, put $p = \binom{e}{2}$, and let V be an e -dimensional complex vector space. An e -plane $R \subset \text{Sym}^2 V$ of quadratic relations defines

$$S_R = \text{Sym}^2 V/R, \quad B_R = \mathbb{C} \oplus V \oplus S_R,$$

where multiplication in degree two is the quotient map and all products of degree at least three vanish. We specify a nonempty invariant open $G_e^\circ \subset \text{Gr}(e, \text{Sym}^2 V)$ in Proposition 2.3. Its points are basepoint-free systems of quadrics on $\mathbb{P}(V^*)$ with smooth ramification hypersurface Y_R . The associated finite morphism

$$\phi_R : \mathbb{P}(V^*) \longrightarrow \mathbb{P}(R^*)$$

has degree 2^{e-1} , and Y_R has degree e . Thus Y_R is a smooth plane cubic for $e = 3$ and a smooth quartic K3 surface for $e = 4$. The class is an open set of a Grassmannian, not a finite list of algebra orbits: the orbit-dimension deficits are at least one and nine, respectively.

On $X_R = \text{Gr}(e, V \oplus S_R)$ let U_R be the open where $K \rightarrow V$ has rank at least $e - 1$, and let $\Delta_R \subset U_R$ be its determinant divisor. The inverse image E_R of Y_R under $K \mapsto \text{im}(K \rightarrow V)$ is

Date: September 23, 2026.

2020 Mathematics Subject Classification. 14M12, 14J28, 13C40, 14B05, 14H50.

Key words and phrases. Multiplication of sections, conductor, finite algebras, primary decomposition, nonreduced schemes.

smooth and integral of codimension two. For every $m \geq 2$, the failure scheme of $\mathrm{Sym}^m(\mathbb{C} \oplus K) \rightarrow B_R$ on *all* of U_R satisfies

$$\mathcal{I}_{D_R} = \mathcal{I}_{\Delta_R} \mathcal{I}_{E_R} = \mathcal{I}_{\Delta_R} \cap \mathcal{I}_{E_R}^2. \quad (1.1)$$

These are its two associated supports. More generally, for every $n \geq 1$,

$$\mathcal{I}_{D_R}^n = \mathcal{I}_{\Delta_R}^n \cap \mathcal{I}_{E_R}^{2n}. \quad (1.2)$$

Both decompositions are irredundant. Theorem 3.2 proves these equalities directly for each fixed tensor, and Theorem 3.3 determines the embedded torsion, its annihilator and its generic length for every n . These are powers of an ideal, not additional symmetric multiplication degrees.

Writing $\mathcal{N}_R$ for the nilradical of $\mathcal{O}_{D_R}$, we have

$$\mathcal{N}_R^2 = 0, \quad \mathcal{N}_R \simeq \mathcal{O}_{E_R}(-\Delta_R), \quad E_R = V(\mathrm{Ann}_{\mathcal{O}_{D_R}} \mathcal{N}_R).$$

The support E_R is a vector bundle over $Y_R \times \mathbb{P}^{p-1}$. It follows that an abstract isomorphism $D_R \simeq D_{R'}$ forces $Y_R \simeq Y_{R'}$ (Theorem 4.2). The reduced divisors, in contrast, are all isomorphic after identifying the underlying vector spaces and have rational smooth projective compactifications. For the explicit elliptic family retained below, distinct j -invariants therefore give nonisomorphic *full failure schemes*, not merely nonisomorphic input algebras.

The link between the two geometries is an equality of cokernel sheaves. For $H = \ker \alpha$, contraction gives

$$\mathrm{coker}(\mathrm{Sym}^2 H \rightarrow S_R) \simeq \mathrm{coker}(R \xrightarrow{c_\alpha} V).$$

On projective space the target on the right carries its natural $\mathcal{O}(1)$ twist. In a basis of quadrics the right-hand map is their Jacobian matrix; its determinant defines the ramification of ϕ_R . This cokernel identity, the full multiplication Fitting ideal, and the intrinsic annihilator are the successive steps of the recovery theorem. They do not follow from the reduced determinant alone.

The primary theorem is also relative over G_e° : the failure schemes and all the specified ideal-power thickenings form flat families (Proposition 3.4). Statements about primary components are made on the complex fibres; no preservation of primarity under arbitrary nonreduced base change is asserted. The omitted locus $\mathrm{rank}(K \rightarrow V) \leq e - 2$ is not part of this theorem. The complete Hilbert-function- $(1, 2, 2)$ calculation, including its extreme-corank locus, and the universal determinantal flag theorem remain separate results with their original scopes.

Sections 2–4 prove the new theorem chain. Sections 5 onward retain the determinantal, conductor, relative and global-section framework in full. The article is standalone; the repository designation A2 imposes no unstated upstream or downstream hypothesis. The literature comparison is separated from proof and is recorded in Section 27.

2. QUADRATIC RELATIONS AND PROJECTIVE RAMIFICATION

All schemes in this section are over $\mathbb{C}$. Put $N = e(e+1)/2$ and $p = N - e$. For $R \in \mathrm{Gr}(e, \mathrm{Sym}^2 V)$ let $\gamma_R : \mathrm{Sym}^2 V \rightarrow S_R$ be the quotient map. We regard elements of $\mathrm{Sym}^2 V$ as homogeneous quadratic functions on V^* . The algebra B_R introduced above is local, with maximal ideal $V \oplus S_R$, square S_R , and cube zero; hence its Hilbert function is $(1, e, p)$.

Let $\mathcal{H}$ be the tautological hyperplane bundle on $P = \mathbb{P}(V^*)$. Contraction is normalized by $c_\alpha(vw) = \alpha(v)w + \alpha(w)v$.

Lemma 2.1 (Restriction–contraction identity). *There is an exact sequence of vector bundles on P*

$$0 \rightarrow \mathrm{Sym}^2 \mathcal{H} \rightarrow \mathrm{Sym}^2 V \otimes \mathcal{O}_P \xrightarrow{c} V \otimes \mathcal{O}_P(1) \rightarrow 0. \quad (2.1)$$

Consequently there is a natural isomorphism

$$\mathrm{coker}(\mathrm{Sym}^2 \mathcal{H} \rightarrow S_R \otimes \mathcal{O}_P) \simeq \mathrm{coker}(R \otimes \mathcal{O}_P \xrightarrow{c_R} V \otimes \mathcal{O}_P(1)). \quad (2.2)$$

Both maps are square, of sizes p and e . Their determinant ideals are equal, and define a degree- e hypersurface whenever the determinant is not identically zero.

Proof. Locally choose α and $\ell \in V$ with $\alpha(\ell) = 1$. Then $V = H \oplus \mathbb{C}\ell$ and

$$\mathrm{Sym}^2 V = \mathrm{Sym}^2 H \oplus H\ell \oplus \mathbb{C}\ell^2.$$

Contraction kills the first summand, maps $h\ell$ to h , and maps ℓ^2 to 2ℓ . This proves exactness, including local splitting; the twist records rescaling of α . Both cokernels in (2.2) are the quotient of $\mathrm{Sym}^2 V \otimes \mathcal{O}_P$ by $R \otimes \mathcal{O}_P + \mathrm{Sym}^2 \mathcal{H}$. Equality of their zeroth Fitting ideals proves the scheme-theoretic determinant assertion. Finally $\det c_R$ is a section of $\det V \otimes (\det R)^* \otimes \mathcal{O}_P(e)$. This also agrees with $\det(\mathrm{Sym}^2 \mathcal{H})^*$, whose nonconstant factor is $\mathcal{O}_P(e)$ because $\det \mathcal{H} = \mathcal{O}_P(-1)$ and $\mathrm{rank} \mathcal{H} = e - 1$. $\square$

Proposition 2.2 (The ramification interpretation). *Suppose R is basepoint-free on P . Its evaluation morphism $\phi_R : P \rightarrow \mathbb{P}(R^*)$ is finite, surjective, and has degree 2^{e-1} . The determinant scheme of $\gamma_R|_{\mathrm{Sym}^2 \mathcal{H}}$ is precisely its ramification scheme*

$$Y_R = V \left(\det \left(\frac{\partial q_i}{\partial a_j} \right)_{i,j=1}^e \right) \subset \mathbb{P}(V^*), \quad (2.3)$$

where $q_1, \dots, q_e$ is any basis of R and $a_1, \dots, a_e$ are linear coordinates on V^* . Moreover

$$R \text{ is basepoint-free} \iff \gamma_R(HV) = S_R \text{ for every hyperplane } H \subset V. \quad (2.4)$$

Proof. Basepoint-freeness gives a morphism with $\phi_R^* \mathcal{O}(1) = \mathcal{O}_P(2)$. A positive-dimensional projective fibre contains a curve; this would make $\mathcal{O}_P(2)$ trivial on that curve, contrary to ampleness. Thus the proper morphism is quasi-finite and hence finite. Its image has dimension $e - 1$ and is the entire target. The intersection of $e - 1$ pulled-back hyperplanes has degree 2^{e-1} , which is the degree of the finite morphism. In characteristic zero it is generically separable.

The differential on the affine cones is the displayed Jacobian J , which is the transpose of the contraction matrix in the chosen bases. Euler's identity is $J(a)a = 2(q_1(a), \dots, q_e(a))$. On a source chart $a_j \neq 0$ and target chart $q_i \neq 0$, choose the radial vectors as the first vectors in the respective tangent frames. The induced map on the radial line is multiplication by a unit. Passing to the quotient by that line gives the differential of the projective morphism. Expansion along this first column shows that its determinant and $\det J$ differ by a unit. Equivalently their cokernels agree after removing this invertible one-dimensional summand. This identifies the ramification Fitting ideal, including its scheme structure. Lemma 2.1 identifies the other determinant.

For the last assertion, a functional on S_R annihilating $\gamma_R(HV)$ pulls back to a functional on $\mathrm{Sym}^2 V$ annihilating both R and HV . If $H = \ker \alpha$, then $(HV)^\perp = \mathbb{C}\alpha^2$. Thus failure of surjectivity is equivalent to $q(\alpha) = 0$ for every $q \in R$, exactly a base point of the quadratic system. This proves both implications. $\square$

Proposition 2.3 (A nonempty open class with moduli). *For $e = 3$ or $e = 4$, the set*

$$G_e^\circ = \{R \in \mathrm{Gr}(e, \mathrm{Sym}^2 V) : R \text{ is basepoint-free and } Y_R \text{ is smooth}\}$$

is nonempty, open, and invariant under $\mathrm{PGL}(V)$. For every $R \in G_e^\circ$, the restriction $\gamma_R|_{\mathrm{Sym}^2 H}$ has rank $p - 1$ on Y_R and rank p off Y_R . The variety Y_R is smooth and integral. It has genus one for $e = 3$, and is a K3 surface for $e = 4$.

Two algebras $B_R, B_{R'}$ are isomorphic if and only if R, R' lie in the same $\mathrm{PGL}(V)$ orbit. In particular

$$\dim G_e^\circ - \max_{R \in G_e^\circ} \dim(\mathrm{PGL}(V) \cdot R) \geq ep - e^2 + 1 = \begin{cases} 1, & e = 3, \\ 9, & e = 4. \end{cases} \quad (2.5)$$

Here the left side is an orbit-dimension deficit; no fine moduli scheme is being asserted.

Proof. We give a parameter-space proof of nonemptiness rather than infer smoothness by specializing a primary decomposition. Fix a p -dimensional space S and let $T = \text{Hom}(\text{Sym}^2 V, S)$. Over $T \times P$, evaluation on $\text{Sym}^2 \mathcal{H}$ is a surjective vector-bundle map to the bundle of p by p matrices. For a fixed H , corank at least two therefore has codimension four in T . Its universal locus in $T \times P$ has dimension $\dim T + e - 1 - 4 < \dim T$. Projection is proper because P is projective, so its image is a proper closed subset of T .

Similarly HV has dimension $p + e - 1$. Restriction to it is a surjective evaluation map to the space of p by $(p + e - 1)$ matrices. Nonsurjectivity has codimension e , so the universal bad locus has dimension $\dim T - 1$. Its proper image is again a proper closed subset. Remove both images, remove nonsurjective γ , and require the restriction at one fixed hyperplane to be invertible. These are nonempty open conditions in the irreducible affine space T , so their intersection T_1 is nonempty. The last condition ensures that the determinant is not the zero section.

On $T_1 \times P$ the universal determinant hypersurface is smooth: evaluation on the square matrix is a submersion, and the determinant hypersurface is smooth at every rank- $p - 1$ matrix. Every fibre is a nonempty degree- e hypersurface. The generic fibre over T_1 is regular, since its local rings are localizations of regular rings, and is geometrically regular in characteristic zero. It is therefore smooth. The nonsmooth locus of the proper morphism to T_1 is closed; its proper image misses the generic point. Deleting this image gives a nonempty open T_2 with all determinant fibres smooth.

The map from the space of surjections $\text{Sym}^2 V \rightarrow S$ to $\text{Gr}(e, \text{Sym}^2 V)$ takes a tensor to its kernel and is the frame bundle of the universal quotient. The preceding conditions depend only on that kernel. Basepoint-freeness and smoothness of the projective hypersurface are open conditions by the same proper-image argument. Thus G_e° is open and contains the image of T_2 ; it is invariant by its intrinsic definition. Smoothness of a nonzero determinant hypersurface excludes matrices of corank at least two, since their cofactors vanish and hence all derivatives of the determinant in local base coordinates vanish.

A smooth hypersurface of positive degree in $\mathbb{P}^{e-1}$ is integral here: for $e - 1 \geq 2$, distinct positive-degree components intersect, and their intersection would be singular. Adjunction gives $\omega_{Y_R} \simeq \mathcal{O}_{Y_R}$. The hypersurface sequence $0 \rightarrow \mathcal{O}_P(-e) \rightarrow \mathcal{O}_P \rightarrow \mathcal{O}_{Y_R} \rightarrow 0$ gives genus one when $e = 3$, and $H^1(Y_R, \mathcal{O}_{Y_R}) = 0$ when $e = 4$. The latter is a K3 surface by the algebraic definition; compare the standard quartic example in [41, Chapter 1, Example 1.3(i)].

An algebra isomorphism preserves the maximal ideal and its square. On the associated graded algebra it induces an isomorphism of V which takes the kernel of quadratic multiplication to the other kernel. Conversely such a linear isomorphism induces a graded algebra isomorphism. Scalars act trivially on relation subspaces, so the equivalence is exactly the stated projective-linear orbit relation. Finally $\dim \text{Gr}(e, N) = e(N - e) = ep$ and every orbit has dimension at most $e^2 - 1$. This proves the estimate and, in particular, excludes a finite-orbit interpretation of the class. $\square$

Remark 2.4. The restrictions $e = 3, 4$ enter the nonemptiness proof at the codimension-four corank-two incidence. For $e \geq 5$ this dimension argument no longer excludes its projection. The universal flag theorem below handles additional restriction-rank strata, but the present smooth-ramification theorem does not silently extend to them. The construction of the finite quadratic morphism, the Jacobian criterion and the quartic K3 calculation are classical ingredients. They are used here to identify what the embedded part of the multiplication-failure scheme intrinsically remembers.

3. PRIMARY POWERS AND THEIR INTRINSIC TORSION

Fix $R \in G_e^\circ$ and abbreviate $S = S_R$, $X = X_R$, $U = U_R$, $\Delta = \Delta_R$. On Δ let $H = \text{im}(K \rightarrow V)$ and $\Lambda = K \cap S$; these are bundles of ranks $e - 1$ and one. Define the closed subscheme

$$E = \Delta \times_{\mathbb{P}(V^*)} Y_R \subset U. \quad (3.1)$$

All ideals in this section are ideals of the displayed ambient scheme, not merely ideals of set-theoretic supports.

Lemma 3.1 (A global two-stratum primary calculation). *Let U be a smooth complex variety, let $\Delta \subset U$ be a smooth integral Cartier divisor, and let $E \subset \Delta$ be a nonempty smooth integral Cartier divisor of Δ . Put $L = \mathcal{I}_\Delta$ and $P = \mathcal{I}_E$. For every $n \geq 1$,*

$$L^n P^n = L^n \cap P^{2n}. \quad (3.2)$$

This is an irredundant primary decomposition with associated points exactly η_Δ, η_E . The specified primary representatives L^n and P^{2n} are intrinsic ordinary powers of the two ideals.

Proof. At a point of E , choose local equations t, f with $L = (t)$ and $P = (t, f)$. They are part of a regular system of parameters. The associated graded ring of P is locally a polynomial ring in two variables over the domain $\mathcal{O}_E$. Multiplication by an element outside P is injective on each of its graded pieces, so induction on the filtration proves that P^a is P -primary for every $a \geq 1$. The initial form of t is one of the two polynomial variables; multiplication by it is injective on the graded ring. It follows, by comparing the first nonzero graded piece, that

$$(P^{2n} : t^n) = P^n.$$

Hence $(t^n) \cap P^{2n} = t^n P^n$. The same argument for the principal smooth divisor proves that (t^n) is (t) -primary. Off E , the equality reduces to $L^n = L^n$. These local identities glue because all displayed ideals are intrinsic.

At η_E , the element t^n lies in L^n but not in P^{2n} , whereas f^{2n} lies in P^{2n} but not in L^n . Neither component can be omitted. A noetherian irredundant primary decomposition with these two distinct radicals has exactly these associated primes. The calculation on affine opens gives the associated points stated globally. In particular, this proof does not replace an equality of ideals by a claim about radicals. $\square$

Theorem 3.2 (Ramification as an embedded primary support). *For every $R \in G_e^\circ$ and every $m \geq 2$, the multiplication failure scheme $D = D_R$ on all of U has ideal*

$$\mathcal{I}_D = \mathcal{I}_\Delta \mathcal{I}_E = \mathcal{I}_\Delta \cap \mathcal{I}_E^2. \quad (3.3)$$

The two supports are smooth and integral, and are its complete set of associated supports on U . There are natural identifications

$$\Delta \simeq \text{Tot}_{\mathbb{P}(V^*) \times \mathbb{P}(S)} \text{Hom}(H, S/\Lambda), \quad (3.4)$$

$$E \simeq \text{Tot}_{Y_R \times \mathbb{P}(S)} \text{Hom}(H, S/\Lambda). \quad (3.5)$$

The vector-bundle rank is $(e - 1)(p - 1)$; $\dim U = ep$, $\dim \Delta = ep - 1$, and $\dim E = ep - 2$. The nilradical is square-zero and fits into the exact sequence

$$0 \longrightarrow \mathcal{O}_E(-\Delta) \longrightarrow \mathcal{O}_D \longrightarrow \mathcal{O}_\Delta \longrightarrow 0. \quad (3.6)$$

Proof. For $K \in \Delta$, its image is a hyperplane H and its intersection with S is a line Λ . The subspace K/Λ is the graph of a unique map $H \rightarrow S/\Lambda$. Conversely the inverse image of any such graph in $H \oplus S$ is an e -plane with the prescribed image and intersection. This construction and its inverse commute with base change; it proves (3.4) as a scheme isomorphism, not just a bijection of points. Restricting to Y_R proves (3.5). Smoothness, integrality, and the dimensions

follow. The rank- $e - 1$ determinantal divisor is smooth on U , since an $(e - 1)$ -minor is invertible there.

We compute the *full* multiplication ideal. Since the augmentation ideal has cube zero, the image of every symmetric power of degree $m \geq 2$ is $\mathbb{C} \cdot 1 + K + K^2$. On a frame chart write $K \rightarrow V \oplus S$ as $\begin{pmatrix} M \\ A_0 \end{pmatrix}$. Its multiplication presentation has scalar, linear and quadratic blocks

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma_R \text{Sym}^2 M \end{pmatrix}.$$

Every nonzero maximal minor uses the scalar column and all e columns meeting the V -rows. Expansion gives

$$\mathcal{I}_D = (\det M) I_p(\gamma_R \text{Sym}^2 M).$$

This argument is valid over the chart's coefficient ring; it is not a fibrewise rank test.

Near Δ , invert an $(e - 1)$ -minor and apply invertible row and column operations to make $M = \text{diag}(I_{e-1}, t)$. The moving basis of V decomposes the quadratic matrix into

$$[A, tB, t^2c], \quad A = \gamma_R|_{\text{Sym}^2 H} \text{ along } t = 0.$$

The condition $\gamma_R(HV) = S$ holds everywhere by Proposition 2.2. Locally $[A, B]$ is therefore onto, so there are matrices U_0, V_0 with $AU_0 + BV_0 = I_p$. It follows that the column module of $[A, tB, t^2c]$ equals that of $[A, tI_p]$: multiplying this identity by t gives one inclusion, and the reverse inclusion is immediate. Zeroth Fitting ideals agree for presentations with the same image.

If A is invertible, the ideal is (t) . If its restriction lies on Y_R , it has rank $p - 1$. Inverting a nonzero $(p - 1)$ -minor and taking the Schur complement reduces the remaining quadratic presentation to a single row with ideal (f, t) , where $f \bmod t$ is a local equation of Y_R . Thus

$$\mathcal{I}_D = (t^2, tf) = t(t, f). \quad (3.7)$$

The equations t, f cut out E and are part of a regular system of parameters. Off Δ , M is invertible, γ_R is onto, and the failure ideal is the unit ideal. This proves $\mathcal{I}_D = \mathcal{I}_\Delta \mathcal{I}_E$ everywhere on U . Lemma 3.1 gives the primary decomposition and its irredundancy and associated points directly on U . Alternatively the same identities descend by Lemma 9.1; the graph descriptions specify all descended supports.

Finally the radical of $\mathcal{I}_D$ is $\mathcal{I}_\Delta$, and this ideal is invertible. Its quotient by $\mathcal{I}_\Delta \mathcal{I}_E$ is $\mathcal{I}_\Delta \otimes \mathcal{O}_E = \mathcal{O}_E(-\Delta)$. Its square vanishes because $\mathcal{I}_\Delta \subset \mathcal{I}_E$. It is nonzero at η_E . This proves the exact sequence and square-zero assertion. $\square$

Theorem 3.3 (All powers and canonical embedded torsion). *Keep the hypotheses of Theorem 3.2. For $n \geq 1$ put $D^{[n]} = V(\mathcal{I}_D^n)$ and $E^{[n]} = V(\mathcal{I}_E^n)$, as subschemes of U . Then*

$$\mathcal{I}_D^n = \mathcal{I}_\Delta^n \mathcal{I}_E^n = \mathcal{I}_\Delta^n \cap \mathcal{I}_E^{2n} \quad (3.8)$$

is an irredundant primary decomposition. There are exactly two associated points, η_Δ and η_E . The generic multiplicity on Δ is n . The nilradical has exact index $2n$ along E and index n on $\Delta \setminus E$.

Let $\mathcal{T}_n$ be the kernel of localization of $\mathcal{O}_{D^{[n]}}$ at its unique minimal associated point. This is an intrinsic coherent module, and

$$\mathcal{T}_n \simeq \mathcal{O}_{E^{[n]}}(-n\Delta), \quad V(\text{Ann}_{\mathcal{O}_{D^{[n]}}} \mathcal{T}_n) = E^{[n]}. \quad (3.9)$$

There is an exact sequence

$$0 \longrightarrow \mathcal{O}_{E^{[n]}}(-n\Delta) \longrightarrow \mathcal{O}_{D^{[n]}} \longrightarrow \mathcal{O}_{n\Delta} \longrightarrow 0. \quad (3.10)$$

At η_E the module $\mathcal{T}_n$ has length $n(n+1)/2$. For $n = 1$ it is exactly the nilradical. Thus the embedded support, its ordinary-power neighbourhoods, and these torsion lengths are canonical, even though an embedded primary representative in an arbitrary decomposition need not be unique.

Proof. The ideal product follows by taking ordinary powers; the intersection and associated points follow from Lemma 3.1. Near η_Δ , f is invertible, so the ideal is (t^n) and its generic multiplicity is n . At a point of E , its radical is (t) and the ideal is $t^n(t, f)^n$. The element t^{2n} lies in this ideal, whereas t^{2n-1} does not, since cancellation of t^n would give $t^{n-1} \in (t, f)^n$. The graded-ring argument in the lemma excludes that membership. Off E the nilradical index is n ; for $n = 1$ the zero nilradical has index one by convention.

On an affine neighbourhood, localization at the minimal prime (t) makes (t, f) the unit ideal. The contraction of the localized failure ideal is (t^n) , which is (t) -primary. The localization kernel is therefore

$$(t^n)/t^n(t, f)^n \simeq \mathcal{O}_U/(t, f)^n$$

with the invertible factor t^n recording the transition functions. This proves the first formula in (3.9) and the exact sequence. Its annihilator is $(t, f)^n/\mathcal{I}_D^n$, because multiplication by t^n can be cancelled in the regular ambient ring. Hence the annihilator defines exactly $E^{[n]}$, with its full scheme structure. The local calculations agree on overlaps, both by their ideal formulas and by the intrinsic definition of the localization kernel. At η_E the ambient ring is regular local of dimension two, with maximal ideal (t, f) . The monomials $t^i f^j$, $i + j < n$, give the successive graded pieces of its quotient by $(t, f)^n$, whose total length is $1 + 2 + \cdots + n$. The claim for $n = 1$ follows also from (3.6). $\square$

Proposition 3.4 (Flat relative primary neighbourhoods). *Over G_e° the construction gives relative schemes $\mathcal{E} \subset \Delta \subset \mathcal{U}$ and $\mathcal{D} \subset \mathcal{U}$. The morphism $\mathcal{E} \rightarrow G_e^\circ$ is smooth. For every $n \geq 1$, $\mathcal{E}^{[n]}$ and $\mathcal{D}^{[n]}$ are flat over G_e° , and (3.10) holds relatively and remains exact after arbitrary base change. Each complex fibre has the primary decomposition (3.8).*

Proof. Use the universal quotient bundle $\mathcal{S}$ on G_e° ; no global trivialization of it is required. The universal smooth ramification hypersurface is proper and smooth over G_e° , and the graph description makes $\mathcal{E}$ a vector bundle over its product with $\mathbb{P}(\mathcal{S})$. It is therefore smooth over the base. The relative determinant divisor is smooth as well. Locally along $\mathcal{E}$, the preceding elimination works over the parameter ring and gives $\mathcal{I}_D = t(t, f)$. The equations t, f are relative regular parameters: t defines the relative smooth divisor and f its smooth relative ramification hypersurface. In particular their conormal module is locally free, and all symmetric powers of it are flat over the parameter base.

The conormal filtration of $\mathcal{O}_{\mathcal{E}^{[n]}}$ has successive quotients $\mathrm{Sym}^j(\mathcal{I}_\mathcal{E}/\mathcal{I}_\mathcal{E}^2)$ for $0 \leq j < n$, viewed on $\mathcal{E}$. Each is base-flat; hence $\mathcal{O}_{\mathcal{E}^{[n]}}$ is base-flat. The analogous principal filtration proves flatness of $\mathcal{O}_{n\Delta}$. Cancellation of t^n gives the relative version of (3.10); its outer terms are flat, so its middle term is flat. Tensoring this sequence with any base algebra is exact because the last term is flat. Relative regularity also identifies the base-changed ideals with the specified products and powers. The primary statement on every complex fibre is supplied by the direct fibre proof in Lemma 3.1, not by an assertion that chosen primary components specialize in arbitrary families. $\square$

4. INTRINSIC RECOVERY OF CURVES AND K3 SURFACES

The exact nilpotent structure has a consequence not available from a transport of the reduced finite-algebra locus to global sections. We first record the birational cancellation used to state it without an ambient marking.

Lemma 4.1 (Cancellation over a non-uniruled base). *Let Y, Y' be smooth projective complex varieties which are not uniruled. If $Y \times \mathbb{P}^r$ and $Y' \times \mathbb{P}^{r'}$ are birational, then Y and Y' are birational. For smooth genus-one curves, or for K3 surfaces, they are then isomorphic.*

Proof. A dominant rational map $Y \times \mathbb{P}^r \dashrightarrow Y'$ must be constant on the general projective-space fibre. Indeed, if it were nonconstant on such fibres, restrictions to lines on these fibres would give a family of nonconstant rational curves covering a dense open of Y' . This would make Y' uniruled. Constancy on the general fibre gives a rational factorization through Y : on function fields, the pullbacks lie in the constant subfield $\mathbb{C}(Y) \subset \mathbb{C}(Y)(z_1, \dots, z_r)$. Apply this to a birational map and its inverse. Their induced maps on Y, Y' compose to the identity, proving birationality. Equivalently one may take the space of lines meeting the domains of the rational maps; a nonconstant generic restriction would supply the excluded covering family directly.

A genus-one curve is not uniruled, and a birational map between smooth projective curves is an isomorphism. A complex K3 surface is not uniruled either: a generically finite dominant map from a ruled surface would inject its nonzero global two-form into the zero space of global two-forms on a resolution of the ruled surface. This is the usual characteristic-zero differential-form argument; see [41, Chapter 9, proof of Corollary 4.4]. For completeness, birational K3 surfaces are isomorphic as follows. Resolve a birational map by point blowups $Y \xleftarrow{p} Z \xrightarrow{q} Y'$. The pullbacks of their nowhere-vanishing two-forms span the same one-dimensional space $H^0(Z, \omega_Z)$. Their zero divisors are therefore equal. By the blowup formula for the canonical divisor, their supports are exactly the p -exceptional and q -exceptional curves, respectively, every coefficient being positive. Thus the two exceptional sets coincide. The morphism q is constant on every fibre of p , and conversely. Descending successively through the point blowups, or using their graphs and connected fibres, gives inverse morphisms $Y \rightarrow Y'$ and $Y' \rightarrow Y$. They are the desired isomorphisms. $\square$

Theorem 4.2 (The ramification variety is intrinsic to the failure scheme). *For $e \in \{3, 4\}$ and $R, R' \in G_e^\circ$, an abstract complex-scheme isomorphism $D_R \simeq D_{R'}$ implies $Y_R \simeq Y_{R'}$. It is not necessary for the isomorphism to respect the Grassmannian, the projection to $\mathbb{P}(V^*)$, the algebra, or any chosen frames.*

The embedded support has the smooth projective compactification

$$\overline{E}_R = \mathbb{P}_{Y_R \times \mathbb{P}(S_R)}(\mathcal{O} \oplus \text{Hom}(H, S_R/\Lambda)). \quad (4.1)$$

For all $i \geq 0$,

$$H^i(\overline{E}_R, \mathcal{O}_{\overline{E}_R}) \simeq H^i(Y_R, \mathcal{O}_{Y_R}). \quad (4.2)$$

In particular $h^1 = 1$ for $e = 3$, whereas $h^1 = 0$ and $h^2 = 1$ for $e = 4$. The smooth projective compactification of the reduced divisor obtained from (3.4) is rational and has no higher coherent cohomology. Its isomorphism type is independent of R after identifying vector spaces of the same ranks. Thus the recovery statement uses the embedded structure of D_R , not an invariant of its reduction.

Proof. An isomorphism of schemes preserves the nilradical and its annihilator. Theorem 3.3, with $n = 1$, consequently recovers E_R as a closed subscheme of D_R and carries it isomorphically to $E_{R'}$. No primary representative is chosen at this step.

The graph-bundle description gives the indicated compactification and shows that

$$E_R \text{ is birational to } Y_R \times \mathbb{P}^{(p-1)+(e-1)(p-1)}.$$

Indeed a vector bundle is trivial on a dense open of its integral base. The same holds for R' . The isomorphism of the embedded supports therefore makes Y_R and $Y_{R'}$ stably birational. By Proposition 2.3 they are smooth genus-one curves or K3 surfaces. Apply Lemma 4.1 to conclude that they are isomorphic as abstract varieties.

For a projective bundle the direct image of its structure sheaf is the structure sheaf of the base and its higher direct images vanish, as is seen on trivializations from the cohomology of projective space. Apply this first to (4.1) and then to $Y_R \times \mathbb{P}(S_R) \rightarrow Y_R$. The Leray sequence proves (4.2). The genus-one and K3 values follow from the hypersurface calculation in Proposition 2.3. Replacing Y_R by $\mathbb{P}(V^*)$ in the same construction compactifies Δ_R . Its base is a product of projective spaces, so it is rational and the same direct-image calculation gives vanishing of all higher coherent cohomology. Neither the vector-bundle description of Δ_R nor this calculation uses R . $\square$

Corollary 4.3 (Nonisomorphic failure schemes with a fixed reduction). *For the family B_λ and open Ω of Theorem 8.1, distinct values of $j(C_\lambda)$ give nonisomorphic abstract failure schemes on the projection-corank-at-most-one open. All their reductions are the same determinant divisor. Their relative family is flat.*

Proof. Here the three-dimensional relation space is spanned by $xy - \lambda z^2$, $yz - \lambda x^2$, and $zx - \lambda y^2$. The proof of Theorem 8.1 verifies basepoint-freeness and smoothness on Ω ; thus these are points of G_3° . Their restriction determinant is C_λ , and its nonconstant j -invariant is computed in (8.9). The recovery theorem gives the asserted nonisomorphism. Flatness is a special case of Proposition 3.4, and also follows from the retained direct proof in Corollary 8.2. $\square$

Remark 4.4. The recovered object is the abstract ramification variety. We do not assert reconstruction of its polarization, the finite quadratic map, or the input algebra from D_R . Nor does the orbit-dimension estimate for the $(1, 4, 6)$ algebras assert that every quartic K3 surface occurs, or compute the dimension of the image in K3 moduli. The theorem asserts recovery for every member of the nonempty open class which is proved to exist. The elliptic subfamily gives an explicit nonconstant invariant of the actual failure schemes. These distinctions separate an intrinsic geometric conclusion from stronger Torelli or dominance statements not used in the proof.

5. PRIMARY FLAGS AND ELLIPTIC BOUNDARIES

An embedded component of a multiplication-failure scheme can carry moduli which disappear after taking the reduced support. We prove this by a uniform primary decomposition over the parameter space of quadratic multiplication tensors, and by a family in which the embedded support is a vector bundle over an elliptic curve times a projective plane. The reduced failure divisor is independent of the elliptic parameter.

Here and in Sections 6–8, the ground field is $\mathbb{C}$. Let V, S have dimensions e, q , and let $\gamma : \text{Sym}^2 V \rightarrow S$. The augmented algebra $B_\gamma = \mathbb{C} \oplus V \oplus S$ has multiplication γ in degree two and zero products involving S and $V \oplus S$. For $K \in \text{Gr}(e, V \oplus S)$ set $W = \mathbb{C} \oplus K$. The scheme D_γ is defined by the zeroth Fitting ideal of the cokernel of $\text{Sym}^m W \rightarrow B_\gamma$; it is independent of $m \geq 2$. These are unital planes. They contain a unit and hence generate the corresponding sheaf, but need not generate B_γ as an algebra. This distinction is essential at a failure point.

Put $b = e - 1$, $p = \binom{e}{2}$, and suppose $1 \leq q \leq p$. Allow γ to vary over all surjections. On the projection-corank-one locus write $H = \text{im}(K \rightarrow V)$ and impose the open condition $\gamma(HV) = S$. If Δ denotes the projection determinant divisor and $Z_j \subset \Delta$ is defined by $\text{rank}(\gamma|_{\text{Sym}^2 H}) < j$, our primary flag theorem gives

$$\mathcal{I}_D = \mathcal{I}_\Delta \cap \bigcap_{j=\max\{1, q-e+2\}}^q \mathcal{I}_{Z_j}^{(q-j+2)}. \quad (5.1)$$

The intersection is irredundant; these are exactly its associated supports. The superscript denotes a symbolic power in the smooth ambient parameter space. The reduced divisor has multiplicity

one. At a point where the restricted quadratic map has rank a , the nilradical has exact index $q - a + 1$. The supports Z_j are normal, and their natural Grassmannian incidences are projective resolutions with explicitly determined fibres (Theorems 7.1 and 7.2). Thus the result is a primary classification on a specified universal open set, in arbitrary embedding dimension; it is not an assertion about unspecified specializations or about the omitted higher projection-corank strata.

The underlying algebra is particularly simple. For a generic q by p matrix A , introduce one independent parameter t . We prove

$$t \sum_{i=0}^q t^{q-i} I_i(A) = (t) \cap \bigcap_{j=1}^q (t, I_j(A))^{(q-j+2)}. \quad (5.2)$$

A coefficientwise symbolic-order argument gives both equality and explicit witnesses to irredundancy. The geometric input identifies the full multiplication ideal with the left side, not merely with its radical. The normality of generic determinantal supports is a classical input, not a new assertion of this paper.

For $e = q = 3$, the family

$$B_\lambda = \mathbb{C}[x, y, z] / ((x, y, z)^3, xy - \lambda z^2, yz - \lambda x^2, zx - \lambda y^2)$$

has Hilbert function $(1, 3, 3)$. For an explicit nonempty open set of parameters, its full failure ideal on the entire projection-corank-at-most-one open is

$$\mathcal{I}_{D_\lambda} = \mathcal{I}_\Delta \cap \mathcal{I}_{E_\lambda}^2, \quad 0 \longrightarrow \mathcal{O}_{E_\lambda}(-\Delta) \longrightarrow \mathcal{O}_{D_\lambda} \longrightarrow \mathcal{O}_\Delta \longrightarrow 0.$$

Here E_λ is smooth and normal, and is the total space of a rank-four bundle over $C_\lambda \times \mathbb{P}^2$, where

$$C_\lambda : \quad \lambda(a^3 + b^3 + c^3) + (1 - 4\lambda^3)abc = 0.$$

The curves C_λ have nonconstant j -invariant. In particular, this is a positive-moduli family of algebras and embedded primary supports, not two orbit types presented with a redundant parameter. The nilradical is an invertible sheaf on E_λ and has square zero. A smooth projective compactification of E_λ has Albanese variety C_λ . The conductor has constant rank one on this boundary, while the quadratic restriction retains the elliptic moduli (Theorem 8.1).

There are three distinct scopes in the article. The coefficient-ring stabilization and conductor comparison apply to general finite locally free algebras. The quadratic factorization applies to split augmented algebras with cube-zero augmentation ideal. The primary flag theorem classifies their universal first-order-spanning corank-one open; the $(1, 2, 2)$ theorem, by contrast, treats that entire fixed Hilbert-function class, including its extreme-corank locus. We retain the complete latter theorem and the higher-contact, relative-incidence, and global section-multiplication results. Their hypotheses are not replaced by the new open conditions.

The article is self-contained as an algebraic-geometric contribution. No theorem from a paper numbered A1, or an unverified application in a later paper, is part of its significance argument. Literature inputs and the precise remaining full-text comparison are separated in Section 26.3. The accompanying complete edition retains all complementary geometry and statistical proofs; these are not premises of the primary flag theorem.

6. A WEIGHTED DETERMINANTAL PRIMARY DECOMPOSITION

For an ideal J , write $I_i(A)$ for the ideal of i -minors of A , with $I_0(A) = (1)$. For a prime P in a noetherian ring R , put $P^{(s)} = P^s R_P \cap R$ for $s \geq 1$, and $P^{(s)} = R$ for $s \leq 0$.

Lemma 6.1 (Coefficientwise symbolic orders). *Let $R_0 = \mathbb{C}[x_1, \dots, x_n]$, let $J \subset R_0$ be prime, and put $P = (t, J) \subset R_0[t]$. For $m \geq 1$ and $f = \sum_{u \geq 0} t^u f_u$, one has*

$$f \in P^{(m)} \iff f_u \in J^{(m-u)} \text{ for every } 0 \leq u < m.$$

Proof. Localize R_0 at J , and denote its maximal ideal by $\mathfrak{m}$. In $R_{0,J}[t]$ the quotient by $(\mathfrak{m}, t)^m$ is an Artinian local ring: it is a finite module over $R_{0,J}/\mathfrak{m}^m$ and t is nilpotent. Hence $(\mathfrak{m}, t)^m$ is $(\mathfrak{m}, t)$ -primary, and contracting it from the localization at $(\mathfrak{m}, t)$ does not enlarge it. This localization is $R_0[t]_P$. Polynomial expansion gives

$$(\mathfrak{m}, t)^m = \bigoplus_{u \geq 0} t^u \mathfrak{m}^{\max\{m-u, 0\}}$$

as an $R_{0,J}$ -submodule. Contracting coefficient by coefficient to R_0 proves the assertion. No assumption about ordinary and symbolic powers of J is involved. $\square$

Lemma 6.2 (Order of a minor at a rank stratum). *Let A be a generic q by p matrix, where $p \geq q$. For $1 \leq j \leq k \leq q$,*

$$I_k(A) \subset I_j(A)^{(k-j+1)}.$$

The ideal $I_j(A)$ is prime and its quotient is normal.

Proof. Primality and normality are the classical generic determinantal results; for instance [39, Theorem 3.5(c)], with $r = j - 1$, gives normality and the domain assertion (the case $j = 1$ is a polynomial quotient). To prove the order statement, invert a $(j - 1)$ -minor not in $I_j(A)$; for $j = 1$ no inversion is needed. Invertible row and column operations replace A by $\text{diag}(I_{j-1}, A')$. At the prime $I_j(A)$, the entries of A' generate its extension. A k -minor must use at least $k - j + 1$ entries from A' . This proves the inclusion after localization at $I_j(A)$, and contraction proves the symbolic inclusion. $\square$

Theorem 6.3 (Weighted determinantal primary flag). *Let A be a generic q by p matrix with $p \geq q \geq 1$, let $R = \mathbb{C}[A, t]$, and set*

$$F = t \sum_{i=0}^q t^{q-i} I_i(A), \quad P_j = (t, I_j(A)).$$

Then an irredundant primary decomposition is

$$F = (t) \cap \bigcap_{j=1}^q P_j^{(q-j+2)}. \quad (6.1)$$

In particular

$$\text{Ass}_R(R/F) = \{(t), P_1, \dots, P_q\}, \quad \sqrt{F} = (t), \quad \text{codim } P_j = 1 + (q - j + 1)(p - j + 1).$$

The reduced divisor has generic multiplicity one. At a point of $t = 0$ where A has rank a , the nilradical of the local failure ring has exact index $q - a + 1$. In particular its maximal index is $q + 1$.

Proof. Every displayed component is primary: P_j is prime and a symbolic power is the contraction of a power of the maximal ideal of R_{P_j} . Membership in (t) requires $f_0 = 0$. For $u \geq 1$ put $k = q + 1 - u$. If $1 \leq k \leq q$, Lemma 6.1 says that the condition supplied by the j th component is

$$f_u \in I_j(A)^{(k-j+1)}.$$

The condition with $j = k$ is exactly $f_u \in I_k(A)$. It implies every condition with $j < k$ by Lemma 6.2; those with $j > k$ are void. For $u \geq q + 1$ all these conditions are void. Consequently the intersection has coefficient ideal $I_{q+1-u}(A)$ in degrees $1 \leq u \leq q$, coefficient ideal (1) in degrees $u \geq q + 1$, and zero constant coefficient. These are exactly the coefficient ideals of F , proving (6.1).

Here are explicit irredundancy witnesses. Choose any nonzero $(j - 1)$ -minor d_{j-1} , with $d_0 = 1$. The polynomial

$$g_j = t^{q-j+1} d_{j-1}^2$$

does not belong to $P_j^{(q-j+2)}$, since d_{j-1} is a unit at $I_j(A)$. It belongs to (t) . For $h > j$ its t -order alone is at least $q - h + 2$. For $h < j$, its order at P_h is at least

$$q - j + 1 + 2(j - h) \geq q - h + 2$$

by Lemma 6.2. It therefore belongs to all the other primary components. Finally, the square of a nonzero q -minor belongs to every $P_j^{(q-j+2)}$, since $2(q - j + 1) \geq q - j + 2$, but not to (t) . Thus no component can be removed, and the standard associated-prime characterization of an irredundant primary decomposition gives the asserted set. The codimension formula is the usual rank-matrix dimension count: rank $j - 1$ matrices have dimension $(j - 1)(q + p - j + 1)$.

At a rank- a point invert an a -minor and perform block elimination. The ideal generated by A and tI_q has a free rank- a image; a presentation of its cokernel is $[A', tI_{q-a}]$, with A' the $(q - a)$ by $(p - a)$ Schur complement. Since

$$I_q([A, tI_q]) = \sum_{i=0}^q t^{q-i} I_i(A),$$

the local failure ideal becomes

$$t \sum_{i=0}^{q-a} t^{q-a-i} I_i(A'). \quad (6.2)$$

All entries of A' vanish at the point. Its coefficient ring is a localized polynomial ring independent of t . Formula (6.2) contains t^{q-a+1} but not t^{q-a} when $q - a > 0$: the latter membership would require $1 \in I_1(A')$. The same nonmembership persists in the local ring at the point, since a denominator not vanishing there has invertible constant term modulo $I_1(A')$. For $a = q$ the local ideal is (t) and the nilradical is zero, whose index we take to be one. This proves both the exact local indices and generic multiplicity. $\square$

Remark 6.4 (What is and is not specialized). The ideal $tI_q([A, tI_q])$ and its coefficient formula can be substituted in any coefficient ring. The primary decomposition (6.1) uses independent matrix entries and is not asserted to remain a primary decomposition under an arbitrary substitution. Polynomial extension and localization do preserve the argument. The geometric application below is first made on the universal tensor space, where independence holds; the elliptic specialization is proved separately by local parameters.

7. UNIVERSAL QUADRATIC FLAGS AND THEIR RESOLUTIONS

Fix $e \geq 2$, $b = e - 1$, $p = \binom{e}{2}$, and $1 \leq q \leq p$. Let V, S be fixed vector spaces of dimensions e, q , and let

$$\Gamma = \text{Surj}(\text{Sym}^2 V, S), \quad \mathcal{X} = \Gamma \times \text{Gr}(e, V \oplus S).$$

The algebra $\mathcal{B} = \mathcal{O} \oplus V \oplus S$ over Γ has the universal quadratic multiplication γ and $S(V \oplus S) = 0$. Write $\mathcal{K}$ for the tautological bundle and D for the failure scheme of $\mathcal{O} \oplus \mathcal{K}$ in any degree $m \geq 2$.

Delete the locus $\text{rank}(\mathcal{K} \rightarrow V) \leq e - 2$. On its determinant divisor Δ there are bundles

$$H = \text{im}(\mathcal{K} \rightarrow V), \quad \Lambda = \mathcal{K} \cap S, \quad \text{rank } H = b, \quad \text{rank } \Lambda = 1.$$

The subspace $HV \subset \text{Sym}^2 V$ is generated by products with one factor in H . Delete also the closed subset of this divisor on which $\gamma(HV) \neq S$. Denote the resulting open scheme by $\mathcal{U}$, and continue to write D, Δ for their restrictions to it. For $1 \leq j \leq q$, define the closed subscheme

$$Z_j = \{(\gamma, K) \in \Delta : \text{rank}(\gamma|_{\text{Sym}^2 H}) < j\} \quad (7.1)$$

using the j -minor ideal on Δ . As a subscheme of $\mathcal{U}$ its ideal includes $\mathcal{I}_\Delta$.

Theorem 7.1 (Universal first-order-spanning primary classification). *Put $j_0 = \max\{1, q - b + 1\}$. The schemes Z_j with $j < j_0$ are empty; each Z_j with $j_0 \leq j \leq q$ is nonempty, integral and normal. On the whole of $\mathcal{U}$,*

$$\mathcal{I}_D = \mathcal{I}_\Delta \cap \bigcap_{j=j_0}^q \mathcal{I}_{Z_j}^{(q-j+2)} \quad (7.2)$$

is an irredundant primary decomposition. Its associated points are exactly the generic point of Δ and the generic points of these Z_j . In particular there are no other associated supports on $\mathcal{U}$.

The divisor Δ is smooth, and its generic multiplicity in D is one. At a point of Δ where $\text{rank}(\gamma|_{\text{Sym}^2 H}) = a$, the nilradical has exact index $q - a + 1$. Its maximal index on $\mathcal{U}$ is $\min\{q, b\} + 1$. The codimension of Z_j in $\mathcal{U}$ is $1 + (q - j + 1)(p - j + 1)$.

Proof. Work on a frame chart of $\mathcal{K}$. Write its inclusion as a stacked matrix with upper block $M : \mathcal{K} \rightarrow V$. The block-minor argument of Theorem 17.1 gives, over the universal coefficient ring,

$$\mathcal{I}_D = (\det M) I_q(\gamma \text{Sym}^2 M).$$

Indeed all products of degree at least three in K vanish, and the image is $\mathcal{O} + K + K^2$. Every maximal minor of the full presentation must use all e linear columns. This supplies the factor $\det M$ before any quadratic minor is taken.

Near a corank-one point, invert a b -minor of M . Block elimination and invertible changes of the source and target bases give $M = \text{diag}(I_b, t)$, with t a local equation for Δ . With respect to $V = H \oplus \langle v \rangle$, let A, B, c be the blocks of γ on $\text{Sym}^2 H, Hv, v^2$. They have sizes $q \times p$, $q \times b$, and $q \times 1$. Up to harmless nonzero scalar factors in the symmetric basis,

$$\gamma \text{Sym}^2 M = [A, tB, t^2c].$$

The condition on $\mathcal{U}$ says that $[A, B]$ is surjective at $t = 0$. Shrink the neighborhood so that it has a right inverse. This gives an equality of column modules over the local coefficient ring:

$$\text{im}[A, tB, t^2c] = \text{im}[A, tI_q]. \quad (7.3)$$

For the nontrivial inclusion, write $I_q = AU + BV$ using the right inverse and multiply by t . The reverse inclusion follows because tB and t^2c are generated by tI_q . Thus

$$\mathcal{I}_D = t \sum_{i=0}^q t^{q-i} I_i(A). \quad (7.4)$$

The entries of A, B, c are independent coordinates of the universal tensor space after the indicated invertible changes of basis; the other frame coordinates add polynomial variables and localizations. Theorem 6.3 therefore applies. Its prime $(t, I_j(A))$ is exactly the intrinsic ideal in (7.1). On the spanning open, a rank- $(j - 1)$ choice of A is possible exactly when $q - j + 1 \leq b$. It is obtained by completing its image with the b columns of B . Hence exactly the primes indexed by $j \geq j_0$ survive. Their generic strata also meet the full-frame open: prescribe $M = \text{diag}(I_b, 0)$ and give its last column a nonzero S component. The witnesses in Theorem 6.3, or its local rank-eliminated form at each surviving generic point, show that the surviving decomposition is irredundant.

For completeness, the global geometry of each support can be read without a frame. The divisor of corank-one planes is the total space of $\text{Hom}(H, S/\Lambda)$ over $\text{Gr}(b, V) \times \mathbb{P}(S)$: a plane with projection H and kernel Λ is exactly the inverse graph of a map $H \rightarrow S/\Lambda$. Over this smooth base the restriction map from all tensors γ to $\text{Hom}(\text{Sym}^2 H, S)$ is a surjective vector-bundle map. Thus Z_j is, locally, a generic determinantal scheme times an affine space, restricted to the nonempty spanning open. The determinantal rank stratum is irreducible; the vector-bundle description shows that these local models form an integral global scheme. Lemma 6.2 gives its normality.

Symbolic powers localize, so the local intersections glue. Alternatively, apply Lemma 9.1 to the intrinsic ideals just defined. It also identifies the associated points on $\mathcal{U}$, rather than only on its frames. Off Δ , M is invertible, so all ideals in the decomposition are unit ideals and D is empty. The local nilpotency calculation is Theorem 6.3. Since $a \geq \max\{0, q - b\}$ on the spanning locus and that lower rank occurs, its maximal index is as stated. The same local models give the codimension and generic multiplicity assertions. $\square$

Theorem 7.2 (Resolutions of the embedded supports). *For $j_0 \leq j \leq q$, let $\tilde{Z}_j$ parametrize triples $((\gamma, K), T)$ with $(\gamma, K) \in \Delta$ and*

$$T \in \mathrm{Gr}(j-1, S), \quad \gamma(\mathrm{Sym}^2 H) \subset T.$$

Then $\tilde{Z}_j$ is smooth and integral, and the forgetful map $\rho_j : \tilde{Z}_j \rightarrow Z_j$ is a projective birational resolution. At a point where $\gamma|_{\mathrm{Sym}^2 H}$ has rank r , its scheme-theoretic fibre is

$$\mathrm{Gr}(j-1-r, S/\gamma(\mathrm{Sym}^2 H)) \simeq \mathrm{Gr}(j-1-r, q-r). \quad (7.5)$$

These constructions are equivariant for changes of bases in V, S and for frame changes of K .

Proof. Start with $\mathrm{Gr}(b, V) \times \mathbb{P}(S) \times \mathrm{Gr}(j-1, S)$, carrying the bundles H, Λ, T . The condition $\gamma(\mathrm{Sym}^2 H) \subset T$ is the kernel of the surjective bundle map

$$\mathrm{Hom}(\mathrm{Sym}^2 V, S) \longrightarrow \mathrm{Hom}(\mathrm{Sym}^2 H, S/T).$$

Consequently its solution space is a vector bundle. Adjoin the independent bundle $\mathrm{Hom}(H, S/\Lambda)$ recording K , and restrict to $\gamma(HV) = S$. This is precisely $\tilde{Z}_j$. It is a nonempty open of a vector bundle over an integral smooth base, hence is smooth and integral. The projection is projective, since it is a closed incidence in the relative Grassmannian over Δ restricted to Z_j . It is surjective and has one-point fibre when the image of $\mathrm{Sym}^2 H$ has dimension $j-1$; over that open the inverse sends a matrix to its image subbundle and is a morphism. This proves birationality. Over a point of rank r , choosing T is exactly choosing a $(j-1-r)$ -subspace of the quotient by that image; the same statement for test schemes proves the fibre formula with its scheme structure. All defining maps are intrinsic, proving equivariance. $\square$

Proposition 7.3 (Conductor and restriction flags). *On the corank-one locus of any fixed B_γ , put $W = \mathbb{C} \oplus K$, $H = \mathrm{im}(K \rightarrow V)$ and $\Lambda = K \cap S$. Then W is a subalgebra exactly when $\gamma(\mathrm{Sym}^2 H) \subset \Lambda$. The largest ideal of B_γ contained in W is*

$$\mathrm{cond}_{B_\gamma}(W) = \ker [K \longrightarrow \mathrm{Hom}(V, S/\Lambda)], \quad k \longmapsto (v \mapsto \gamma(\bar{k}, v)). \quad (7.6)$$

On geometric fibres its dimension is $1 + \dim\{h \in H : \gamma(h, V) \subset \Lambda\}$. The kernel commutes with base change on every exact-rank stratum of the displayed map. Thus the restriction-rank flag defining the primary supports and the conductor flag are computed by two specified maps of the same quadratic tensor; neither is replaced by a numerical conductor dimension.

Proof. The multiplication image is $\mathbb{C} + K + \gamma(\mathrm{Sym}^2 H)$, proving the closure assertion. A conductor element with nonzero constant coefficient would be a unit of the local algebra and would force $B_\gamma \subset W$, which is impossible. Over coefficient rings the same exclusion follows when $q > 1$ by multiplying with the locally free quotient S/Λ ; when $q = 1$, multiply with V/H , which is a free line and receives the scalar coefficient of multiplication by a lift of V/H . For $k \in K$, multiplication by S is zero and multiplication by a lift of v is $\gamma(\bar{k}, v)$. Such a product is in K exactly when it is in Λ . This proves the kernel formula, including over coefficient rings. Finally $K \rightarrow H$ has kernel Λ , and on a scheme-theoretic exact-rank locus the indicated matrix has locally split kernel and image. The dimension and base-change assertions follow. $\square$

8. AN ELLIPTIC FAMILY OF EMBEDDED PRIMARY BOUNDARIES

The universal theorem has a global consequence in a non-isotrivial family of algebras of Hilbert function $(1, 3, 3)$. In this section the parameter tensor is specialized, so its primary decomposition is proved directly; no compatibility of primary components with arbitrary base change is presumed.

Let

$$B_\lambda = \mathbb{C}[x, y, z]/((x, y, z)^3, xy - \lambda z^2, yz - \lambda x^2, zx - \lambda y^2). \quad (8.1)$$

The elements $1, x, y, z, x^2, y^2, z^2$ are a basis, also over the polynomial parameter ring. Thus this is a finite free rank-seven family. Write $V = \langle x, y, z \rangle$ and $S = \langle x^2, y^2, z^2 \rangle$. Work over the nonempty open

$$\Omega = \{\lambda : \lambda \neq 0, \lambda^3 \neq 1, (4\lambda^3 - 1)^3 \neq 27\lambda^3\}. \quad (8.2)$$

For example $\lambda = 1/2$ belongs to Ω . Let $X = \text{Gr}(3, V \oplus S)$ and $U = \{K : \text{rank}(K \rightarrow V) \geq 2\} \subset X$. Its determinant divisor Δ has projection rank exactly two and is independent of λ . On $\mathbb{P}(V^*)$ put

$$C_\lambda : F_\lambda(a, b, c) = \lambda(a^3 + b^3 + c^3) + (1 - 4\lambda^3)abc = 0. \quad (8.3)$$

Denote by $E_\lambda \subset \Delta$ its inverse image under $K \mapsto H = \text{im}(K \rightarrow V)$.

Theorem 8.1 (Elliptic embedded boundary). *For every $\lambda \in \Omega$ the curve C_λ is smooth and has genus one. For all $m \geq 2$, the full failure scheme D_λ of $\text{Sym}^m(\mathbb{C} \oplus K) \rightarrow B_\lambda$ on all of U has the irredundant primary decomposition*

$$\mathcal{I}_{D_\lambda} = \mathcal{I}_\Delta \mathcal{I}_{E_\lambda} = \mathcal{I}_\Delta \cap \mathcal{I}_{E_\lambda}^2. \quad (8.4)$$

Its only associated points are the generic points of Δ and E_λ . The reduced divisor is smooth and has generic multiplicity one. The embedded support is smooth and normal, with

$$E_\lambda \simeq \text{Tot}_{C_\lambda \times \mathbb{P}(S)} \text{Hom}(H, S/\Lambda), \quad (8.5)$$

where H is the tautological hyperplane bundle and Λ is the tautological line of $\mathbb{P}(S)$. This is a rank-four bundle, so $\dim E_\lambda = 7$ and $\text{codim}_U E_\lambda = 2$.

The nilradical has square zero and is the line bundle $\mathcal{O}_{E_\lambda}(-\Delta)$, viewed as an $\mathcal{O}_{D_\lambda}$ -module supported on E_λ . More precisely there is an exact sequence

$$0 \longrightarrow \mathcal{O}_{E_\lambda}(-\Delta) \longrightarrow \mathcal{O}_{D_\lambda} \longrightarrow \mathcal{O}_\Delta \longrightarrow 0. \quad (8.6)$$

For every $K \in \Delta$, the conductor of $\mathbb{C} \oplus K$ in B_λ is exactly $\Lambda = K \cap S$.

The family has genuine moduli: the elliptic curves have nonconstant j -invariant, and distinct j -values give nonisomorphic algebras B_λ . A smooth projective compactification of E_λ is a projective bundle over $C_\lambda \times \mathbb{P}^2$ and has Albanese variety C_λ .

Proof. A hyperplane with equation $ax + by + cz = 0$, on the chart $c \neq 0$, has basis $u = (c, 0, -a)$ and $v = (0, c, -b)$. The restriction of the quadratic multiplication, in the product basis u^2, uv, v^2 , is

$$A_\lambda = \begin{pmatrix} c^2 & -\lambda ac & -2\lambda bc \\ -2\lambda ac & -\lambda bc & c^2 \\ a^2 & ab + \lambda c^2 & b^2 \end{pmatrix}, \quad \det A_\lambda = -c^3 F_\lambda(a, b, c). \quad (8.7)$$

Expansion along the first row proves the determinant identity. Cyclic charts give the same equation up to the transition unit of $\det(\text{Sym}^2 H)^*$. Since $\det(\text{Sym}^2 H) = (\det H)^3$, this determinant is a section of $\mathcal{O}_{\mathbb{P}(V^*)}(3)$. Thus (8.3) defines its global degeneracy curve, not a component seen only on one chart.

Divide F_λ by λ and set

$$k = \frac{4\lambda^3 - 1}{3\lambda}.$$

This gives the Hesse cubic $a^3 + b^3 + c^3 - 3kabc = 0$. Its three partial derivatives vanish simultaneously at a projective point exactly when $k^3 = 1$: no coordinate of such a point is zero, and multiplication of $a^2 = kbc$, $b^2 = kac$, $c^2 = kab$ forces $k^3 = 1$; the converse follows by solving these equations. Condition (8.2) therefore makes C_λ smooth. It is integral, since distinct components of a reducible plane cubic intersect and give singularities. Adjunction gives genus one, and $[1 : -1 : 0]$ is a point on every fibre. Moreover A_λ has rank exactly two along the curve. If its rank were at most one, all its cofactors, hence all derivatives of its determinant in the local hyperplane coordinates, would vanish, contrary to smoothness.

We next check the first-order-spanning hypothesis at every hyperplane, not only at a general point of the curve. A nonzero functional of S annihilating $\gamma(HV)$ would give a quadratic form in the dual net vanishing on HV . Such a form must be a square of the equation of H , up to a scalar. The dual net consists of

$$\alpha(X^2 + 2\lambda YZ) + \beta(Y^2 + 2\lambda ZX) + \chi(Z^2 + 2\lambda XY).$$

A square $(aX + bY + cZ)^2$ belongs to it only if

$$ab = \lambda c^2, \quad bc = \lambda a^2, \quad ca = \lambda b^2.$$

When $\lambda \neq 0$, a nonzero solution has all coordinates nonzero. Their product gives $\lambda^3 = 1$, excluded on Ω . It follows that $\gamma(HV) = S$ for every H .

Now fix $K \in \Delta$. The block elimination of (7.4) is valid for this particular γ . If $H \notin C_\lambda$, the restriction has rank three, and the local ideal is (t) . If $H \in C_\lambda$, invert a nonzero two-minor of A_λ . Eliminating it leaves a one-row presentation, so the local ideal is

$$(t^2, tf), \tag{8.8}$$

where t is a local equation for Δ and f is the remaining Schur entry, a unit times F_λ in local hyperplane coordinates. Since C_λ is smooth and the map $\Delta \rightarrow \mathbb{P}(V^*)$ is smooth, t, f are part of a regular system of parameters. Off Δ , M is invertible and the failure scheme is empty. This proves the ideal calculation on all of U without specializing a universal primary decomposition.

The intrinsic graph description of corank-one planes gives (8.5), so E_λ is smooth and integral. Let $P = (t, f)$ locally. Its associated graded ring is a polynomial ring in two variables over the integral ring of E_λ . Consequently P^2 is P -primary and $(P^2 : t) = P$. It follows that $(t) \cap P^2 = tP$, proving (8.4) and its irredundancy. The two intrinsic ideals glue, and Lemma 9.1 supplies an alternative frame-to-scheme verification. The ideal $\mathcal{I}_\Delta$ is invertible, and therefore

$$\mathcal{I}_\Delta / (\mathcal{I}_\Delta \mathcal{I}_{E_\lambda}) \simeq \mathcal{I}_\Delta \otimes \mathcal{O}_{E_\lambda}.$$

This is the kernel in (8.6); its square vanishes because $\mathcal{I}_\Delta \subset \mathcal{I}_{E_\lambda}$. It is nonzero along E_λ . This proves all nilradical and multiplicity assertions.

For the conductor, multiplication by a vector $(x, y, z) \in V$ has matrix

$$T(x, y, z) = \begin{pmatrix} x & \lambda z & \lambda y \\ \lambda z & y & \lambda x \\ \lambda y & \lambda x & z \end{pmatrix}.$$

For nonzero (x, y, z) this matrix has rank at least two on Ω . Indeed its principal two-minors imply, if its rank is at most one, $xy = \lambda^2 z^2$, $yz = \lambda^2 x^2$, and $zx = \lambda^2 y^2$. No coordinate can then vanish. The minor in rows 1, 2 and columns 1, 3 also gives $x^2 = \lambda yz$. Combining the latter equality with $yz = \lambda^2 x^2$ gives $\lambda^3 = 1$, a contradiction. Thus no nonzero element of H multiplies all of V into a line Λ . Proposition 7.3 gives $\text{cond}(W) = \Lambda$. Notice also that W is not a subalgebra on this locus, since $\text{rank } \gamma(\text{Sym}^2 H) \geq 2$. Here the conductor means the largest ideal contained in W ; it does not presuppose that W is a subalgebra.

Finally the Hesse Weierstrass calculation gives

$$j(C_\lambda) = \frac{27k^3(k^3 + 8)^3}{(k^3 - 1)^3}, \quad k = \frac{4\lambda^3 - 1}{3\lambda}. \quad (8.9)$$

For normalization, substitute $(u_0, u_1) = (1, -k/2)$ in [40, Section 2, equations (4)–(5)] and use $j = 1728 4A^3/(4A^3 + 27B^2)$ for its Weierstrass coefficients. This yields (8.9). It is a nonconstant rational function of λ . An isomorphism of local algebras induces isomorphisms of $\mathfrak{m}/\mathfrak{m}^2$ and $\mathfrak{m}^2$ intertwining γ . It therefore carries the hyperplane restriction degeneracy curve projectively to the corresponding curve. Distinct j -values give nonisomorphic algebras.

The compactification

$$\overline{E}_\lambda = \mathbb{P}_{C_\lambda \times \mathbb{P}(S)}(\mathcal{O} \oplus \text{Hom}(H, S/\Lambda))$$

contains E_λ as the affine-bundle chart. It is smooth and projective. A morphism from a projective space to an abelian variety is constant; applying this first on its projective-bundle fibres and then on the $\mathbb{P}(S)$ fibres shows that its Albanese is the Jacobian of C_λ , identified with C_λ by the specified point. Equivalently, projective-bundle cohomology gives $h^1(\mathcal{O}_{\overline{E}_\lambda}) = 1$. This gives a global geometric invariant carried by the embedded support although the reduced determinant divisor stays fixed. $\square$

Corollary 8.2 (A non-isotrivial family of full failure schemes). *Over Ω , the construction above gives relative schemes $\mathcal{E} \subset \Omega \times \Delta$ and $\mathcal{D} \subset \Omega \times U$ with*

$$\mathcal{I}_\mathcal{D} = \mathcal{I}_{\Omega \times \Delta} \mathcal{I}_\mathcal{E}, \quad 0 \rightarrow \mathcal{O}_\mathcal{E}(-\Delta) \rightarrow \mathcal{O}_\mathcal{D} \rightarrow \mathcal{O}_{\Omega \times \Delta} \rightarrow 0.$$

The morphisms $\mathcal{E} \rightarrow \Omega$ and $\mathcal{D} \rightarrow \Omega$ are flat; the former is smooth. Each geometric fibre has exactly the two associated supports in Theorem 8.1.

Proof. The cubic equation defines a smooth proper family of plane curves on Ω , since every fibre has nonvanishing relative gradient. The graph construction is a vector bundle over its product with $\mathbb{P}(S)$, hence $\mathcal{E} \rightarrow \Omega$ is smooth. The local proof of (8.8) works over the parameter ring: its two-minor is invertible and the residual equation f cuts out that smooth family. It gives the displayed ideal and exact sequence before specialization. Both outer terms of the sequence are flat over Ω , so the middle term is flat. Tensoring the sequence with any residue field is exact, and the fibre proof above gives the stated primary decomposition. Flatness here is a conclusion of this specific family, not a general assertion about Fitting schemes. $\square$

9. PRIMARY DESCENT FROM FRAMES

We record explicitly the descent used in the global primary statements. Its hypotheses are verified on Zariski frame charts; no claim about arbitrary regular-fibre morphisms is needed.

Lemma 9.1 (Frame descent of primary decompositions). *Let X be a noetherian scheme, let $\mathcal{K}$ be a vector bundle, and let $f : F = \text{Fr}(\mathcal{K}) \rightarrow X$ be its frame torsor. Suppose $\mathcal{I}, \mathcal{Q}_1, \dots, \mathcal{Q}_s$ are coherent ideals of $\mathcal{O}_X$ and $Y_1, \dots, Y_s \subset X$ are distinct integral closed subschemes. Assume that on F the pulled-back ideals give a primary decomposition*

$$f^*\mathcal{I} = \bigcap_i f^*\mathcal{Q}_i, \quad \sqrt{f^*\mathcal{Q}_i} = \mathcal{I}_{f^{-1}Y_i},$$

with each nonempty local component primary. Then $\mathcal{I} = \bigcap_i \mathcal{Q}_i$ on X , each $\mathcal{Q}_i$ is primary along Y_i , and the associated points of $\mathcal{O}_X/\mathcal{I}$ are the generic points of the components in its irredundant decomposition. The intersection on X is irredundant if and only if the intersection on F is irredundant.

The hypotheses can be checked by polynomial extension and localization of affine primary decompositions, provided every claimed associated stratum meets the frame open. For an invariant

symbolic-power ideal, its intrinsic description on X is its symbolic power at the generic point of the indicated integral support.

Proof. On an affine trivializing open $U = \operatorname{Spec} A$, one has

$$f^{-1}U = \operatorname{Spec} A[z_{ij}, \det(z)^{-1}].$$

This ring is faithfully flat over A , and the inclusion has the retraction $z_{ij} \mapsto \delta_{ij}$. Flatness commutes with a finite intersection of ideals, as follows by tensoring the kernel sequence defining the intersection. Faithful flatness detects ideal equality, so the displayed intersection contracts to the asserted one on U .

If $ab \in \mathcal{Q}_i(U)$ and $b \notin \mathcal{Q}_i(U)$, faithfulness implies that b remains outside its extended ideal. Primarity upstairs therefore implies that a power of a lies in the extended ideal, and contraction puts that power in $\mathcal{Q}_i(U)$. The radical contracts to the ideal of Y_i . When a trivializing open requires several affine subcharts of F , the same argument is local: an element nonzero in the quotient remains nonzero on some such chart; the prime support specifies its unique generic point. This proves the local primary assertion. Conversely polynomial extension of a primary ideal is primary, and localization preserves primarity unless the ideal becomes the unit ideal. One may verify the polynomial assertion by the usual zero-divisor criterion for polynomials over a primary ring, or by contracting from the localization at the extended prime.

If a component can be omitted upstairs, faithful flatness contracts the resulting equality. If it can be omitted downstairs, flatness extends the equality. This proves the irredundancy equivalence. On an affine chart the associated primes of an irredundant primary decomposition with distinct radicals are exactly those radicals; applying this statement and gluing gives the associated points on X . In particular the assertion is not deduced solely from a statement about fibre regularity.

For symbolic powers, polynomial extension followed by localization at the extended prime commutes with taking the contraction of a power of the maximal ideal. Equivalently apply the defining membership test at the generic point on each trivialization. Thus the ideals described this way agree on overlaps with the intrinsic symbolic-power sheaves. All assertions are compatible with further Zariski localization. $\square$

Corollary 9.2 (The global ideals in Hilbert function $(1, 2, 2)$). *In Theorem 17.2, let $\mathcal{D} = \mathcal{I}_\Delta$ and let $\mathcal{P}, \mathcal{Q}$ denote the ideals of the factor surfaces Y_L (one surface in the square case, two in the distinct factor case). In the latter case put*

$$\begin{aligned} \mathcal{N} &= \mathcal{P} + \mathcal{Q} = \mathcal{I}_{\{K=S_2\}}, & \mathcal{J} &= (\mathcal{I}_D : \mathcal{D}^2), \\ \mathcal{Q}_0 &= \mathcal{J} + \mathcal{P}^2 + \mathcal{Q}^2, & \mathcal{Q}_* &= \mathcal{D}^2 \mathcal{Q}_0 + \mathcal{N}^7. \end{aligned}$$

These are global coherent ideals on $\operatorname{Gr}(2, V \oplus S_2)$. The square decomposition is $\mathcal{D}^2 \cap \mathcal{P}^4$; the distinct-factor decomposition is $\mathcal{D}^2 \cap \mathcal{P}^3 \cap \mathcal{Q}^3 \cap \mathcal{Q}_$. Both are irredundant globally and their associated points are exactly those stated in Theorem 17.2.*

Proof. The surface ideal is the zero ideal of the bundle map $K \rightarrow V/L$, and the point ideal is the zero ideal of $K \rightarrow V$. Thus the ideals $\mathcal{P}, \mathcal{Q}, \mathcal{N}$ are intrinsic; on frames they are $(a, c), (b, d), (a, b, c, d)$ after choosing the factor lines as axes. The determinant ideal is intrinsic as well. The ideal quotient by its square commutes with flat pullback: locally that square is generated by a nonzerodivisor, and the quotient is the kernel of multiplication by that generator on $\mathcal{O}_X/\mathcal{I}_D$. This identifies its pullback with the ideal $\mathcal{J}$ used in the affine calculation. The stated sums and products therefore pull back to $\mathcal{Q}_0, \mathcal{Q}_*$ exactly.

The affine primary equalities are proved in Theorem 17.2. Polynomial extension by the lower frame block and restriction to the full-frame open preserve them. Every associated stratum survives this restriction: the lower block can be the identity matrix, independently of the upper block. The strata $\Delta, Y_L, \{K = S_2\}$ are integral, and their inverse images under a frame torsor are

integral. Apply Lemma 9.1 to obtain equality, primarity, irredundancy and the exact associated-point set on the Grassmannian. Permuting the two factor lines permutes the two surface ideals and leaves the unordered decomposition unchanged. Thus no labelled choice is needed for the asserted geometric collection. $\square$

10. FINITE-ALGEBRA AND GLOBAL CONDUCTOR FRAMEWORK

The multiplication map of a linear series can fail to be surjective with a scheme structure which changes even when its support does not. We study the actual multiplication cokernel on generating Grassmannians of finite algebras, and transport it to global sections of projective schemes. The central link is a codimension bound over the coefficient ring, followed by a conductor incidence which retains the local rings of the failure scheme.

Let $W \subset B$ contain 1 and have locally free quotient of rank r . Theorem 13.1 proves $W^{r+1} = \mathcal{O}[W]$ over every $\mathbb{Q}$ -algebra. The proof compresses the commuting multiplication operators to B/W . Their commutators land in the quadratic residual image; polarized Cayley–Hamilton then bounds the residual filtration without discarding nilpotents on the base. In codimension two the entire stable failure scheme is therefore defined by two by two minors of one degree-three residual matrix (Theorem 16.1). Its entry ideal defines a different scheme: the extreme-corank subalgebra locus.

Off that locus the full failure scheme is an open relative Grassmannian over $\text{Hilb}^2(B)$. This is an isomorphism of schemes, not only a parameterization of geometric points. The proper incidence extending it has fibres controlled by the conductor of the codimension-two subalgebra. When the conductor action has rank one, the fibre is the length-two quotient scheme of a rank-three algebra, expressed by a binary cubic. When the action has rank two, it is the spectrum of a quadratic algebra (Proposition 16.2). The rank-three and rank-four alternatives of Theorem 15.4 thus govern one incidence over the full multiplication failure scheme; they do not by themselves classify every primary component on its extreme-corank locus.

The incidence yields a systematic non-Cartier family. For $B_h = \mathbb{C}[x, y]/(x^2, xy, y^h)$, $h \geq 5$, the full failure scheme of unital $(h-1)$ -planes has local rings which are localized polynomial extensions of

$$\mathbb{C}[b, u, v]/(F_h, vF_{h-1}, bv, b^2u), \quad F_1 = 1, \quad F_{j+1} = uF_j + vF_{j-1}.$$

Their exact primary decomposition is

$$(u, v) \cap (b^2, bv, F_h, vF_{h-1}).$$

The minimal support has codimension $h-3$, an embedded component lies over $b = u = v = 0$, and the nilradical has exact index $2h-3$ (Theorem 18.1). The mixed equations bv and b^2u couple the extra tangent direction to the higher contact coefficients. Corollary 18.2 also supplies genuine three-generator examples. The unit-cover description places these local rings on smooth covers of the full generating Grassmannians, rather than on selected linear slices.

A general conductor theorem completes the passage to global geometry. If I_Z is the saturated ideal of any zero-dimensional scheme in $\mathbb{P}^e$, then $n \geq 2 \text{reg } I_Z - 1$ suffices to identify the global and finite multiplication cokernels in every degree $m \geq 2$ (Theorems 19.1 and 19.2). It applies to nonmonomial local schemes, unions, and finite flat families with a uniform regularity bound. The proof fills the conormal layer and the ordinary ideal-square layer. The square-regularity bound is the classical input of [35, Theorem 1.8]; the exact cokernel comparison is proved here. Corollary 19.3 transports the preceding embedded components to actual section multiplication on $\mathbb{P}^2$.

These results extend, rather than replace, the preceding calculations. The stable generated algebra still carries its representation on the multiplication defect; its exact-action-rank schemes

define conductor flags $C \subset Q$ compatible with base change on each stratum (Theorem 15.1). Their rank transitions can themselves be nonreduced. The following determinant-power family provides a complementary, pure-primary case of the theory.

Let Z be the order- h fat point of $\mathbb{P}^e$, with $e \geq 2$ and $h \geq 3$. For every $n \geq 2h - 1$, take all series containing $H^0(\mathcal{I}_Z(n))$ and restricting to a generating $(e + 1)$ -plane on Z . In symmetric degrees $m \geq h - 1$ their failure ideal is

$$\mathcal{I}_\Delta^{\binom{e+h-1}{e+1}}, \quad (10.1)$$

where Δ is the integral determinant divisor of the map from the augmentation kernel of the restricted series to $\mathfrak{m}/\mathfrak{m}^2$. This is the full primary ideal, and all its powers remain primary (Corollary 21.3). The proof separates a filtered substitution determinant from a triangular global conductor argument. It gives three-plane results on every $\mathbb{C}[x, y]/(x, y)^h$ and works in arbitrary embedding dimension; no diagonal-ideal theorem is used. The global range depends on the fat-point order, rather than its binomially growing length.

Let $D \subset \mathbb{P}^1$ have length d , put $E_{n,D} = H^0(\mathcal{O}_D(n))$ and $W_D = H^0(\mathcal{O}(n)(-D))$, and let $A \subset E_{n,D}$ be a generating k -plane. Write $U_A = \text{res}_D^{-1}(A)$. For $1 \leq k < d$ our conductor range is $n \geq 2d - k$. In every symmetric degree $m \geq 2$ it gives

$$\text{coker}(\text{Sym}^m \mathcal{U}_A \rightarrow V_{mn}) \simeq \text{coker}(\text{Sym}^m \mathcal{A} \rightarrow E_{mn,D}). \quad (10.2)$$

The isomorphism holds on the entire relative generating Grassmannian and after arbitrary base change. The range is sharp uniformly in D, A, m : at $n = 2d - k - 1$ an explicit monomial series already violates it in degree two. The exact fibrewise obstruction is the first cohomology of an evaluation kernel. On a curve of genus g , intrinsic cohomological hypotheses give the same transport; the uniform sufficient bound is $\deg L \geq 2d + 2g - 1$.

The defect-one structural theorem concerns generating hyperplanes in an arbitrary finite locally free algebra, not only pencils or powers of one element. Its failure scheme in all degrees $m \geq 2$ is the space of rank-two quotient algebras with a generating line. The identification is an isomorphism of functors on arbitrary, including nonreduced, test schemes. It follows from the conductor of a unital hyperplane subalgebra and a cyclic presentation of the multiplication cokernel (Theorem 14.2).

For $D = \sum_i d_i p_i$ this gives a complete primary classification of all generating $(d - 1)$ -planes. The components supported on $2p_i$ have transverse algebra

$$G_{d_i} = \mathbb{C}[u, v]/(F_{d_i}, F_{d_i+1}), \quad F_0 = 0, \quad F_1 = 1, \quad F_{j+1} = uF_j + vF_{j-1};$$

those supported on $p_i + p_j$ have transverse algebra $\mathbb{C}[x, y]/(x^{d_i}, y^{d_j})$. Their lengths are $\binom{d_i}{2}$ and $d_i d_j$, but their exact nilpotency indices are $2d_i - 3$ and $d_i + d_j - 1$. The reduced components are disjoint smooth curves on the generating open. The moving scheme is smooth and is a generating open of a projective-line bundle over $\text{Sym}^2 C \times \text{Sym}^{d-2} C$. These results apply to global multiplication on $\mathbb{P}^1$ for $n \geq d + 1$. They classify the whole contact-hyperplane family for every multiplicity partition, not a collection of transverse examples.

There is also a complete stable description for generating three-planes on a reduced divisor. On unit frames its ideal is

$$\bigcap_{i < j} (x_i - x_j, y_i - y_j).$$

The equality, and the analogous equality for every power of the ideal, use Haiman's diagonal-ideal theorem explicitly. The step specific to multiplication is an exact total-degree truncation and the conductor comparison; the radical-ideal theorem is a classical input, not a new assertion here.

The pencil family and the distinction between fibres and total spaces. The earlier contact-pencil classification remains part of the article, now in the improved range. We record its complete statement to fix the bases of its primary assertions.

Theorem 10.1 (Contact-pencil failure schemes). *Let $d \geq 4$ and $n \geq 2d - 2$. On the generating open $G_D^\circ \subset \text{Gr}(2, E_{n,D})$, let $\mathcal{J}_m$ be the zeroth Fitting ideal of the global multiplication cokernel. For $2 \leq m < d - 1$, this ideal is zero: the failure scheme is the whole parameter space. For $m \geq d - 1$ it is a Cartier divisor, and, on a fixed divisor $D = \sum_i d_i p_i$, its complete primary decomposition is*

$$\mathcal{J}_m = \bigcap_{i: d_i \geq 2} \mathcal{I}_{T_i}^{\binom{d_i}{2}} \cap \bigcap_{i < j} \mathcal{I}_{E_{ij}}^{d_i d_j}. \quad (10.3)$$

Here T_i and E_{ij} are failure to separate $2p_i$ and $p_i + p_j$. These are distinct smooth irreducible divisors; there are no embedded associated points. The exact local equations, all coranks and infinitesimal layers are given in Theorems 23.2 and 23.3. Over a moving multiplicity stratum, primary descent is governed by Lemma 22.1.

With D unrestricted, the pencil failure divisor is integral, and its normalization is the smooth finite incidence marking a scalar length-two subdivisor. The normalization and its singular-support test are given in Theorem 24.2 and Proposition 24.3.

On a unit pencil frame, with $D = (g)$ and ratio v , the determinant is the classical index form $\Delta(g, v) = \det[1, v, \dots, v^{d-1}]$. Its relation to the characteristic polynomial χ_v is

$$\text{disc}(\chi_v) = \text{disc}(g)\Delta(g, v)^2.$$

The equation of failure is $\Delta = 0$, not the discriminant on the left: the latter also vanishes when g has repeated roots, even if v generates the algebra. The quotient and index-form constructions are related to the schemes of generators in [29]. The Grassmannian cohomology presentation of G_r is likewise classical. What must be proved here is the exact identification with the multiplication schemes, including primary fibres and base change.

For the new non-Cartier results the dependence is

$$\begin{aligned} \text{codimension stabilization} &\implies \text{full degree-three Fitting scheme,} \\ \text{hyperplane conductor incidence} &\implies \text{embedded multigenerator local rings,} \\ \text{regularity and the conormal layer} &\implies \text{global embedded failure schemes.} \end{aligned}$$

The first arrow is valid over nonreduced coefficient rings, the second is a scheme isomorphism on the specified open, and the last is a coherent-cokernel isomorphism. These are different kinds of assertions.

The dependence of the retained arguments is

$$\begin{aligned} \text{evaluation kernel} &\implies \text{conductor cokernel,} \\ \text{unital hyperplane conductor} &\implies \text{rank-two quotients,} \\ \text{rank-two quotients} &\implies \text{primary fibres and smooth total scheme,} \\ \text{index form} &\implies \text{pencil primary fibres and normalization,} \\ \text{polarized Cayley-Hamilton + Haiman} &\implies \text{three-plane schemes.} \end{aligned}$$

None of these arguments depends on the statistical interpretation, the quadratic excess classification, or the unrestricted hyperplane embedded structure. Those developments, including all their proofs and hypotheses, are retained in the full complementary manuscript. The present family is not identified with the unrestricted ambient Grassmannian. Section 26 gives the precise classical inputs and the remaining limitation in the nearest-source comparison.

Primary geometry across the conductor boundary. The principal boundary theorem is a classification of an entire Hilbert-function class, rather than a further isolated local example. Every complex local algebra with Hilbert function $(1, 2, 2)$ is isomorphic to $\mathbb{C}[x, y]/((x, y)^3, y^2)$ or $\mathbb{C}[x, y]/((x, y)^3, xy)$. For all its unital three-planes, in every degree at least two, Theorem 17.2 determines the full zeroth-Fitting scheme and every associated stratum, including the extreme-corank boundary. On a smooth frame cover put $\delta = ad - bc$, $P = (a, c)$ and $Q = (b, d)$. In the square case the ideal is $\delta^2 P^2 = (\delta^2) \cap P^4$. In the two-factor case it is $\delta^2(ab, ad + bc, cd)$, with embedded primes $P, Q, P + Q$ and the explicit primary decomposition (17.7). Both schemes have a normal reduced determinant divisor, generic multiplicity two, and nilradical index four.

The factor lines of the quadratic relation index the embedded surfaces. A double factor gives a surface contained entirely in the subalgebra locus; distinct factors give two surfaces meeting that locus and an additional embedded point at the quadratic layer. Theorem 17.3 computes the conductor kernel on these strata, over their actual coefficient rings. This is the boundary information not supplied by the corank-one incidence isomorphism alone. Proposition 17.4 places both types in one finite flat algebra family, with an exact formula for the full Fitting ideal before specialization.

Three levels of statement should be distinguished. For an arbitrary finite algebra, Theorem 16.1 is a global presentation and an incidence theorem. For cube-zero augmentation ideals, Theorem 17.1 factors the full multiplication ideal through the quadratic multiplication tensor. For Hilbert function $(1, 2, 2)$, Theorem 17.2 converts this factorization into a complete primary classification. The connected higher-contact family of Theorem 18.1 supplies a complementary phenomenon: unbounded nilpotent complexity. Corollary 17.5 transports the complete length-five classification to section multiplication on the projective plane.

Finally, Proposition 13.2 proves optimality of the stabilization bound in every codimension and every generating rank at least two. Its elementary field-level interpretation is separated from the coefficient-ring assertion. The arguments in Sections 16.1 and 19.1 give the full functorial and cohomological details needed for arbitrary, possibly nonreduced, base change.

11. SHARP CONDUCTOR REDUCTION AND A CURVE CRITERION

All schemes in this article are over $\mathbb{C}$. Put $V_r = H^0(\mathbb{P}^1, \mathcal{O}(r))$, with $V_r = 0$ for $r < 0$. For a subspace U of sections, U^j denotes its multiplication image, not its abstract symmetric power.

Lemma 11.1 (The evaluation-kernel bound). *Let $U \subset V_n$ be base-point-free, of codimension c , and write $M_U = \ker(U \otimes \mathcal{O} \rightarrow \mathcal{O}(n))$. There are integers $a_i \geq 1$, with $\sum_i a_i = n$, such that $M_U = \bigoplus_{i=1}^{n-c} \mathcal{O}(-a_i)$. If $a = \max_i a_i$, then for $t \geq 0$,*

$$\text{coker}(U \otimes V_t \rightarrow V_{n+t}) \simeq H^1(M_U(t)), \quad UV_t = V_{n+t} \iff t \geq a - 1.$$

In particular $a \leq c + 1$, so $UV_t = V_{n+t}$ for $t \geq c$.

Proof. The splitting theorem for vector bundles on $\mathbb{P}^1$ applies to the locally free evaluation kernel. Its degree is $-n$ and its rank is $n - c$. Moreover $H^0(M_U) = 0$, because $U \rightarrow H^0(\mathcal{O}(n))$ is injective. Thus all its summands have negative degree. Twisting the evaluation sequence by $\mathcal{O}(t)$ and using $H^1(\mathcal{O}(t)) = 0$ gives the asserted cokernel. The formula for $H^1(\mathcal{O}(t - a_i))$ gives the equivalence. Finally the other $n - c - 1$ positive integers have sum at least $n - c - 1$, whence $a \leq n - (n - c - 1) = c + 1$. $\square$

Theorem 11.2 (Sharp uniform conductor range). *Let $D \subset \mathbb{P}^1$ have length $d \geq 2$, let $1 \leq k < d$, and suppose*

$$n \geq 2d - k. \tag{11.1}$$

Let $A \subset H^0(\mathcal{O}_D(n))$ be a generating k -plane, and put $W = H^0(\mathcal{O}(n)(-D))$ and $U = \text{res}_D^{-1}(A)$. For every $m \geq 2$,

$$WU^{m-1} = H^0(\mathcal{O}(mn)(-D)). \quad (11.2)$$

On the entire relative generating Grassmannian over $\text{Hilb}^d(\mathbb{P}^1)$, restriction gives

$$\text{coker}(\text{Sym}^m \mathcal{U} \rightarrow V_{mn} \otimes \mathcal{O}) \simeq \text{coker}(\text{Sym}^m \mathcal{A} \rightarrow \pi_* \mathcal{O}_D(mn)). \quad (11.3)$$

These are isomorphisms of coherent sheaves after arbitrary base change; all their Fitting ideals agree. For a particular geometric fibre, the weaker condition $n - d \geq \max_i a_i(U) - 1$ suffices for the same conclusion. Write $X_{d,k}^\circ$ for this relative generating Grassmannian. The bound (11.1) is sharp as a bound uniform in D, A and simultaneously in all $m \geq 2$. For $k = d$ the cokernel assertion is immediate from $U = V_n$ in the range $n \geq d - 1$; the saturation assertion in that case holds for $n \geq d$.

Proof. Write $D = (g)$, so $W = gV_{n-d}$. The series U is base-point-free: W generates away from D and A generates on D . Its codimension in V_n is $c = d - k$. Lemma 11.1 and $n - d \geq c$ give $WU = gV_{2n-d}$. Further multiplication by U uses input degrees $2n - d, 3n - d, \dots$, all at least c , and proves (11.2). The same argument uses $a - 1$ in place of c on a particular fibre.

All restriction targets and kernels are vector bundles: their first cohomology groups vanish in this range. The map $\mathcal{W} \otimes \text{Sym}^{m-1} \mathcal{U} \rightarrow \pi_* I_D(mn)$ is onto on every geometric fibre, hence is a surjection of vector bundles. It remains onto after every base change. Locally split $0 \rightarrow \mathcal{W} \rightarrow \mathcal{U} \rightarrow \mathcal{A} \rightarrow 0$. The kernel of the map on symmetric powers is the image of $\mathcal{W} \otimes \text{Sym}^{m-1} \mathcal{U}$. Quotienting the restriction square by these kernels proves (11.3). Fitting base change now applies [25].

For sharpness take $n = 2d - k - 1$, $D = d\{0\}$, and, in the affine coordinate t , take the degree- n homogenizations of

$$U = \langle 1, t, \dots, t^{k-1}, t^d, t^{d+1}, \dots, t^n \rangle.$$

This is base-point-free and is the inverse image of a generating k -plane on D . Exponents of its quadratic products lie in

$$[0, 2k - 2] \cup [d, n + k - 1] \cup [2d, 2n].$$

None is $2d - 1$. Thus $t^{2d-1} \notin U^2$, although it belongs to $H^0(I_D(2n))$. Its nonzero cokernel class restricts to zero on D , so even the cokernel comparison fails at this boundary in degree two. This does not assert a sharp threshold separately for each $m > 2$. $\square$

The numerical bound can be replaced by a cohomological hypothesis on a curve. Here and below an ideal such as I_D is the ideal on the curve, whereas a Fitting ideal is on the parameter space.

Proposition 11.3 (An exact cohomological test). *Let C be a smooth projective curve, let L be a line bundle, and let $U \subset H^0(L)$ be base-point-free. Put $W = H^0(L(-D))$. If $H^1(L(-D)) = 0$, then*

$$\text{coker}(W \otimes U \rightarrow H^0(L^2(-D))) \simeq H^1(M_U \otimes L(-D)).$$

More generally, for $B_j = U^j \subset H^0(L^j)$ and its evaluation kernel M_{B_j} , the obstruction to $WB_j = H^0(L^{j+1}(-D))$ is exactly $H^1(M_{B_j} \otimes L(-D))$.

Proof. Tensor the evaluation sequence for B_j by $L(-D)$ and take cohomology. The next term is $B_j \otimes H^1(L(-D)) = 0$. $\square$

Theorem 11.4 (Conductor transport on curves). *Let C have genus g , and let D be effective of length $d \geq 2$. Suppose $L(-D)$ is globally generated, $\text{Sym}^2 H^0(L(-D)) \rightarrow H^0(L^2(-2D))$ is surjective, and*

$$H^1(L^j(-2D)) = 0 \quad (j \geq 1). \quad (11.4)$$

For every generating $A \subset H^0(L|_D)$ and every $m \geq 2$, $U = \text{res}_D^{-1}(A)$ satisfies

$$WU^{m-1} = H^0(L^m(-D)), \quad \text{coker}(\text{Sym}^m U \rightarrow H^0(L^m)) \simeq \text{coker}(\text{Sym}^m A \rightarrow H^0(L^m|_D)).$$

In particular these assertions hold if

$$\deg L \geq 2d + 2g - 1. \quad (11.5)$$

For the relative assertion, let $f : C \rightarrow S$ be smooth and projective of relative dimension one, with S locally Noetherian over $\mathbb{C}$, let D be a relative effective Cartier divisor finite flat of degree d , and let L be invertible. Assume the preceding hypotheses on every geometric fibre. For each $j \geq 1$ and $b = 0, 1, 2$, put $E_{j,b} = f_* L^j(-bD)$. These are locally free, $R^1 f_* L^j(-bD) = 0$, and their formation commutes with arbitrary base change. The same base-change property holds for the finite direct images of $L^j|_D$ and $L^j(-D)|_D$. On the relative generating Grassmannian form $\mathcal{U}$ as the inverse image of $\mathcal{A}$ in $E_{1,0}$. The displayed saturation and cokernel assertions then hold as maps of these bundles and sheaves, after every base change.

Proof. The hypotheses imply $H^1(L^j(-D)) = 0$ by the exact sequence with quotient on D . They make U base-point-free. It contains a pencil without base points, obtained by avoiding the finitely many zeros of one section. The pencil sequence, tensored by $L^j(-2D)$, shows that multiplication onto $H^0(L^{j+1}(-2D))$ is surjective for $j \geq 2$: its obstruction is $H^1(L^{j-1}(-2D))$. Starting with the assumed surjectivity for W^2 , this proves $W^2 U^{m-2} = H^0(L^m(-2D))$.

The map $W \rightarrow H^0(L(-D)|_D)$ is onto by (11.4). Choose $a \in A$ a unit on D and lift it to $u \in U$. Multiplication by u^{m-1} identifies the first normal quotients in these degrees. Consequently Wu^{m-1} fills that quotient in $H^0(L^m(-D))$. Together with the double-vanishing space just proved, this gives the assertion about WU^{m-1} . The locally split restriction square gives the cokernel isomorphism as before.

Under (11.5), the line bundle $L(-D)$ has degree at least $2g + 1$. The classical normal-generation theorem in this degree range gives the required quadratic surjectivity and global generation; see [14, p. 73, Introduction]. Serre duality gives (11.4), since $\deg L(-2D) \geq 2g - 1$. For the relative assertion, the assumed fibrewise H^1 -vanishing for $b = 2$ implies the same vanishing for $b = 1, 0$ by restriction to D . Cohomology and base change for these flat line bundles on a proper flat relative curve gives the named direct images, with no higher cohomology; finite flat pushforward gives the two restriction bundles. In particular $0 \rightarrow E_{1,1} \rightarrow E_{1,0} \rightarrow f_*(L|_D) \rightarrow 0$ is a locally split exact sequence. The quadratic map $\text{Sym}^2 E_{1,1} \rightarrow E_{2,2}$ and each map $E_{1,1} \otimes \text{Sym}^{m-1} \mathcal{U} \rightarrow E_{m,1}$ are surjective on every geometric fibre by the fixed-curve proof. They are thus surjections of vector bundles and remain so under arbitrary pullback. The same locally split restriction square proves the relative cokernel assertion; no base-change assumption on an unflattened image U^j is needed. $\square$

For example on any elliptic curve, a line bundle of degree at least $2d + 1$ transports all the finite-contact schemes below. The curve statement is not a sharpness assertion in positive genus. Its intrinsic hypotheses, and the exact obstruction in Proposition 11.3, distinguish a failure of regularity from a failure of the finite-contact multiplication.

12. GENERATING OPENS AND FINITE PRESENTATIONS

Here “generating” refers to generation as an ideal or invertible module. It does not assert generation as a unital algebra.

Lemma 12.1 (The relative generating open). *Let B/S be a finite locally free commutative algebra over a $\mathbb{C}$ -scheme, and let $A \subset B$ be a subbundle. The following conditions are equivalent: $B \otimes A \rightarrow B$ is surjective; on every geometric fibre A has nonzero image in every residue field of B ; and the open $A^\times$ of sections of A which are units of B is smooth and surjective over S .*

These conditions define an open in the Grassmannian. The assertions have the same form for a subbundle of an invertible B -module, using generating sections in place of units.

Proof. Surjectivity is equivalent to its geometric-fibre version by Nakayama's lemma. On a finite algebra over an algebraically closed field, an ideal is the unit ideal exactly when its images in all residue fields are nonzero. These evaluations cut out finitely many proper hyperplanes in A . Their complement is nonempty over an infinite field and consists precisely of units. On the total space of the vector bundle A , invertibility is the open condition $\det(b \mapsto ab) \neq 0$. This open is smooth over S , and the preceding fibre calculation proves surjectivity. Conversely a unit section after a faithfully flat cover generates the unit ideal, which descends. The Grassmannian open is the complement of the support of the cokernel of $B \otimes A \rightarrow B$. Local trivializations of an invertible module prove the last assertion. $\square$

Lemma 12.2 (Total-degree truncation). *Let B be a rank- d finite locally free algebra over a $\mathbb{C}$ -algebra R , and let $b_1, \dots, b_q \in B$. The R -subalgebra they generate is spanned as an R -module by the monomials of total degree at most $d - 1$. This assertion commutes with base change, with polynomial reduction coefficients on a free chart.*

Proof. Set $w = \sum_j t_j b_j$ in $B[t_1, \dots, t_q]$ and apply Cayley–Hamilton to multiplication by w . The coefficient of t^α , for $|\alpha| = d$, in w^d is $\binom{d}{\alpha} b^\alpha$. All other terms have B -monomial degree less than d : coefficients of the characteristic polynomial are scalars in $R[t]$. The multinomial coefficient is invertible in R . This expresses each degree- d monomial using lower degrees. Multiplying these identities and inducting reduces every higher monomial. The adjugate identity for Cayley–Hamilton makes the coefficients polynomial in the multiplication table. $\square$

13. A CODIMENSION BOUND OVER THE COEFFICIENT RING

The degree needed to generate a finite algebra can be bounded by the codimension of the initial subspace, rather than by the length of the algebra. A fibrewise dimension argument suggests this bound but does not prove the assertion over a nonreduced parameter scheme. The following argument proves an equality of submodules over the coefficient ring.

Theorem 13.1 (Codimension stabilization). *Let B be a commutative finite locally free algebra over a $\mathbb{Q}$ -scheme, and let $W \subset B$ be a subbundle containing 1, with B/W locally free of rank r . Write W^j for the image of $\text{Sym}^j W \rightarrow B$. Then*

$$W^{r+1} = \mathcal{O}[W], \quad W^j = W^{r+1} \quad (j \geq r + 1). \quad (13.1)$$

These equalities remain true after every base change, without a flatness hypothesis on W^j .

For a generating rank- k subbundle A of a rank- d algebra over a $\mathbb{C}$ -scheme, all Fitting ideals of the multiplication cokernel are consequently independent of the degree $j \geq d - k + 1$. The generating opens and normalization by units are those of Lemma 12.1. This is a statement about ideal equalities; the unnormalized cokernels need not have a preferred isomorphism in different degrees.

Proof. The case $r = 0$ is immediate. Work otherwise on a free chart and choose a splitting $B = W \oplus M$, where $\text{rank } M = r$. For $w \in W$ write multiplication by w in blocks

$$L_w = \begin{pmatrix} A_w & D_w \\ \beta_w & C_w \end{pmatrix}.$$

Thus $\beta_w : W \rightarrow M$ and $C_w : M \rightarrow M$. Let $N_0 = \sum_{w \in W} \text{im } \beta_w$, and, for $j \geq 0$, let N_j be the sum of the images of N_0 under all words of length at most j in the operators C_w . A finite local basis

of W suffices in these definitions. Since $1 \in W$, every W^m contains W . Projection to M and subtraction of its W component therefore give, as submodules of the chosen splitting,

$$W^{j+2} = W \oplus N_j. \quad (13.2)$$

Indeed the initial case is the definition of N_0 , and multiplication by W changes the residual module N into $N_0 + \sum_w C_w N$.

Commutativity of the full multiplication operators gives the identity

$$[C_w, C_v] = \beta_v D_w - \beta_w D_v, \quad \text{im}[C_w, C_v] \subset N_0. \quad (13.3)$$

Consider a word of length r applied to N_0 . Interchanging two adjacent factors changes it by a word with a commutator in those positions. The commutator sends its entire argument into N_0 ; at most $r - 2$ factors remain to its left. The change thus belongs to N_{r-2} , and in particular to N_{r-1} . For $r = 1$ no interchange is needed. All permutations of a length- r word therefore agree modulo N_{r-1} when applied to N_0 .

Apply Cayley–Hamilton to the r by r matrix $C(t) = \sum_{i=1}^r t_i C_{w_i}$ and extract the coefficient of $t_1 \cdots t_r$. The leading term is the sum of the $r!$ ordered products of the C_{w_i} . Every other term is a scalar linear combination of words of length at most $r - 1$. Applied to N_0 these other terms lie in N_{r-1} . The preceding interchange argument and the invertibility of $r!$ now show that every length- r word sends N_0 into N_{r-1} . Hence $N_r = N_{r-1}$. This equality says that N_{r-1} is invariant under all C_w , so the residual filtration is constant thereafter. Equation (13.2) proves (13.1).

This calculation used no division except by $r!$ and no passage to residue fields. It is valid over every base-changed coefficient ring; in particular, it does not assert that an unsplit image is a vector bundle. Finally choose a unit $a \in A$ smoothly locally and put $W = a^{-1}A$. Multiplication by a^j identifies W^j with A^j , and is an automorphism of B . The Fitting ideals are thus constant in the stated range on a faithfully flat cover and hence on the original parameter scheme. $\square$

Combining this theorem with Lemma 12.2 gives the stable range $j \geq \min\{d - 1, d - k + 1\}$ for $d \geq 2$. The improvement is largest when the starting subspace has small codimension. In particular, a codimension-two multiplication problem is determined in degree three, however large d is. Section 16 uses this ring-level assertion to retain the entire failure scheme, including the closed locus where the defect is two.

13.1. Sharpness and the algebra-length problem. For a unital subspace W over a field, let $\ell(W)$ be the least j such that W^j is the algebra generated by W . This is the usual length of a generating subspace. Algebra-length bounds are studied in the literature of Pappacena and Markova; see [36, 37] and the more recent discussion in [38, Sections 1 and 6]. No novelty is claimed here for the following field-level dimension argument, which works even without commutativity. If $W^j = W^{j+1}$, multiplication by W stabilizes that space and the sequence is constant thereafter. Until then its dimension strictly increases. Starting with dimension k in an algebra of dimension d gives $\ell(W) \leq d - k + 1$. Theorem 13.1, by contrast, proves an equality of submodules over a coefficient ring without assuming the intervening images flat. A dimension count on residue fields does not establish that equality. Its application here is stability of all Fitting ideals after arbitrary base change.

Proposition 13.2 (Sharpness in every codimension). *For every pair of integers $r \geq 1$, $k \geq 2$, there is a connected commutative local algebra B of dimension $r + k$ and a unital k -plane $W \subset B$ such that*

$$W^r \neq W^{r+1} = B.$$

Thus $r + 1$ is the optimal universal bound for codimension r , even when the generating rank k is prescribed.

Proof. Take

$$B = \mathbb{C}[z, \epsilon_1, \dots, \epsilon_{k-2}] / (z^{r+2}, z\epsilon_i, \epsilon_i\epsilon_j : 1 \leq i, j \leq k-2)$$

and $W = \text{span}(1, z, \epsilon_1, \dots, \epsilon_{k-2})$. When $k = 2$ the lists involving ϵ_i are empty. A basis of B is $1, z, \dots, z^{r+1}, \epsilon_1, \dots, \epsilon_{k-2}$. For $1 \leq j \leq r+1$, a basis of W^j is $1, z, \dots, z^j, \epsilon_1, \dots, \epsilon_{k-2}$. The assertions follow, with z^{r+1} witnessing strictness. $\square$

The result is an optimal bound depending on the initial codimension, not a claim to improve every bound involving the minimal-polynomial degrees or the internal structure of a particular algebra. The relative proof uses commutativity of the full multiplication operators; it does not assert that their compressed blocks commute.

14. GENERATING HYPERPLANES AND LENGTH-TWO QUOTIENTS

We now solve a multigenerator problem without assuming that the finite algebra is monogenic, reduced, or constant. If B is a finite locally free commutative algebra of rank d over a scheme S , let $\text{Hilb}^2(B/S)$ denote the functor of quotient algebras $B_T \rightarrow Q$ that are locally free of rank two over T . It is the relative Hilbert scheme of two points of $\text{Spec}_S B$. A generating line in Q means a line subbundle whose Q -span is Q .

Lemma 14.1 (The conductor of a unital hyperplane). *Unital subalgebras $S_0 \subset B_T$ which are hyperplane subbundles are naturally equivalent to rank-two quotient algebras of B_T . The maps are*

$$S_0 \longmapsto B_T/J, \quad J = \{b \in B_T : bB_T \subset S_0\}, \quad (B_T \rightarrow Q) \longmapsto \text{preimage}(\mathcal{O}_T \cdot 1).$$

The conductor J is an ideal subbundle of rank $d-2$, and its formation commutes with arbitrary base change.

Proof. Work locally with $T = \text{Spec } R$, trivialize B/S_0 , and write it as the target of $\phi : B \rightarrow R$. Choose e with $\phi(e) = 1$. Since $1 \in S_0$, the bilinear form $\beta(b, c) = \phi(bc)$ has $\beta|_{S_0 \times S_0} = 0$ and $\beta(1, e) = 1$. An element of its radical first lies in S_0 , by pairing with 1; it then lies in the kernel of the split surjection

$$S_0 \longrightarrow R, \quad s \longmapsto \phi(se),$$

which sends 1 to 1. Thus the radical has rank $d-2$, and B/J has basis the classes of $1, e$. Associativity shows that the radical is an ideal: $\beta(jb, c) = \beta(j, bc) = 0$. It is precisely the displayed conductor. This split-kernel description proves base change and gives $S_0/J = R \cdot 1$.

Conversely, in a rank-two locally free algebra Q , the unit is a unimodular vector, so $Q/R \cdot 1$ is a line bundle. The inverse image of the scalars is a unital hyperplane subalgebra. No nonzero ideal of Q can be contained in $R \cdot 1$: in a basis $1, e$, if $c \cdot 1$ belongs to such an ideal, then ce must be scalar, so $c = 0$. Its conductor is therefore exactly the original kernel. The two constructions are inverse and are independent of all trivializations. $\square$

Theorem 14.2 (The universal hyperplane failure scheme). *Let B/S be finite locally free of rank $d \geq 3$ over a $\mathbb{C}$ -scheme. On $X = \text{Gr}_S(d-1, B)^\circ$, the generating open, let $\mathcal{A}$ be the tautological hyperplane and put*

$$\mathcal{C}_m = \text{coker}(\text{Sym}^m \mathcal{A} \rightarrow B), \quad Y_m = V(\text{Fitt}_0 \mathcal{C}_m).$$

For every $m \geq 2$ the ideal of Y_m is the same. The common scheme Y is naturally isomorphic to

$$\{(B \twoheadrightarrow Q, \ell) : \text{rank } Q = 2, \ell \subset Q \text{ a generating line}\}. \quad (14.1)$$

The map to X sends (Q, ℓ) to the inverse image of ℓ . If $i : Y \hookrightarrow X$ and $B_Y \twoheadrightarrow Q \supset \ell$ is the universal quotient with its generating line, the canonical formula is

$$\mathcal{C}_m \simeq i_*(Q/\ell^{(m)}), \quad \ell^{(m)} = \text{im}(\ell^{\otimes m} \longrightarrow Q). \quad (14.2)$$

Both $\ell^{(m)}$ and its quotient are line bundles on Y . In particular all positive-index Fitting ideals are the unit ideal. The isomorphism of schemes and all these assertions commute with arbitrary base change. The same statement holds for an invertible B -module in place of B , with target its m th tensor power.

Proof. Locally for the smooth topology, choose a unit $a \in \mathcal{A}$. Such choices cover X : on each geometric fibre avoid the finitely many evaluation hyperplanes. Replace $\mathcal{A}$ by $A' = a^{-1}\mathcal{A}$, and the target by multiplication by a^{-m} . Now $1 \in A'$. Split $B = A' \oplus Re$ locally. Since $(A')^m$ contains A' , elementary row and column operations present the cokernel by a single row. For $m = 2$ its entries are exactly the coefficients of the products ss' in B/A' , for $s, s' \in A'$. Call their ideal I . Thus Y_2 represents the condition that A' is a subalgebra, with its full scheme structure, not just its support. Modulo I all higher products lie in A' . Conversely every quadratic product is an m -fold product by inserting 1's. Hence the residual ideals equal I for every $m \geq 2$, and the cokernel is locally R/I .

Lemma 14.1 supplies the unique rank-two quotient on Y . Its kernel J also lies in $\mathcal{A} = aA'$, since J is an ideal and $aJ = J$. The image of $\mathcal{A}$ is a generating line in Q . The construction does not depend on a . Indeed another unit $b \in \mathcal{A}$ gives $b/a \in A'$, and its inverse belongs to A' by Cayley–Hamilton in B (the constant coefficient is a unit). Thus $(b/a)^{-1}A' = A'$ and the conductor is unchanged. This also proves descent from the smooth cover. Conversely the inverse image of a generating line in a rank-two quotient becomes the inverse image of the scalars after division by a unit, so it satisfies precisely the residual equations. These constructions are inverse on every test scheme, proving the scheme isomorphism. On Y , the normalized subalgebra A' has conductor J , and $J \subset a^m A' = (\mathcal{A})^m$ since a is a unit and J an ideal. Passing to Q identifies $(\mathcal{A})^m/J$ with $\ell^{(m)}$. This gives the canonical map in (14.2); the local cyclic presentations identify it as an isomorphism. Multiplication by a local unit spanning ℓ makes $\ell^{(m)}$ a direct-summand line, so the quotient is indeed a line bundle. For an invertible B -module P , put $P_Q = P \otimes_B Q$; the same formula is $i_*(P_Q^{\otimes Q^m}/\ell^{(m)})$, where now $\ell \subset P_Q$. Trivialization of P gives the local proof and this intrinsic expression supplies its gluing. $\square$

14.1. Every fixed multiplicity partition. For $r \geq 2$ define polynomials

$$F_0 = 0, \quad F_1 = 1, \quad F_2 = u, \quad F_{j+1} = uF_j + vF_{j-1}, \quad G_r = \mathbb{C}[u, v]/(F_r, F_{r+1}). \quad (14.3)$$

For positive r, s put $H_{r,s} = \mathbb{C}[x, y]/(x^r, y^s)$.

Theorem 14.3 (Complete primary schemes for contact hyperplanes). *Let $D = \sum_i d_i p_i$ have length $d \geq 3$. Consider the entire generating open $\text{Gr}(d-1, H^0(\mathcal{O}_D))^\circ$. For every $m \geq 2$, its multiplication failure scheme is the disjoint union*

$$Y_D \simeq \coprod_{i: d_i \geq 2} (\mathbb{A}^1 \times \text{Spec } G_{d_i}) \amalg \coprod_{i < j} (\mathbb{G}_m \times \text{Spec } H_{d_i, d_j}). \quad (14.4)$$

The identifications use local coordinates on D and trivializations of the restricted line bundle. Each displayed summand is a primary component on this fixed-divisor parameter space. There are no embedded associated points. All reduced components are smooth and pairwise disjoint. Their transverse lengths and exact nilradical indices are, respectively,

length-two support	length	nilpotency index
$2p_i$	$\binom{d_i}{2}$	$2d_i - 3$
$p_i + p_j$	$d_i d_j$	$d_i + d_j - 1$.

The convention for a reduced component is index one. Every failure fibre has cokernel dimension one. On binary inverse-image subseries these are exactly the multiplication failure schemes whenever $n \geq d + 1$; on curves they are the same schemes under Theorem 11.4.

Proof. A degree-two quotient of the product algebra $\prod_i \mathbb{C}[z_i]/(z_i^{d_i})$ decomposes according to the ranks of its idempotent factors. These ranks are locally constant even on a nonreduced base. Its possibilities are rank two at a single factor, or rank one at each of two distinct factors. This is a decomposition into open and closed Hilbert subschemes, not merely a decomposition of closed points.

For rank two in $\mathbb{C}[z]/(z^r)$, the elements $1, z$ form a basis in every geometric quotient, hence on every family by Nakayama. Write $z^2 = uz + v$. Then

$$z^j = F_j(u, v)z + vF_{j-1}(u, v).$$

The relation $z^r = 0$ is equivalent to $F_r = vF_{r-1} = 0$, equivalently $F_r = F_{r+1} = 0$. Thus this Hilbert scheme is $\text{Spec } G_r$. A rank-one quotient at a factor of length r is $\text{Spec } \mathbb{C}[x]/(x^r)$, so the two-factor case is $\text{Spec } H_{r,s}$. A generating line in the first rank-two algebra is uniquely spanned by $1 + \tau z$; in the split two-factor algebra it is uniquely spanned by $(1, t)$, with t a unit. Theorem 14.2 now proves (14.4) over arbitrary test schemes.

The algebra G_r is Artin local: if $v \neq 0$, two consecutive zero values in the recurrence force $F_1 = 0$, and if $v = 0$ they force $u = 0$. The two equations form a regular sequence in $\mathbb{C}[u, v]$. With $\deg u = 1$, $\deg v = 2$, its Hilbert series is

$$\frac{(1 - t^{r-1})(1 - t^r)}{(1 - t)(1 - t^2)}. \quad (14.5)$$

It has length $\binom{r}{2}$ and weighted socle degree $2r - 4$. To see the exact ordinary nilpotency, identify u and $-v$ with the elementary symmetric functions in two variables. Then $F_j = h_{j-1}$ and

$$G_r \simeq H^{2*}(\text{Gr}(2, r), \mathbb{C}), \quad (14.6)$$

where the cohomological degree is divided by two. This is the classical presentation by h_{r-1}, h_r , not a new presentation of Grassmannian cohomology; the rectangular Schur basis is recalled in [31, Theorem 2.7 and the following discussion]. Successive multiplication by $u = s_{(1)}$ adds one box. The only partition of size $2r - 4$ in the 2 by $r - 2$ rectangle is $(r - 2, r - 2)$; the ballot-path count of its standard tableaux is $\frac{1}{r-1} \binom{2r-4}{r-2}$. Consequently

$$u^{2r-4} = \frac{1}{r-1} \binom{2r-4}{r-2} (-v)^{r-2} \neq 0.$$

Every monomial of ordinary degree $2r - 3$ has weighted degree at least $2r - 3$ and vanishes. This proves the index $2r - 3$, including $r = 2$. In $H_{r,s}$ the top monomial is $x^{r-1}y^{s-1}$, giving length rs and index $r + s - 1$.

Each summand is a complete intersection with reduced support a smooth integral curve, so it is primary and has no embedded points. Equivalently, over a local Artin coefficient algebra, a polynomial with nonzero reduction is a nonzerodivisor, as one checks on the successive powers of the maximal ideal. The same holds after inverting t . This proves the primary assertions also globally on the indicated affine curves. The cokernel assertion follows from Theorem 14.2. The two transports follow from Theorems 11.2 and 11.4. $\square$

Theorem 14.4 (The smooth moving scheme). *Let $d \geq 3$. On the relative generating Grassmannian of $(d - 1)$ -planes over $\text{Hilb}^d(C)$, for a smooth connected projective curve C , the finite-contact failure scheme is a generating open of a projective-line bundle over*

$$\text{Sym}^2 C \times \text{Sym}^{d-2} C, \quad (Z, Q) \mapsto D = Z + Q.$$

It is smooth, irreducible, and of dimension $d + 1$. Its projection to $\text{Sym}^d C$ is flat of relative dimension one. The finite map of divisor spaces has degree $\binom{d}{2}$; the primary lengths in Theorem 14.3

sum to this degree on every fixed fibre. Under either conductor theorem, these are statements about the original global multiplication scheme.

Proof. A nested pair $Z \leq D$ of effective divisors on a smooth curve is uniquely $D = Z + Q$. This holds in families: a finite flat subscheme of a smooth relative curve is a relative Cartier divisor, and inclusions of their invertible ideals give the residual divisor. The incidence in Theorem 14.2 is therefore exactly the following canonical open projective bundle. On $T = \text{Sym}^2 C \times \text{Sym}^{d-2} C$ let $\mathcal{Z} \subset C \times T$ be the universal degree-two divisor and set $F = (\pi_{\mathcal{Z}})_* \mathcal{O}_{\mathcal{Z}}$. Use the projectivization $\mathbb{P}_T^{\text{lines}}(F)$ parametrizing line subbundles of F , and retain the open where the tautological line ℓ satisfies $F\ell = F$. This is the claimed bundle, not an unspecified local form. For a restricted line bundle replace F by $(\pi_{\mathcal{Z}})_*(L|_{\mathcal{Z}})$ and use generation as an F -module. The open is nonempty on every fibre of its smooth irreducible base, and hence is smooth and irreducible of dimension $d + 1$.

The map of divisor spaces is finite (proper with finite geometric fibres). Both spaces are smooth of dimension d ; the finite source is Cohen–Macaulay and the target regular, so the map is flat. Over a reduced divisor it has $\binom{d}{2}$ points. This proves the degree and flatness assertions; alternatively, the fibre lengths computed above give $\sum_i \binom{d_i}{2} + \sum_{i < j} d_i d_j = \binom{d}{2}$. The open projective bundle is smooth of relative dimension one, so its composite projection is flat of that relative dimension. $\square$

14.2. A direct residual presentation. For comparison with the multiplication matrix, normalize a unit and write, on the unital hyperplane chart in $\mathbb{C}[z]/(z^r)$,

$$A' = \langle 1, z^i - c_i z : 2 \leq i \leq r-1 \rangle.$$

Put $c_1 = 1$ and $c_j = 0$ for $j \geq r$. After eliminating the pivot columns, the remaining entries of the quadratic matrix are

$$c_{i+j} - c_i c_{j+1} - c_j c_{i+1} + c_2 c_i c_j \quad (2 \leq i \leq j \leq r-1). \quad (14.7)$$

For $r \geq 4$, the relations with $i = 2$ give $c_{j+2} = u c_{j+1} + v c_j$, with $u = c_2$, $v = c_3 - c_2^2$. They eliminate $c_4, \dots, c_{r-1}$ and leave F_r, F_{r+1} . Every other relation follows from the recurrence identity

$$F_{i+j} = F_i F_{j+1} + F_j F_{i+1} - u F_i F_j,$$

proved by induction in j . For $r = 3$ the single residual entry is c_2^3 , agreeing with $G_3 \simeq \mathbb{C}[u]/(u^3)$. For $r = 4$ one obtains

$$G_4 = \mathbb{C}[u, v]/(u^3 + 2uv, u^2v + v^2), \quad \text{length } G_4 = 6, \quad \mathfrak{m}^5 = 0 \neq \mathfrak{m}^4.$$

Thus the multigenerator transverse scheme is not a powered smooth divisor, even though its length equals the corresponding pencil weight. The smooth moving scheme has these nonreduced fibres; nonreducedness of a fibre is not an embedded component of the total scheme. Nor does this determine the embedded primary ideal along the evaluation curve in the unrestricted, non-generating hyperplane problem retained in the complementary material.

14.3. A noncurvilinear algebra and an embedded component. The general finite-algebra theorem is not restricted to the primary patterns arising from divisors on a smooth curve.

Proposition 14.5 (An exact embedded chart). *Let $B = \mathbb{C}[x, y]/(x^2, y^2)$. On the unit-normalized hyperplane chart*

$$A' = \langle 1, y - ax, xy - bx \rangle \subset B \otimes \mathbb{C}[a, b],$$

the multiplication failure ideal, for every $m \geq 2$, is

$$I = (ab, b^2) = (b) \cap (a, b)^2.$$

Its associated primes are (b) and (a, b) . Its nilradical is a one-dimensional square-zero module supported at the embedded point. These equations give the pullback of the full failure scheme on the generating Grassmannian, not only its reduced support.

Proof. Use the quotient functional $\phi(1) = 0$, $\phi(x) = 1$, $\phi(y) = a$, $\phi(xy) = b$. Products involving 1 are pivots. The other quadratic products have residual values $-2ab, -b^2, 0$. This gives I ; Theorem 14.2 gives every higher degree. The displayed primary decomposition follows by intersecting the monomial ideals. The nilradical is generated by b , whose annihilator modulo I is (a, b) . The normalization by a unit is the same smooth-cover calculation used in that theorem. In particular adding the freely chosen unit does not remove the embedded associated prime. $\square$

Thus the quotient description distinguishes smooth moving contact schemes from finite-algebra families with embedded structure. The absence of embedded points in Theorem 14.3 is a consequence of the curvilinear contact algebra and its explicit Hilbert schemes, not a general assertion about finite algebras.

15. STABLE IMAGES AND HIGHER-DEFECT CONDUCTOR STRATA

The rank-two quotient in Theorem 14.2 is specific to a hyperplane. In larger codimension the appropriate object is a quotient algebra together with a subalgebra acting faithfully on the quotient module. The rank of this action is an additional stratification invariant. We keep its scheme structure throughout.

For a map u of vector bundles write $I_j(u)$ for its ideal of j -minors. Its *exact-rank s locus* means $V(I_{s+1}(u)) \setminus V(I_s(u))$, with this locally closed scheme structure. On it the image and cokernel are vector bundles and commute with arbitrary base change: after inverting a pivot, elementary operations reduce the matrix to $\text{diag}(I_s, 0)$. This is stronger than requiring constant rank on geometric points.

Theorem 15.1 (Stable image and conductor stratification). *Let B be a commutative finite locally free algebra of rank $d \geq 2$ over a locally Noetherian $\mathbb{C}$ -scheme S , and let $\mathcal{A}$ be the universal bundle on $X = \text{Gr}_S(k, B)^\circ$, with $1 \leq k \leq d$. Put $\mu_m : \text{Sym}^m \mathcal{A} \rightarrow B_X$. All Fitting ideals of $\text{coker } \mu_m$ are independent of $m \geq d - 1$; Theorem 13.1 also gives stability for $m \geq d - k + 1$. We use $d - 1$ below to retain a uniform notation for the stable-image strata. On the exact-rank s locus X_s of μ_{d-1} there is a canonical unital subalgebra bundle $E \subset B_{X_s}$ of rank s . There is an invertible E -module $L \subset B_{X_s}$, also a subbundle over X_s , such that*

$$E = \mathcal{O}[a^{-1}\mathcal{A}], \quad L = E\mathcal{A} = aE, \quad \text{im } \mu_m = L^m \quad (m \geq d - 1) \quad (15.1)$$

on any local unit cover $a \in \mathcal{A} \cap B^\times$.

Suppose $r = d - s > 0$ and put $M = B/E$. The action

$$\theta : E \longrightarrow \text{End}_{\mathcal{O}}(M)$$

defines exact-rank strata $X_{s,t}$, with $1 \leq t \leq r^2$. On such a stratum the conductor and its quotient flag are

$$J = \ker \theta = \{b \in B : bB \subset E\}, \quad C = E/J \subset Q = B/J, \quad \text{rank } C = t, \quad \text{rank } Q = r + t. \quad (15.2)$$

Here J is an ideal subbundle of B , and $C \rightarrow \text{End}(Q/C)$ is a subbundle injection. Each of these constructions commutes with arbitrary base change within the specified stratum.

More generally, the functor of unital codimension- r subalgebra bundles with action of exact rank t is equivalent to the functor of flags $B \twoheadrightarrow Q \supset C$ of these ranks, where the first map is a surjective morphism of unital commutative algebras, Q is locally free of rank $r + t$, $C \subset Q$ is a unital subalgebra subbundle of rank t , Q/C is locally free of rank r , and the specified multiplication

action $C \rightarrow \mathcal{E}nd(Q/C)$ is a locally split subbundle injection. The inverse sends the flag to the preimage of C .

Proof. Choose a unit a smoothly locally, as in Lemma 12.1, and set $W = a^{-1}\mathcal{A}$. Then $1 \in W$. Lemma 12.2 says that $W^{d-1} = \mathcal{O}[W]$ and that $W^m = W^{d-1}$ for $m \geq d-1$, over the coefficient ring itself. Multiplication by a^m identifies the original image with $a^m\mathcal{O}[W]$. The resulting cokernels are locally isomorphic in these degrees, proving equality of every Fitting ideal before any rank restriction.

On X_s , the pivot description makes $E = \mathcal{O}[W]$ a subbundle with locally free quotient. If b is a second unit in $\mathcal{A}$, then $b/a \in E$ is a unit of B . Cayley–Hamilton for multiplication by b/a , whose constant coefficient is a unit, puts its inverse in E . Thus $\mathcal{O}[b^{-1}\mathcal{A}] \subset E$; reversing a, b gives equality. This proves descent of E . The image $E\mathcal{A}$ is locally aE and therefore descends to the stated invertible E -module and subbundle L . It also proves (15.1) universally.

Because E is a subalgebra, it acts on $M = B/E$. Its action kernel consists of the $e \in E$ with $eB \subset E$. An element b of the displayed conductor already lies in E , by multiplication by 1; hence the two kernels agree. If $jB \subset E$, then for $b, c \in B$ one has $(jb)c = j(bc) \in E$, so J is an ideal of B . On an exact-rank t stratum, pivot reduction for θ makes J locally free of rank $s - t$ and C a subbundle of $\mathcal{E}nd(M)$. The exact sequence $0 \rightarrow C \rightarrow B/J \rightarrow M \rightarrow 0$ proves that Q is locally free of rank $r + t$. All the kernels and images just used are split as $\mathcal{O}$ -modules locally; this proves their base-change assertions. The unit acts as the identity on M , so $t \geq 1$; the other bound comes from $\text{rank } \mathcal{E}nd(M) = r^2$.

Conversely, given a flag as in the statement, its inverse image E is a subalgebra bundle and $B/E \simeq Q/C$. The kernel of its action is precisely the kernel of $B \rightarrow Q$: a larger kernel would give an element of C annihilating Q/C , contrary to the subbundle injection. Thus the conductor construction recovers the flag, including over nonreduced test schemes. These mutually inverse constructions respect all pullbacks within the exact-rank functors. $\square$

The word “stratum” cannot be removed from the base-change assertion for J . The multiplication cokernel is a universally defined coherent sheaf on X ; its conductor need not be a bundle on X . Example 15.5 below exhibits the distinction.

Proposition 15.2 (The extreme-corank failure scheme). *On the Grassmannian of rank- $(d - r)$ subbundles $W \subset B$ containing 1, with $1 \leq r < d$, the unital subalgebra scheme is*

$$V(\text{Fitt}_{r-1} \text{coker}(\text{Sym}^m W \rightarrow B)), \quad m \geq 2.$$

The defining ideal is independent of m , including its nonreduced structure. Its conductor exact-rank t subschemes are the quotient-flag functors of Theorem 15.1.

Proof. Since $1 \in W$, the image W^m contains W . Split $B = W \oplus M$ locally and remove the identity pivot columns. The residual presentation has r rows, so its $(r - 1)$ st Fitting ideal is the ideal of all entries. In degree two these entries are precisely the coefficients of $W \cdot W$ in B/W . They vanish if and only if W is a subalgebra, on every test scheme. Modulo their ideal, every higher product remains in W . Conversely each quadratic product is an m -fold product by insertion of 1’s. The entry ideals are equal. The last assertion is the functorial equivalence just proved. $\square$

For $r = 1$ this is the usual zeroth-Fitting failure scheme. For $r > 1$ it is the *extreme-corank* locus, not the whole locus where multiplication is nonsurjective. This distinction prevents a false extension of the hyperplane theorem by changing only the quotient rank.

15.1. The two quotient mechanisms in codimension two.

Lemma 15.3 (Relative cyclic vectors). *Let M be a rank-two vector bundle, and let $C \subset \mathcal{E}nd(M)$ be a unital rank-two algebra subbundle over a $\mathbb{C}$ -scheme. Then M is an invertible C -module locally on the base. More explicitly, write $C = \langle 1, T \rangle$ and $T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ after trivializing. The opens*

$$D(c), \quad D(b), \quad D(c + d - a - b)$$

cover the base and have cyclic vectors $(1, 0)$, $(0, 1)$, and $(1, 1)$, respectively. These opens and the resulting module isomorphisms commute with base change.

Proof. On every geometric fibre, the subbundle injection makes T nonscalar. If all three displayed functions vanish, then $b = c = d - a = 0$, contradicting that assertion. Their principal opens therefore cover. The determinant of the map $C \rightarrow M$, $q \mapsto qv$, is respectively c , $-b$, and $c + d - a - b$ for the three chosen vectors. On the relevant open it is a unit, so the map is a C -linear isomorphism. The same invertible matrices prove the assertion after any pullback. $\square$

Theorem 15.4 (Codimension-two quotient flags). *Let $E \subset B$ be a unital subalgebra bundle with $M = B/E$ of rank two over a locally Noetherian $\mathbb{C}$ -scheme. For its action θ one has $I_3(\theta) = 0$ scheme-theoretically. The exact-rank-one closed locus and the exact-rank-two open locus have the following functorial descriptions.*

On the first, E is the inverse image of the scalar subalgebra in a locally free rank-three quotient algebra of B . On the second, E is the inverse image of C in a quotient $B \twoheadrightarrow Q$, where C is a rank-two algebra over the base and Q is a locally free rank-two algebra over C . In particular Q has rank four over the base. In both cases the kernel of $B \rightarrow Q$ is the conductor, and these descriptions commute with arbitrary base change on their respective strata.

Proof. Locally trivialize M . Since 2 is invertible, write every matrix as a scalar plus a trace-zero matrix. The trace-zero parts of the commuting matrices $\theta(E)$ still commute. The bracket map

$$\bigwedge^2 \mathfrak{sl}_2 \longrightarrow \mathfrak{sl}_2$$

is an isomorphism over $\mathbb{C}$: for the usual basis H, U, V its values are $2U, -2V, H$. Thus the wedge of any two trace-zero columns is zero over the whole coefficient ring, not just over its residue fields. Adjoining the scalar identity column proves $I_3(\theta) = 0$. The identity also gives $I_1(\theta) = \mathcal{O}$.

On the exact-rank-one locus the image is the scalar algebra. Theorem 15.1 gives $C = \mathcal{O}$ and $\text{rank } Q = 3$. Conversely the scalar line in any rank-three algebra has faithful action on its rank-two quotient module: the action is ordinary scalar multiplication. Its inverse image has exactly this conductor.

On the exact-rank-two locus, $C \subset \mathcal{E}nd(M)$ is a rank-two subalgebra bundle. Locally write $C = \langle 1, T \rangle$. Lemma 15.3 gives an explicit cyclic-vector open cover and proves that M is locally free of rank one as a C -module. The exact sequence of C -modules $0 \rightarrow C \rightarrow Q \rightarrow M \rightarrow 0$ consequently splits locally, so Q is locally free of rank two over C .

Conversely, if Q is a locally free rank-two C -algebra, its unit is unimodular as a C -section. Hence Q/C is locally free of rank one over C . Its regular C -action is faithful and is a subbundle injection into its endomorphism bundle over the original base (check on geometric fibres and use a pivot). The quotient-flag equivalence therefore gives exactly the second stratum. All arguments are functorial and use split modules on the indicated strata. $\square$

Remark 15.5. Consider $R = \mathbb{C}[s]$, $B = R[z]/(z^5)$, and the direct summand

$$E = \langle 1, sz^2 + z^3, z^4 \rangle_R.$$

It is a subalgebra since $(sz^2 + z^3)^2 = s^2z^4$. Modulo E take the basis $\bar{z}, \bar{z}^2$; then $\bar{z}^3 = -s\bar{z}^2$. The action of $sz^2 + z^3$ sends $\bar{z}$ to $-s^2\bar{z}^2$ and $\bar{z}^2$ to zero, while z^4 acts as zero. Thus the exact-rank-one subscheme is $s^2 = 0$, not merely $s = 0$. Over R the conductor is Rz^4 ; at the reduced fibre $s = 0$ it

is $\langle z^3, z^4 \rangle$. Quotient rank drops from four to three. This simultaneously records the nonreduced rank stratum and the failure of conductor formation to commute with unrestricted specialization. The fibre subalgebras belong to the classical codimension-two polynomial-algebra classification of [34, Theorem 39]; the calculation here retains the relative action and its determinantal thickening.

16. THE FULL CODIMENSION-TWO FAILURE SCHEME

Fix a finite locally free commutative algebra B/S of rank $d \geq 4$ over a locally Noetherian $\mathbb{C}$ -scheme. Let

$$X = \mathrm{Gr}_S(d-3, B/\mathcal{O} \cdot 1),$$

viewed as the Grassmannian of rank- $(d-2)$ subbundles $W \subset B$ containing 1. This is a unital parameter space, not an open subset of the Grassmannian of all generating planes. We explain the passage to the latter at the end of the section.

Write $\mu_j : \mathrm{Sym}^j W \rightarrow B_X$, and let D denote the stable zeroth-Fitting failure scheme. Let $T \subset X$ denote the unital subalgebra scheme. In general T is a proper closed subscheme of D ; its ideal is the first, rather than the zeroth, Fitting ideal.

Theorem 16.1 (Equations and the corank-one incidence). *The stable scheme D is defined already by μ_3 . On a splitting $B_X = W \oplus M$, with $\mathrm{rank} M = 2$, use the block notation of (13.3). Let β be the matrix with columns $\beta_{w_i}(w_j)$, for a local basis (w_i) of W , and set*

$$\mathcal{K} = [\beta \mid C_{w_1}\beta \mid \cdots \mid C_{w_{d-2}}\beta]. \quad (16.1)$$

For every $m \geq 3$ there are equalities

$$\begin{aligned} \mathrm{coker} \mu_m &\simeq \mathrm{coker} \mathcal{K}, \\ \mathcal{I}_D &= I_2(\mathcal{K}), \\ \mathcal{I}_T &= I_1(\beta) = I_1(\mathcal{K}) = \mathrm{Fitt}_1 \mathrm{coker} \mu_m. \end{aligned} \quad (16.2)$$

The cokernel isomorphism uses the chosen splitting; the ideals glue canonically. These are equations for the full schemes, not just for their radicals or their exact-rank strata.

Let $H = \mathrm{Hilb}_S^2(B)$ be the scheme of locally free rank-two quotient algebras. Over H , let $E_H \subset B_H$ be the preimage of the scalar subalgebra in the universal quotient. Put

$$Z = \mathrm{Gr}_H(d-3, E_H/\mathcal{O} \cdot 1), \quad Z^\circ = \{W : W^2 \longrightarrow E_H \text{ is onto}\}.$$

Then the forgetful morphism induces an isomorphism

$$Z^\circ \xrightarrow{\sim} D \setminus |T|. \quad (16.3)$$

It holds on arbitrary test schemes, including nonreduced ones. On this open the stable generated algebra is the hyperplane E_H , and the normalized multiplication cokernel is its quotient line B/E_H . The proper morphism $Z \rightarrow X$ factors through D and is surjective on geometric points of D .

Proof. Theorem 13.1 gives $W^3 = \mathcal{O}[W]$. By (13.2), its residual image in M is $N_1 = \mathrm{im} \mathcal{K}$. The identity columns supplied by $W \subset W^3$ may be removed from a presentation of B/W^3 . This proves the first two assertions in (16.2). Every entry of $C_{w_i}\beta$ is a linear combination of entries of β , so their entry ideals are equal. Vanishing of these entries is exactly $W^2 \subset W$, which is the subalgebra functor. This proves the last line.

Work now where a column v of β is unimodular in M . Its lift is a quadratic product modulo W , and $E = W + \mathcal{O}v$ is a hyperplane subbundle of B . Pivot on the unit coordinate of v . The two by two minors of $\mathcal{K}$ then say precisely that all its columns belong to $\mathcal{O}v$. In the quotient ring by those minors, $W^2 = E$ and $WE \subset E$. Moreover $v \in W^2$, so multiplication by v preserves E : write v as a sum of products of elements of W and use $WE \subset E$ twice. Thus E is a unital subalgebra. Equivalently its equality with the stable image follows directly from Theorem 13.1.

The conductor equivalence of Lemma 14.1 associates to E a unique rank-two quotient of B . It recovers a point of Z° . Conversely, a point of Z° has stable image E_H , since $W^2 = E_H$ and E_H is an algebra. Its residual multiplication rank is exactly one, so it factors through $D \setminus |T|$. The two constructions are inverse on every test scheme and are defined by split pivots and the hyperplane conductor. They therefore glue to (16.3), including its nilpotent structure. The same description gives the cokernel line.

The relative Hilbert scheme is projective: it is the closed ideal-quotient subscheme of the Grassmannian of rank-two module quotients of B . Hence $Z \rightarrow X$ is proper. Every multiplication image on Z lies in the hyperplane algebra E_H , so its maximal minors vanish and the map factors through D . Surjectivity away from T is already proved. Surjectivity on geometric points of T follows from the two fibre descriptions in Proposition 16.2 below. $\square$

The theorem gives both an equation for D along T and a complete functorial normal form off T . In particular, it is not obtained by replacing the ideal $I_2(K)$ with the entry ideal $I_1(\beta)$. Nor does it assert that the corank-one open is dense in every component: for a particular algebra a component may be contained in T .

Proposition 16.2 (Fibres over the conductor strata). *Restrict T to an exact-action-rank stratum and write $J = \text{cond}_B(W)$, $C = W/J$, and $Q = B/J$. The fibre product $Z \times_X T$ has the following descriptions, with arbitrary base change within the indicated stratum.*

If the action rank is one, $C = \mathcal{O}$ and Q has rank three. The fibre product is $\text{Hilb}^2(Q)$. More explicitly, on $\mathbb{P}(Q/\mathcal{O})$, in the convention of lines, it is the zero scheme of the canonical binary cubic

$$\ell^{\otimes 3} \longrightarrow \bigwedge^2(Q/\mathcal{O}), \quad v^{\otimes 3} \longmapsto \bar{v} \wedge \overline{v^2}. \quad (16.4)$$

The cubic may vanish identically; no relative degree-three flatness is asserted without a nonvanishing hypothesis.

If the action rank is two, C has rank two and Q is a rank-two C -algebra. The fibre product is canonically $\text{Spec } C$, a finite locally free double cover of the stratum, possibly nonreduced.

Proof. A point of the fibre product is a hyperplane algebra $E \subset B$ containing W . Since J is an ideal of B contained in W , it is contained in the conductor of E . Thus the problem is exactly to choose a unital hyperplane subalgebra of Q containing C . Conversely its inverse image supplies E .

In the first case this is the unital-hyperplane functor of a rank-three algebra, hence $\text{Hilb}^2(Q)$ by Lemma 14.1. A line $\ell \subset Q/\mathcal{O}$ has an algebra as its preimage if and only if its square remains in that preimage. This is exactly (16.4). Changing a lift v by a scalar changes v^2 by a scalar and a multiple of v , so the displayed wedge is independent of the lift. This also proves its scheme-theoretic meaning over a nonreduced base.

In the second case put $M = Q/C$. By Lemma 15.3, M is locally an invertible C -module. A hyperplane extension gives a C -stable subbundle $\ell \subset M$ of rank one over the original base. The induced action on the quotient line M/ℓ is a character $C \rightarrow \mathcal{O}$. Conversely such a character, with kernel I , gives $\ell = IM$; it is a subbundle and the quotient is a line. These constructions are inverse, as can be checked after trivializing M over C . They are the functor of $\text{Spec } C$. There is no additional multiplication equation. Indeed locally write $Q = C \oplus Cz$, with $z^2 = \alpha + \gamma z$, $\alpha, \gamma \in C$. The extension is $C \oplus Iz$, which is an algebra because I is an ideal of C . This proves the assertion over every test scheme.

For completeness both fibres are nonempty over an algebraically closed field. A rank-three commutative algebra has a quotient of length two, by an ideal composition series; a rank-two algebra has a character. This proves the geometric surjectivity used above. $\square$

For example, the rank-three algebra $\mathbb{C}^3$ gives three points in (16.4), whereas $\mathbb{C}[u, v]/(u, v)^2$ gives the whole projective line. Mere invariance under the scalar action would incorrectly give a projective line in both cases. The square condition in (16.4) is essential. In contrast, on the rank-two action stratum it follows from the quadratic-algebra tower. The conductor stratification thus controls the exceptional fibres of one proper incidence over the full failure scheme. We do not identify this morphism with a normalization or a blow-up.

Remark 16.3 (Unit covers and unrestricted generating planes). Let $G = \text{Gr}_S(d-2, B)^\circ$ be the full generating Grassmannian, and let $\tilde{G}$ parametrize pairs (A, a) with $a \in A \cap B^\times$. The morphism $\tilde{G} \rightarrow G$ is smooth and surjective, and

$$\tilde{G} \simeq B^\times \times_S X, \quad (A, a) \longmapsto (a, a^{-1}A).$$

Multiplication by a^{-m} identifies the pulled-back cokernel with the unital one. All the ideal identities above therefore give the full failure scheme on this smooth cover of G and descend. Over $\mathbb{C}$, local primary structures discussed below may equally be read on this cover; the assertion is not that a special linear slice of G determines its ambient scheme structure.

16.1. The quadratic incidence fibre as a functor. In Theorem 16.1, the notation $D \setminus |T|$ means the open complement of the support of the possibly nonreduced closed scheme T . Nothing is subtracted from its structure sheaf. We supply a full functorial description of the rank-two action fibre in Proposition 16.2.

Let $C \subset Q$ be the conductor flag on that exact-action-rank stratum. Then C is a rank-two algebra, Q is a rank-two locally free C -algebra, and $M = Q/C$ is an invertible C -module. The incidence fibre parametrizes rank-three subalgebras $E' \subset Q$ containing C , with Q/E' a line bundle over the base.

A morphism $U \rightarrow \text{Spec } C$ is an algebra map $\chi : C_U \rightarrow \mathcal{O}_U$. Write $I = \ker \chi$; it is a line subbundle of C_U , since the unit splits χ . Tensoring $0 \rightarrow I \rightarrow C_U \rightarrow \mathcal{O}_U \rightarrow 0$ with the invertible C_U -module M_U gives

$$0 \longrightarrow IM_U \longrightarrow M_U \longrightarrow M_U \otimes_{C_U, \chi} \mathcal{O}_U \longrightarrow 0.$$

The last term is a line bundle over U . The middle term has rank two over U , so IM_U is a line subbundle and this sequence is locally split over $\mathcal{O}_U$. Let E'_χ be its inverse image in Q_U . It is a rank-three subbundle containing C_U . To check multiplicative closure, one may work faithfully flat locally where M_U has a C_U -generator lifted to $e \in Q_U$. Then $Q_U = C_U \oplus C_U e$, $e^2 = \alpha + \beta e$, and $E'_\chi = C_U + Ie$. Products stay in this submodule because I is an ideal and $I^2 \subset I$. The closure condition descends, and the definition of E'_χ did not use the generator.

Conversely let E' be such a subalgebra, and put $N = E'/C_U$. The exact sequence

$$0 \longrightarrow N \longrightarrow M_U \longrightarrow Q_U/E' \longrightarrow 0$$

exhibits N as a line subbundle. Since $C_U \subset E'$, the submodule N is C_U -stable. Its quotient line $L' = Q_U/E'$ therefore defines an algebra map

$$\chi : C_U \longrightarrow \text{End}_{\mathcal{O}_U}(L') = \mathcal{O}_U.$$

The equality on the right is canonical for an invertible module. Let $I = \ker \chi$. The quotient map factors through $M_U \otimes_{C_U, \chi} \mathcal{O}_U \rightarrow L'$. This is a surjective map of line bundles, hence an isomorphism. Thus $N = IM_U$ and $E' = E'_\chi$. Conversely the action on the quotient constructed from χ recovers χ itself. All the quotient sequences are locally split as sequences of base modules, and tensoring an invertible C -module is exact. The two constructions therefore commute with every pullback and are inverse on every test scheme, not just on geometric points. This proves the asserted fibre isomorphism with $\text{Spec } C$ by the functor of points.

17. PRIMARY GEOMETRY AT A CONDUCTOR BOUNDARY

The distinction between a presentation of the full failure scheme and its primary geometry is essential along the subalgebra locus. We now resolve that geometry for every complex local algebra with Hilbert function $(1, 2, 2)$. The proof begins with a factorization valid for all algebras whose augmentation ideal has cube zero. In the classified case the quadratic relation determines the embedded associated strata, including a stratum supported entirely at the extreme-corank boundary. No restriction to the quadratic-generation open is made.

17.1. The quadratic-layer factorization. Let B be a finite local $\mathbb{C}$ -algebra, with maximal ideal $\mathfrak{m}$ satisfying $\mathfrak{m}^3 = 0$. Put $V = \mathfrak{m}/\mathfrak{m}^2$, $S_2 = \mathfrak{m}^2$, and write $e = \dim V$, $q = \dim S_2$. Multiplication induces a surjection

$$\gamma : \text{Sym}^2 V \longrightarrow S_2.$$

Choose a splitting $\mathfrak{m} = V \oplus S_2$. The unital parameter space of $(e+1)$ -planes is $X = \text{Gr}(e, \mathfrak{m})$: its tautological plane is $W = \mathcal{O} \cdot 1 \oplus K$. On its frame bundle write

$$\begin{pmatrix} M \\ H \end{pmatrix} : \mathcal{O}^e \longrightarrow V \oplus S_2, \quad M \in \text{Mat}_{e \times e}, \quad H \in \text{Mat}_{q \times e}. \quad (17.1)$$

The stacked matrix has rank e . Its entries, not just the entries of M , are the frame coordinates.

Theorem 17.1 (Quadratic-layer factorization). *For every degree $m \geq 2$, the full zeroth-Fitting ideal of $\text{Sym}^m W \rightarrow B$ pulls back to*

$$\mathcal{I}_D = (\det M) I_q(\gamma \text{Sym}^2 M). \quad (17.2)$$

The identity holds on the whole frame bundle and after arbitrary base change, including substitution of coefficients in a nonreduced ring. In particular it also gives equations at $M = 0$ whenever that stratum meets the frame bundle. The variables H are unrestricted except for the frame condition; the formula is not a transverse slice.

Proof. Products of three elements of K vanish. Because $1 \in W$, the multiplication image in every degree $m \geq 2$ is $\mathcal{O} \cdot 1 + K + K^2$. After deleting zero and repeated columns, a presentation of its cokernel is given by

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & H & \gamma \text{Sym}^2 M \end{pmatrix}.$$

Every nonzero maximal minor must use the first column and all e columns of the middle block: no other column has an entry in the V rows. Expansion in the first $e+1$ rows leaves exactly $\det M$ times a q by q minor of the last block. These are all the maximal minors. This is a polynomial identity in the universal coefficient ring. It proves the assertion before taking a radical, localizing at a tangent minor, or passing to residue fields. $\square$

The same calculation applies to an augmented locally free algebra $\mathcal{O} \oplus V \oplus S_2$ over a base scheme, with locally free layers, $S_2(V \oplus S_2) = 0$, and multiplication γ . Formula (17.2) therefore describes families of quadratic relations as well as families of generating planes.

17.2. The complete class of Hilbert function $(1, 2, 2)$. Every local algebra in this class has $\mathfrak{m}^3 = 0$. The kernel of $\gamma : \text{Sym}^2 V \rightarrow S_2$ is a line of binary quadratic forms. Over $\mathbb{C}$ its generator is either a square or a product of two independent linear forms. Thus the complete list of isomorphism types is

$$\begin{aligned} B_{\text{sq}} &= \mathbb{C}[x, y]/((x, y)^3, y^2), \\ B_{\text{tf}} &= \mathbb{C}[x, y]/((x, y)^3, xy). \end{aligned} \quad (17.3)$$

Indeed a linear splitting of $\mathfrak{m} \rightarrow V$ identifies B with $\mathbb{C} \oplus V \oplus S_2$, with multiplication determined entirely by γ ; there are no further extension coefficients when $\mathfrak{m}^3 = 0$.

In both cases $X = \text{Gr}(2, \mathfrak{m})$ parametrizes unital three-planes. On its frame bundle put

$$M = \begin{pmatrix} a & c \\ b & d \end{pmatrix}, \quad \delta = ad - bc, \quad R = \mathbb{C}[a, b, c, d], \quad P = (a, c), \quad Q = (b, d), \quad \mathfrak{n} = (a, b, c, d).$$

The four entries of H are independent additional variables. All the following ideals are extended to $R[H]$ and restricted to the open where the stacked matrix (17.1) has rank two. A component absent from an individual affine chart is simply omitted there; the decomposition is irredundant on the whole frame bundle.

Theorem 17.2 (Complete primary decomposition for Hilbert function $(1, 2, 2)$). *For every $m \geq 2$ the full multiplication failure scheme has the following primary decompositions.*

For B_{sq} ,

$$I_{\text{sq}} = \delta^2 P^2 = (\delta^2) \cap P^4, \quad \text{Ass}(R/I_{\text{sq}}) = \{(\delta), P\}. \quad (17.4)$$

The prime P is embedded. More generally, for every $j \geq 1$,

$$I_{\text{sq}}^j = (\delta^{2j}) \cap P^{4j}. \quad (17.5)$$

For B_{tf} , set

$$\begin{aligned} J &= (ab, ad + bc, cd), \\ Q_0 &= J + P^2 + Q^2, \\ Q_* &= \delta^2 Q_0 + \mathfrak{n}^7. \end{aligned} \quad (17.6)$$

Then Q_0 and Q_* are $\mathfrak{n}$ -primary and

$$\begin{aligned} I_{\text{tf}} &= \delta^2 J = (\delta^2) \cap P^3 \cap Q^3 \cap Q_*, \\ \text{Ass}(R/I_{\text{tf}}) &= \{(\delta), P, Q, \mathfrak{n}\}. \end{aligned} \quad (17.7)$$

The last three primes are embedded. The displayed Q_* is a specified primary representative, not a canonical choice of an embedded primary component.

These identities describe the entire schemes on X , including the extreme-corank locus T . Intrinsically, an embedded surface is indexed by a linear factor $L \subset V$ of the quadratic relation: its support is

$$Y_L = \{K \subset \mathfrak{m} : K \rightarrow V/L \text{ is zero}\} = \text{Gr}(2, S_2 + L) \simeq \mathbb{P}^2. \quad (17.8)$$

There is one such surface for a square and two for distinct factors. In the two-factor case there is also the embedded point $K = S_2$. There are no further associated strata.

The reduced support in both cases is the same integral normal divisor

$$\Delta = \{\text{rank}(K \rightarrow V) \leq 1\} \subset X.$$

Its singular locus is the point $K = S_2$. The generic multiplicity of D along Δ is two, whereas the nilradical of $\mathcal{O}_D$ has exact nilpotency index four. Neither full failure scheme is Cartier.

Proof. Choose bases in the quadratic layer, and rescale the middle symmetric basis vector where necessary. The two quadratic product matrices are

$$L_{\text{sq}} = \begin{pmatrix} a^2 & 2ac & c^2 \\ ab & ad + bc & cd \end{pmatrix}, \quad L_{\text{tf}} = \begin{pmatrix} a^2 & 2ac & c^2 \\ b^2 & 2bd & d^2 \end{pmatrix}.$$

For the first matrix the target basis may be taken to be $x^2, 2xy$; for the second it is x^2, y^2 . Their two-minor ideals are, respectively, δP^2 and δJ . Theorem 17.1 gives the asserted full failure ideals. Notice the extra factor δ supplied by the linear layer; dropping that layer would produce a different primary problem.

For the square case, the P -adic associated graded ring of R is $\mathbb{C}[b, d][a, c]$, a domain. The P -adic order of δ is one. Consequently

$$(P^t : \delta^u) = P^{\max\{t-u, 0\}} \quad (t, u \geq 0).$$

It follows that $(\delta^2) \cap P^4 = \delta^2 P^2$, and the same calculation with $2j, 4j$ proves (17.5). The polynomial δ is prime, and powers of P are primary. The two components are irredundant: $\delta^2 \notin P^4$ and $a^4 \notin (\delta^2)$. This proves the associated-prime assertion.

For the second case we first prove

$$J = P \cap Q \cap Q_0. \quad (17.9)$$

Since $P \cap Q = PQ$, its quotient by J is spanned by the class of δ : the relations set $ab = cd = 0$ and $ad = -bc$. Every element of $\mathfrak{n}$ annihilates this class, and the class is nonzero by linear independence in degree two. In R/Q_0 a basis is

$$1, a, b, c, d, ad,$$

with $bc = -ad$ and all other quadratic monomials zero. Thus $\mathfrak{n}^3 \subset Q_0$, Q_0 is $\mathfrak{n}$ -primary, and $\delta = 2ad$ remains nonzero in this quotient. The natural map $PQ/J \rightarrow R/Q_0$ is therefore injective, proving (17.9). It is irredundant; the embedded prime $\mathfrak{n}$ is also witnessed directly by $\text{ann}_{R/J}(\delta) = \mathfrak{n}$.

The same associated-graded argument as before gives

$$(P^3 : \delta^2) = P, \quad (Q^3 : \delta^2) = Q.$$

We claim that $(Q_* : \delta^2) = Q_0$. For a homogeneous polynomial of degree j , membership of its product with δ^2 in $\mathfrak{n}^7$ is equivalent to $j \geq 3$. More generally the homogeneous decomposition, and injectivity of multiplication by δ^2 in R , give

$$(\delta^2 Q_0 + \mathfrak{n}^7 : \delta^2) = Q_0 + \mathfrak{n}^3 = Q_0.$$

Intersecting with (δ^2) and using (17.9) proves (17.7). Since $Q_* \subset \mathfrak{n}$ and contains $\mathfrak{n}^7$, it is $\mathfrak{n}$ -primary. For completeness multiplication by δ^2 gives an injection $R/J \hookrightarrow R/I_{\text{tf}}$. Hence its three associated primes $P, Q, \mathfrak{n}$ remain associated. The prime (δ) is the unique minimal prime of I_{tf} . This proves irredundance and excludes additional associated primes.

All these decompositions survive polynomial extension by H and localization, and every associated stratum meets the frame open by taking H to be the identity matrix. The global ideals, irredundancy, and associated points are established formally in Corollary 9.2, using Lemma 9.1. That corollary uses only the affine primary equalities just proved; it does not assume the global conclusion of the present theorem. In particular no specialization assertion or unexpanded regular-fibre argument is needed here.

The reduced equation on the frame cover is $\delta = 0$. It is an integral hypersurface whose singular locus is $M = 0$, of codimension three within that hypersurface. A hypersurface is Cohen–Macaulay; regularity in codimension one and Serre’s criterion give normality. Integrality on X also follows from the surjection $\text{Gr}_{\mathbb{P}(V)}(2, S_2 + \mathcal{L}) \rightarrow \Delta$ from an integral scheme. At a general point of Δ the residual ideal P^2 or J is a unit ideal, so the generic multiplicity is two.

In either case write $I = \delta^2 J'$, where $J' = P^2$ or J . One has $\delta \notin J'$ and $\delta^2 \in J'$; in the latter case use $\delta^2 = (ad + bc)^2 - 4abcd$. Therefore $\delta^4 \in I$ and $\delta^3 \notin I$. Since $\sqrt{I} = (\delta)$, the nilradical has exact index four. Nonvanishing is retained on the chart where H is invertible. Finally a nonzero principal ideal in the regular scheme X cannot have these embedded associated primes. Thus D is not Cartier. $\square$

17.3. Conductor equations on the embedded strata. The preceding theorem computes local rings across T , not just the fibres of the incidence over T . We now locate the conductor strata inside these primary supports. This also specifies which embedded components lie entirely in the extreme-corank locus.

Theorem 17.3 (Conductor position of the primary strata). *On the exact rank-one locus of $K \rightarrow V$, write $L = \text{im}(K \rightarrow V)$ and $H_0 = K \cap S_2$. Both are line bundles. The subalgebra locus T there is precisely the zero scheme of*

$$\gamma|_{\text{Sym}^2 L} : \text{Sym}^2 L \longrightarrow S_2/H_0. \quad (17.10)$$

On that scheme the conductor is the actual kernel

$$\text{cond}_B(W) = \ker(K \longrightarrow \text{Hom}(V, S_2/H_0)), \quad k \longmapsto (v \longmapsto \gamma(\bar{k}, v)). \quad (17.11)$$

If the induced map $L \rightarrow \text{Hom}(V, S_2/H_0)$ has exact rank zero, the conductor is K and the conductor action has rank one. If it has exact rank one, the conductor is H_0 and the action has rank two. These kernel statements commute with base change on the specified scheme-theoretic rank strata.

For B_{sq} the embedded surface $Y_{\langle y \rangle}$ is contained in T . On its rank-one part the action rank is one exactly when $H_0 = \langle xy \rangle$, and is two otherwise. For B_{tf} the two embedded surfaces are $Y_{\langle x \rangle}$ and $Y_{\langle y \rangle}$. Their rank-one intersections with T are, respectively, $H_0 = \langle x^2 \rangle$ and $H_0 = \langle y^2 \rangle$; the conductor action has rank one on both intersections. Elsewhere on the rank-one part of T it has rank two.

At $K = S_2$, in both types, $W = \mathbb{C} \oplus S_2$ has conductor S_2 and quotient $B/S_2 = \mathbb{C} \oplus V$ with square-zero augmentation ideal. The incidence fibre there is $\mathbb{P}(V)$, whereas the action-rank-two fibres above are spectra of the dual numbers. The extra embedded point at $K = S_2$ occurs precisely in the two-factor type. Thus numerical action rank alone does not determine associated primes: the quadratic relation in the conductor data is indispensable.

Proof. Since $S_2 \mathfrak{m} = 0$, one has $W^2 = \mathcal{O} \cdot 1 + K + \gamma(L^2)$. The quotient S_2/H_0 is a line bundle, so closure under multiplication is exactly (17.10), including its nonreduced zero-scheme structure.

An element $\lambda \cdot 1 + k \in W$ in the conductor must have $\lambda = 0$: multiplication by S_2 and projection to the free line S_2/H_0 is scalar multiplication by λ . For $k \in K$, multiplication by S_2 is zero, and multiplication by a lift of $v \in V$ is $\gamma(\bar{k}, v)$. It belongs to W exactly when it belongs to H_0 . This proves (17.11) over every coefficient ring on the indicated rank locus. Its restriction to the exact-rank strata is a kernel of constant rank, so the sequences split locally and pull back. The conductor has rank two or one accordingly; as $\text{rank } W = 3$, the action ranks are one or two.

In the square type, $\gamma(y^2) = 0$ and $\gamma(y, V) = \langle xy \rangle$. This proves the assertion about the entire embedded surface and its action-rank jump. If $L \neq \langle y \rangle$, then $\gamma(L, V) = S_2$, so action rank one cannot occur on its rank-one subalgebra locus. In the two-factor type, multiplication by x has image $\langle x^2 \rangle$ and multiplication by y has image $\langle y^2 \rangle$. For any other line both quadratic basis vectors occur and the image is S_2 . These observations prove all the rank-one assertions. They are equations in the universal line and quotient bundles, rather than an appeal to continuity of ranks.

At $K = S_2$ the conductor computation is immediate. The action-rank-one incidence is the scheme of length-two quotients of $\mathbb{C} \oplus V$, hence the projective line: its binary cubic square obstruction vanishes identically. On the action-rank-two locus the rank-two conductor algebra is local with square-zero augmentation line, since every quadratic product in K lies in H_0 . Proposition 16.2 therefore gives the spectrum of the dual numbers on geometric fibres. The assertion about the embedded point follows from Theorem 17.2, not from the incidence fibre list. $\square$

On the rank-one, action-rank-one loci just described, the residual rank-three algebra is curvilinear: the surviving tangent vector has nonzero square. Thus its length-two quotient scheme has

length three. The change from these cubic fibres, or the quadratic fibres, to the projective line at $K = S_2$ is accompanied by the local equations (17.4) and (17.7). Those equations, rather than the dimensions of the fibres, determine the nilpotent boundary.

Proposition 17.4 (Collision of the quadratic factors). *The algebras*

$$\mathcal{B}_\tau = \mathbb{C}[\tau, x, y] / ((x, y)^3, y^2 - \tau xy)$$

form a finite free rank-five family over $\mathbb{C}[\tau]$. On the full unital three-plane frame space its failure ideal in every degree $m \geq 2$ is

$$\delta^2(a(a + \tau b), 2ac + \tau(ad + bc), c(c + \tau d)). \quad (17.12)$$

Its fibre at $\tau = 0$ is the square-type scheme, and every fibre at $\tau \neq 0$ is the two-factor-type scheme. Thus the primary strata in Theorem 17.2 occur in one family of full Fitting schemes with fixed algebra length and Hilbert function. No flatness of that failure family, or compatibility of a chosen primary decomposition with specialization, is asserted.

Proof. The elements $1, x, y, x^2, xy$ are a free basis over $\mathbb{C}[\tau]$; the displayed relation is the only elimination in degree two and all terms of degree at least three vanish. The quadratic matrix, in the product basis without rescaling the middle column, is

$$\begin{pmatrix} a^2 & ac & c^2 \\ 2ab + \tau b^2 & ad + bc + \tau bd & 2cd + \tau d^2 \end{pmatrix}.$$

Its minors are δ times the three displayed generators. Theorem 17.1 proves the ideal identity over $\mathbb{C}[\tau]$ and after specialization. The quadratic relation is $y(y - \tau x)$, proving the assertions about the fibres. $\square$

Corollary 17.5 (Projective realization of the complete boundary). *Let $Z \subset \mathbb{P}^2$ have homogeneous ideal $((X, Y)^3, Y^2)$ or $((X, Y)^3, XY)$ in $\mathbb{C}[T, X, Y]$. For every $n \geq 5$, the inverse-image linear series restricting to all generating three-planes on Z have, in each symmetric degree $m \geq 2$, exactly the primary schemes of Theorem 17.2, on their smooth unit covers. In particular the embedded conductor-boundary components are components of actual global section-multiplication failure.*

Proof. The ideals are saturated because T is a nonzerodivisor on the quotient. The affine algebras have top graded degree two, so their homogeneous ideals have regularity three. Apply Theorem 19.1, using $n \geq 2 \cdot 3 - 1$, and trivialize $\mathcal{O}_Z(n)$ by T^n . The cokernel isomorphism transfers the full Fitting ideal. Finally Remark 16.3 transfers the equations from unital planes to a smooth cover of the unrestricted generating Grassmannian. No boundary is removed in either step. $\square$

18. EMBEDDED PRIMARY STRATA FOR CONNECTED LOCAL ALGEBRAS

The incidence of Theorem 16.1 transfers more than the support of a quotient scheme. On its quadratic-generation open it transfers its local rings, and hence its primary structure. We apply it to a connected noncurvilinear family in which the full multiplication failure scheme has an embedded component.

For $h \geq 5$ set

$$B_h = \mathbb{C}[x, y] / (x^2, xy, y^h), \quad \dim_{\mathbb{C}} B_h = h + 1. \quad (18.1)$$

Let $F_0 = 0$, $F_1 = 1$, and $F_{j+1} = uF_j + vF_{j-1}$, and put

$$\begin{aligned} J_h &= (F_h, vF_{h-1}) \subset \mathbb{C}[u, v], \\ I_h &= (F_h, vF_{h-1}, bv, b^2u) \subset \mathbb{C}[b, u, v], \\ P_h &= (u, v), \quad Q_h = (b^2, bv, F_h, vF_{h-1}). \end{aligned} \quad (18.2)$$

The equality $J_h = (F_h, F_{h+1})$ follows from the recurrence.

Theorem 18.1 (A connected embedded-primary family). *Consider the unital Grassmannian of $(h-1)$ -planes in B_h , and its full zeroth-Fitting multiplication failure scheme D_h in any degree $m \geq 3$. At the plane*

$$W_0 = \langle 1, x, y^2, y^3, y^5, \dots, y^{h-1} \rangle \quad (\text{the term } y^4 \text{ is omitted}), \quad (18.3)$$

there is a Zariski neighbourhood in D_h isomorphic to

$$\text{Spec}((\mathbb{C}[b, u, v]/I_h)[t_1, \dots, t_{h-2}])_f, \quad f(0) = 1. \quad (18.4)$$

Here the variables t_i are actual Grassmannian coordinates, and f is the coefficient of a specified quadratic product in the one-dimensional quotient of the conductor hyperplane by W . This is a local description of the full failure scheme, not of a transverse test family.

The ideal in (18.2) has the irredundant primary decomposition

$$I_h = P_h \cap Q_h. \quad (18.5)$$

Its associated primes are (u, v) and (b, u, v) . The first is the minimal prime and the second is embedded. The same decomposition, with the ideals extended and localized, describes (18.4). The reduced support has codimension $h-3 \geq 2$ in the unital Grassmannian, so this failure scheme is not Cartier. Its nilradical has exact nilpotency index $2h-3$. For the displayed choice of embedded primary ideal,

$$\dim_{\mathbb{C}} \mathbb{C}[b, u, v]/Q_h = \binom{h}{2} + h - 1, \quad \dim_{\mathbb{C}} \sqrt{I_h}/I_h = \binom{h}{2} + h - 3. \quad (18.6)$$

The second formula concerns the coefficient ring before adjoining the Grassmannian variables. It records a nilradical supported at the embedded point, rather than multiplicity along the generic point of a Cartier divisor.

Proof. We first compute an entire open chart of $\text{Hilb}^2(B_h)$. Choose the chart on which $1, y$ is a basis of the universal quotient. Write $y^2 = uy + v$ and $x = a + by$. The relation $xy = 0$ gives $a = -bu$ and $bv = 0$. After this substitution, $x^2 = 0$ is equivalent to $b^2u = 0$: its constant equation follows from $bv = 0$ and $b^2u = 0$. Finally

$$y^h = F_h y + v F_{h-1}$$

shows that $y^h = 0$ is equivalent to the two generators of J_h . These calculations give a quotient algebra free on $1, y$ over $\mathbb{C}[b, u, v]/I_h$, and conversely every such quotient has these coefficients. They prove an isomorphism of the chart functor with $\text{Spec } \mathbb{C}[b, u, v]/I_h$ over arbitrary coefficient rings. No radical has been taken.

Here is a direct primary calculation. Write $A_h = \mathbb{C}[u, v]/J_h$ and $\mathfrak{n} = (u, v)A_h$. It is an Artin local algebra, with residue field $\mathbb{C}$. In $A_h[b]$, coefficient comparison in the variable b gives

$$\mathfrak{n}A_h[b] \cap (b^2, bv) = (bv, b^2\mathfrak{n}) = (bv, b^2u).$$

Lifting to $\mathbb{C}[b, u, v]$ proves (18.5). The ideal P_h is prime; the quotient by Q_h is an Artin local ring supported at (b, u, v) , so Q_h is primary. The decomposition is irredundant: $b^2 \in Q_h \setminus P_h$, whereas $u \in P_h \setminus Q_h$. The asserted associated primes follow. Polynomial extension and localization at a function nonzero at the specified point retain both components.

For clarity, the numerical invariants can be read without any primary-decomposition algorithm. The ring A_h is the contact algebra G_h of Theorem 14.3. It has dimension $\binom{h}{2}$, and

$$A_h/(v) \simeq \mathbb{C}[u]/(u^{h-1}), \quad \mathfrak{n}^{2h-3} = 0, \quad u^{2h-4} \neq 0.$$

These last assertions also follow from its two-row Schur basis: $\deg u = 1$, $\deg v = 2$, the top weighted degree is $2h-4$, and the coefficient of the top Schur class in u^{2h-4} is nonzero. This is

the classical basis input recorded in [31]. As vector spaces the quotient by Q_h is $A_h \oplus b(A_h/(v))$, proving the first length formula. In the quotient by I_h , the nilradical is

$$\mathfrak{n} \oplus b(u\mathbb{C}[u]/(u^{h-1})).$$

All coefficients of b^j for $j \geq 2$ are scalars. This proves the second length formula. The inclusion of the constant coefficient ring A_h is injective, so the upper bound $2h - 3$ on nilpotency is attained. Localization at $f(0) = 1$ does not kill its nonzero top class: it is supported at the retained maximal ideal.

It remains to prove that this chart is realized by the full multigenerator failure scheme. At $b = u = v = 0$, the conductor hyperplane is

$$E_0 = \langle 1, x, y^2, \dots, y^{h-1} \rangle.$$

The plane W_0 in (18.3) is a hyperplane in E_0 , and $W_0^2 = E_0$, since $(y^2)^2 = y^4$. In the universal quotient chart the conductor hyperplane has the basis

$$1, \quad x - by, \quad y^j - F_j y \quad (2 \leq j \leq h-1).$$

Take the Grassmannian chart in which the vector $y^4 - F_4 y$ is the omitted direction, and add its multiples t_i to each of the other nonconstant basis vectors. This gives $h - 2$ free coordinates on $\text{Gr}(h - 2, E/\mathcal{O})$. Let q_2 be the resulting lift of the y^2 basis vector. Its square belongs to E , because E is an algebra. The coefficient f of q_2^2 in E/W , in the omitted basis direction, equals 1 at the origin. On $f \neq 0$ one has $W^2 = E$. Theorem 16.1 now identifies this localized polynomial chart with an open neighbourhood of W_0 in D_h , proving (18.4).

The unital Grassmannian has dimension $2(h - 2)$, whereas the support of (18.4) has dimension $1 + (h - 2) = h - 1$. Their difference is $h - 3$. This proves the non-Cartier assertion and finishes the proof. $\square$

For $h = 5$ the theorem gives four-planes in a connected length-six algebra. Its coefficient ideal is explicitly

$$(u^4 + 3u^2v + v^2, u^3v + 2uv^2, bv, b^2u).$$

The reduced support is the b -axis, the chosen embedded primary quotient has length 14, and the nilradical has dimension 12 and index 7. Increasing h changes both the higher-jet equations and the embedded nilpotent algebra; it does not merely change the exponent of a fixed determinant divisor. The parameter b couples the extra square-zero tangent direction x to the higher contact coefficients u, v through bv and b^2u .

Corollary 18.2 (A three-generator member after adjoining a reduced factor). *For every $s \geq 1$, let $B = (\mathbb{C}[x, y]/(x^2, y^2)) \times \mathbb{C}^s$. The full stable failure scheme of unital codimension-two planes has a chart, meeting an embedded associated stratum, of the form*

$$\text{Spec}((\mathbb{C}[a, b]/(ab, b^2))[t_1, \dots, t_{s+1}])_g, \quad g(0) = 1.$$

Its primary decomposition is the extension of $(ab, b^2) = (b) \cap (a, b)^2$. In particular $s = 1$ gives three multiplication generators and an embedded component on the full failure scheme, rather than on a selected slice.

Proof. Use the open-and-closed part of $\text{Hilb}^2(B)$ supported in the first factor. Proposition 14.5 gives the coefficient chart $\mathbb{C}[a, b]/(ab, b^2)$ there. At its origin the conductor hyperplane is $\langle 1, y, xy \rangle \times \mathbb{C}^s$. Let e_i denote the standard idempotent of the i th reduced factor. The plane generated by

$$1_B, \quad (y, e_1), \quad (xy, 0), \quad e_2, \dots, e_s$$

has rank $s + 2 = \dim B - 2$. Its square contains $e_1 = (y, e_1)^2$, so its quadratic image is the entire conductor hyperplane. The relative Grassmannian chart has $s + 1$ parameters. Localize at the

quadratic-generation minor nonzero at this point and apply Theorem 16.1. The displayed primary identity is retained under this polynomial extension and localization. $\square$

The connected family of Theorem 18.1 and the three-generator corollary serve different purposes. The first gives unbounded connected local complexity; the second supplies a three-generator instance of the same incidence mechanism. Both also describe smooth local covers of the unrestricted generating Grassmannians by Remark 16.3. Neither uses a claim that primary decomposition survives arbitrary specialization.

19. CONDUCTOR TRANSPORT FOR ARBITRARY FINITE PROJECTIVE SCHEMES

The finite-algebra results apply to arbitrary zero-dimensional projective schemes once the conductor kernel is filled. This can be done by a normal-layer argument in which Castelnuovo–Mumford regularity replaces the monomial calculation for a homogeneous fat point. We distinguish the regularity of a homogeneous ideal from that of its associated sheaf; the distinction is essential for its ordinary square.

Theorem 19.1 (Regularity-controlled conductor transport). *Let $Z \subset \mathbb{P}_{\mathbb{C}}^e$, $e \geq 1$, be a nonempty zero-dimensional closed subscheme, and let $I \subset R = \mathbb{C}[X_0, \dots, X_e]$ be its saturated homogeneous ideal. Put*

$$r = \operatorname{reg} I, \quad s = \operatorname{reg}(I^2),$$

where I^2 is the ordinary homogeneous ideal square. If

$$n \geq \max\{r, s - 1\}, \tag{19.1}$$

then every generating subspace $A \subset H^0(Z, \mathcal{O}_Z(n))$, with $U = \operatorname{res}_Z^{-1} A$ and $K_t = H^0(\mathbb{P}^e, \mathcal{I}_Z(t))$, satisfies

$$K_n U^{m-1} = K_{mn} \quad (m \geq 2). \tag{19.2}$$

Consequently restriction induces a canonical isomorphism

$$\operatorname{coker}(\operatorname{Sym}^m \mathcal{U} \rightarrow H^0(\mathbb{P}^e, \mathcal{O}(mn)) \otimes \mathcal{O}) \simeq \operatorname{coker}(\operatorname{Sym}^m \mathcal{A} \rightarrow H^0(Z, \mathcal{O}_Z(mn)) \otimes \mathcal{O}) \tag{19.3}$$

on the entire generating Grassmannian, for every $m \geq 2$. In particular all Fitting ideals agree.

The simpler sufficient range

$$n \geq 2 \operatorname{reg} I - 1 \tag{19.4}$$

always implies (19.1). It has no restriction on the local embedding dimension, the support partition, or a monomial presentation of Z .

Proof. We use the following classical regularity facts. An ideal of regularity r is generated in degrees at most r . If an ideal has regularity s , then it agrees with its saturation in degrees at least s , and the first cohomology of its associated sheaf vanishes in degrees at least $s - 1$; see [35, Theorem 1.3]. Moreover

$$\operatorname{reg}(I^2) \leq 2 \operatorname{reg} I \tag{19.5}$$

for a zero-dimensional projective scheme, by [35, Theorem 1.8]. This is a bound for the ordinary ideal square, not merely its sheafification. It proves the implication from (19.4). We prove the sharper assertion under (19.1).

Choose $g \in U$ whose restriction generates $\mathcal{O}_Z(n)$ as an invertible $\mathcal{O}_Z$ -module. Such a section exists by Lemma 12.1. Let $t \geq n$. Since $t \geq s - 1$, the conormal sequence gives a surjection

$$K_t \longrightarrow H^0(Z, (\mathcal{I}_Z/\mathcal{I}_Z^2)(t)). \tag{19.6}$$

Multiplication by g is an isomorphism between the conormal twists t and $t + n$, since its restriction is a unit after trivializing $\mathcal{O}_Z(n)$. Thus, for any $f \in K_{t+n}$, one can choose $q \in K_t$ for which $f - gq$ lies in $H^0(\mathbb{P}^e, \mathcal{I}_Z^2(t + n))$.

The ideal I is generated in degrees at most r , and $n, t \geq r$. Factoring coefficient monomials between two chosen ideal generators therefore gives the elementary homogeneous equality

$$(I^2)_{t+n} = I_n I_t. \quad (19.7)$$

Indeed for generators of degrees $a, b \leq r$, a coefficient monomial of degree $t + n - a - b$ can be split into factors of degrees $n - a$ and $t - b$. This proves both containments. Since $s \geq 2$ and $n \geq s - 1$, one has $t + n \geq s$. Agreement with saturation in that degree identifies the preceding sheaf section space with $(I^2)_{t+n}$. Also $K_t = I_t$, because I is saturated. We have proved

$$K_{t+n} = gK_t + K_n K_t = UK_t. \quad (19.8)$$

The first equality means equality of the sum with the whole kernel; the last follows from $g, K_n \subset U$, with the reverse inclusion clear from vanishing on Z . Iterating for $t = n, 2n, \dots$ proves (19.2).

Restriction in degrees n and mn is surjective by regularity. The restriction sequence $0 \rightarrow K_n \rightarrow U \rightarrow A \rightarrow 0$ is locally split on the parameter Grassmannian. The kernel on symmetric powers is generated by $K_n \otimes \text{Sym}^{m-1} U$, whose multiplication image is the full target restriction kernel by (19.2). Taking cokernels of the restriction square proves (19.3). Its map is restriction itself, so the local choices of g disappear from the result. The relative version below supplies, in particular, the asserted scheme-level statement over this whole Grassmannian. $\square$

Theorem 19.2 (Relative finite schemes and base change). *Let S be a locally Noetherian $\mathbb{C}$ -scheme and $\mathcal{Z} \subset \mathbb{P}_S^e$ a finite locally free family. Suppose the saturated homogeneous ideal I_z of every geometric fibre has regularity at most r . For $n \geq 2r - 1$, the kernel identity and cokernel comparison of Theorem 19.1 hold on each relative generating Grassmannian over S and commute with arbitrary base change. The same conclusion holds under uniform bounds $\text{reg } I_z \leq r$, $\text{reg}(I_z^2) \leq s$ and $n \geq \max\{r, s - 1\}$.*

No hypothesis that formation of a conductor ideal or of a primary decomposition commutes with specialization is used. The assertion is about the actual coherent multiplication cokernels.

Proof. Let $p : \mathbb{P}_S^e \rightarrow S$. For $t \geq n$, regularity on geometric fibres and cohomology and base change give split exact sequences locally on S ,

$$0 \longrightarrow p_* \mathcal{I}_{\mathcal{Z}}(t) \longrightarrow p_* \mathcal{O}(t) \longrightarrow p_* \mathcal{O}_{\mathcal{Z}}(t) \longrightarrow 0.$$

Their terms are vector bundles and their formation commutes with base change. One may also obtain this directly from fibrewise surjectivity of restriction, the local freeness of its two outer bundles, and Nakayama's lemma. The resulting kernel has the required fibre by the exact sequence on each fibre.

Over the relative Grassmannian, form the multiplication morphism

$$p_* \mathcal{I}_{\mathcal{Z}}(n) \otimes \text{Sym}^{m-1} \mathcal{U} \longrightarrow p_* \mathcal{I}_{\mathcal{Z}}(mn).$$

It is surjective on every geometric fibre by the preceding theorem, using (19.5) if only the bound r is specified. Its coherent cokernel therefore vanishes by Nakayama's lemma. The same symmetric-power restriction square proves the relative cokernel isomorphism. The restriction bundles and their split sequences pull back, and a surjective morphism remains surjective after every pullback. Hence the construction proves arbitrary base change, not just flat base change. Fitting ideals commute with base change for finitely presented modules. This argument does not require flatness of the first infinitesimal neighbourhood of $\mathcal{Z}$ or base change for a selected primary component. $\square$

The bound recovers the fat-point range $n \geq 2h - 1$, since the homogeneous ideal of an order- h fat point has regularity h . The direct triangular proof in Theorem 21.2 is retained, as is its sharp boundary example. The present proof also applies to nonmonomial local schemes, unions with unequal local lengths, and finite flat deformations with a uniform regularity bound. A uniform

bound is available on a fixed projective Hilbert scheme by boundedness; sharpness of $2r - 1$ is not asserted for each individual finite scheme or each generating rank.

Corollary 19.3 (Global embedded multiplication failure). *Let $h \geq 5$ and let $Z_h \subset \mathbb{P}^2$ have saturated ideal*

$$I = (X^2, XY, Y^h) \subset \mathbb{C}[T, X, Y].$$

For every $n \geq 2h - 1$, take all inverse-image series of generating $(h - 1)$ -planes on Z_h . In every degree $m \geq 3$, their global multiplication failure scheme has, on a smooth unit cover near the plane (18.3), the local primary description (18.4)–(18.5). It has the embedded component and exact nilradical index stated there. The coherent cokernel comparison persists in any finite flat family with the stated regularity bound.

Proof. The variable T is a nonzerodivisor on $\mathbb{C}[T, X, Y]/I$, so I is saturated and the scheme is supported at $[1 : 0 : 0]$. Its affine algebra is B_h . The standard monomials in $\mathbb{C}[X, Y]/I$ are $1, X, Y, \dots, Y^{h-1}$, and its top degree is $h - 1$; thus $\text{reg } I = h$. For comparison, the minimal monomial generators of I^2 are

$$X^4, X^3Y, X^2Y^2, XY^{h+1}, Y^{2h},$$

and its top quotient degree is $2h - 1$, so $\text{reg}(I^2) = 2h$. The conductor theorem applies at $n \geq 2h - 1$. Trivialize $\mathcal{O}_{Z_h}(n)$ by T^n and apply Theorem 18.1 and the unit-cover construction. The equality of coherent cokernels gives the same Fitting scheme globally. The final assertion is the relative theorem, not a claim that the displayed embedded primary ideal persists unchanged under every deformation. $\square$

19.1. Cohomology and arbitrary base change in conductor transport. We make explicit the bundle assertions used in Theorem 19.2. Write $p : \mathbb{P}_S^e \rightarrow S$ and $q : \mathcal{Z} \rightarrow S$. The sheaf $\mathcal{I}_{\mathcal{Z}}$ is S -flat: it is the kernel of a surjection between S -flat sheaves $\mathcal{O}$ and $\mathcal{O}_{\mathcal{Z}}$. For $t \geq \max\{0, r - 1\}$ the two modules

$$E_t = p_*\mathcal{O}(t), \quad F_t = q_*\mathcal{O}_{\mathcal{Z}}(t)$$

are locally free and commute with arbitrary base change. The first is the familiar homogeneous polynomial bundle; for the second use that q is finite locally free and $\mathcal{O}_{\mathcal{Z}}(t)$ is invertible. The fibre regularity bound gives $H^1(\mathbb{P}_S^e, \mathcal{I}_{\mathcal{Z}_S}(t)) = 0$ in this range. Hence the restriction map $E_t \rightarrow F_t$ is surjective on each geometric fibre and, by Nakayama's lemma, on S .

Its kernel K_t is locally free because the quotient F_t is locally free, so

$$0 \longrightarrow K_t \longrightarrow E_t \longrightarrow F_t \longrightarrow 0$$

is locally split. The long exact sequence of direct images identifies $K_t = p_*\mathcal{I}_{\mathcal{Z}}(t)$ and gives $R^1p_*\mathcal{I}_{\mathcal{Z}}(t) = 0$. In fact all higher direct images vanish in this range: $\mathcal{Z}$ is finite over S , and $\mathcal{O}(t)$ on projective space has no higher direct images for $t \geq 0$.

After any base change $S' \rightarrow S$ the displayed split sequence stays exact. The corresponding sequence of sheaves on projective space also stays exact because the original family is flat. Pushforward identifies the new kernel with $p'_*\mathcal{I}_{\mathcal{Z}'}(t)$; restriction is still surjective and its first higher direct image is still zero. This proves the arbitrary base-change assertion directly, without assuming S' reduced, flat over S , or Noetherian.

Now form the relative generating Grassmannian and let $\mathcal{U}$ be the inverse image of its tautological restricted series. For $m \geq 2$ the map

$$K_n \otimes \text{Sym}^{m-1} \mathcal{U} \longrightarrow K_{mn}$$

is a morphism of locally free sheaves. If the uniform bounds are $\text{reg } I_{\bar{s}} \leq r$ and $\text{reg}(I_{\bar{s}}^2) \leq s$, the single-fibre conductor proof applies for $n \geq \max\{r, s - 1\}$. If only r is given, Sidman's ordinary-square bound supplies $s = 2r$. These bounds are used on the geometric fibres only: no

flatness or base change is required for $\mathcal{I}_Z^2$ or its conormal quotient. Thus the displayed morphism is fibrewise surjective, and its finitely presented cokernel is zero. It remains surjective after every pullback.

Finally the locally split sequence $0 \rightarrow K_n \rightarrow \mathcal{U} \rightarrow \mathcal{A} \rightarrow 0$ shows that the kernel of $\mathrm{Sym}^m \mathcal{U} \rightarrow \mathrm{Sym}^m \mathcal{A}$ is generated by $K_n \otimes \mathrm{Sym}^{m-1} \mathcal{U}$. Its multiplication image is exactly the target restriction kernel K_{mn} . The induced isomorphism of multiplication cokernels is therefore canonical restriction and commutes with arbitrary base change. This explains all relative assertions without replacing a coherent cokernel by its fibre dimensions.

20. THREE GENERATORS ON A REDUCED CONTACT DIVISOR

This theorem realizes Haiman's diagonal ideal as the stable Fitting ideal of an actual multiplication map. The all-powers identity is Haiman's input, not a theorem deduced here from multiplication. The resulting global-section failure schemes and their coherent cokernels are obtained using the conductor comparison.

Theorem 20.1 (The complete stable three-plane ideal). *Let $D = p_1 + \cdots + p_d$ be reduced, $d \geq 4$, and let A vary on the entire generating open $\mathrm{Gr}(3, H^0(\mathcal{O}_D))^\circ$. On the smooth unit-frame cover write $A = \langle a, ax, ay \rangle$; thus the matrix with rows $(1, x_i, y_i)$ has rank three. For $m \geq d - 1$, the zeroth Fitting ideal J_m is independent of m , and*

$$J_m = \bigcap_{i < j} P_{ij}, \quad P_{ij} = (x_i - x_j, y_i - y_j). \quad (20.1)$$

For every $q \geq 1$ one has the stronger identity

$$J_m^q = \bigcap_{i < j} P_{ij}^q. \quad (20.2)$$

These are complete primary decompositions, including all component intersections. Thus J_m is radical, all its components are smooth of codimension two on this cover and on the fixed- D Grassmannian, and neither it nor any of its powers has embedded associated points. The corank is d minus the number of distinct pairs (x_i, y_i) . For global binary subseries the same statements hold for $n \geq 2d - 3$.

Proof. After dividing the target by a^m , the multiplication matrix has as columns all monomials in x, y of total degree at most m , evaluated at the d labelled points. Lemma 12.2 identifies, over the entire coefficient ring, the ideal of its d -minors for $m \geq d - 1$ with the ideal generated by determinants of arbitrary monomial columns. Those determinants span the alternating polynomials in the d pairs of variables: antisymmetrize a monomial and use linearity. Haiman's theorem [30, Corollary 3.8.3] states that the ideal generated by these alternants has every power equal to the intersection of the corresponding powers of the pair-diagonal ideals. It gives (20.1) and (20.2) before any localization; hence they hold on the rank-three open. In that corollary Haiman's n is our number d of labelled points, and Haiman's exponent d is our power q . These two indices are independent; no restriction on q is imposed. This use of the diagonal-ideal theorem is essential.

Each P_{ij} is a smooth prime ideal of codimension two; its powers are primary (use two normal coordinates in a regular local ring). The intersections therefore have exactly the indicated associated primes. Smooth faithful flat descent gives the fixed- D assertions. Polynomial functions separate any finite set of distinct affine points, so their image dimension is the number of distinct pairs. The truncation lemma gives the stated corank in this finite degree. Finally apply Theorem 11.2 with $k = 3$. $\square$

The stable moving scheme over the reduced-divisor stratum is integral, though it need not be smooth or unibranched. Its normalization marks one equal pair and is smooth and finite. Indeed the ordered configuration cover is etale and the permutation group is transitive on pairs. Upstairs the reduced scheme is the union in (20.1); its normalization is the disjoint union of the smooth pair diagonals. This construction descends. A point is smooth precisely when only one pair coincides; two or more failed pairs give distinct local branches after etale base change. No assertion about nonreduced divisors is inferred from this etale calculation; those have their own quotient-algebra structure in Section 14 when $k = d - 1$.

21. NONREDUCED MULTIGENERATOR SCHEMES AND FAT-POINT TRANSPORT

We next determine a multiplication failure scheme with arbitrarily large embedding dimension and nilpotence order. It is not a diagonal arrangement and its proof does not use Haiman's theorem. Fix integers $e \geq 2$ and $h \geq 3$, and put

$$B_{e,h} = \mathbb{C}[z_1, \dots, z_e]/(z_1, \dots, z_e)^h, \quad d = \binom{e+h-1}{e}, \quad N = \binom{e+h-1}{e+1}. \quad (21.1)$$

Write $\mathfrak{m}$ for its augmentation ideal. On $X = \text{Gr}(e+1, B_{e,h})^\circ$ the augmentation of the universal bundle $\mathcal{A}$ is surjective. Its kernel K has rank e . Let $\Delta \subset X$ be the determinant divisor of the canonical map

$$K \longrightarrow (\mathfrak{m}/\mathfrak{m}^2) \otimes \mathcal{O}_X. \quad (21.2)$$

Thus this divisor is defined without a choice of unit or coordinates.

Theorem 21.1 (Exact primary law for truncated local algebras). *For every $m \geq h - 1$, on the entire generating Grassmannian X ,*

$$\text{Fitt}_0 \text{ coker}(\text{Sym}^m \mathcal{A} \rightarrow B_{e,h}) = \mathcal{I}_\Delta^N. \quad (21.3)$$

The divisor Δ is nonempty, integral, and Cartier. The displayed ideal is primary, its exact nilradical index is N , and its generic transverse length is N . For every $q \geq 1$ its q th ordinary power has the complete primary description $\mathcal{I}_\Delta^{Nq}$; there are no embedded associated points. On a smooth unit-frame cover, Δ is a generic e by e determinant, and its singular locus is given by matrix rank at most $e - 2$.

More precisely, choose a unit $a \in \mathcal{A}$ and write $\mathcal{A} = \langle a, ax_1, \dots, ax_e \rangle$, with $x_i \in \mathfrak{m}$. After multiplying the target by a^{-m} , the cokernel in (21.3) is the cokernel of the square substitution map

$$T_x : B_{e,h} \longrightarrow B_{e,h}, \quad z_i \longmapsto x_i.$$

If M is the matrix of linear coefficients of the x_i , then $\det T_x = (\det M)^N$. These presentation and ideal identities hold after arbitrary base change. The primary assertions refer to the universal parameter space X over $\mathbb{C}$, not to every specialization. For $m < h - 1$ the zeroth Fitting ideal is zero: the failure scheme is all of X , not the empty scheme. Thus the stated stable threshold is sharp for this family.

Proof. A unit exists locally because the augmentation of $\mathcal{A}$ is onto; subtracting constants from the remaining normalized basis vectors puts all x_i in $\mathfrak{m}$. The normalized multiplication matrix consists of the columns x^α with $|\alpha| \leq m$. Columns of total degree at least h vanish. When $m \geq h - 1$ the remaining columns are exactly the d columns of T_x ; all other columns are zero. This proves the cokernel assertion, over any coefficient algebra.

The map T_x preserves the $\mathfrak{m}$ -adic filtration and its block on $\mathfrak{m}^j/\mathfrak{m}^{j+1}$ is $\text{Sym}^j M$. Hence

$$\det T_x = \prod_{j=1}^{h-1} \det(\text{Sym}^j M) = (\det M)^{\sum_{j=1}^{h-1} \binom{e+j-1}{e}} = (\det M)^N.$$

For completeness, the middle identity follows by taking M diagonal: among all degree- j monomials each diagonal entry occurs with total exponent $\frac{j}{e} \binom{e+j-1}{e-1} = \binom{e+j-1}{e}$. The identity holds for arbitrary matrices because both sides are polynomial functions and diagonalizable matrices with distinct eigenvalues form a dense open over $\mathbb{C}$. It is consequently a polynomial identity and remains valid on nonreduced bases. No condition on the coefficients of degree at least two was used.

The determinant of (21.2) differs from $\det M$ by a unit: the vectors ax_i form a basis of K , and their linear terms are the linear terms of x_i multiplied by the nonzero augmentation of a . Thus the ideal equality descends canonically. To check its geometric structure, send A to $K = A \cap \mathfrak{m}$. This identifies X with the total space of the quotient bundle $\mathfrak{m}/K$ over $\mathrm{Gr}(e, \mathfrak{m})$: a lift of the augmented line has the unique form $1 + v$ modulo K . The projection of its universal e -plane to $\mathfrak{m}/\mathfrak{m}^2$ vanishes in determinant precisely on the standard Schubert divisor

$$\Sigma = \{K \in \mathrm{Gr}(e, \mathfrak{m}) : K \cap \mathfrak{m}^2 \neq 0\}.$$

This is the integral Schubert divisor for the fixed projection with kernel $\mathfrak{m}^2$, and Δ is its pullback along the quotient-bundle projection. This identifies the divisor globally. For an elementary verification of integrality, its frame cover has independent linear coefficients M and independent higher coefficients, subject only to full column rank of the combined matrix. The generic determinant is irreducible. Its zero scheme remains nonempty on the frame open, as witnessed by

$$x_1 = z_1, \dots, x_{e-1} = z_{e-1}, \quad x_e = z_1^2.$$

The standard integral determinant hypersurface on that open descends to Σ , and its vector-bundle pullback is integral and Cartier.

Locally its equation is a prime element in a regular local ring. Every positive power is primary; alternatively, a hypersurface in a regular scheme has no embedded associated points and its support here has a single generic point. The radical of (δ^N) is (δ) , and δ^{N-1} is nonzero modulo δ^N at its generic point. This proves the exact nilradical index and length, and the same argument proves the assertion for all powers. The partial derivatives of a determinant are its cofactors, giving the stated singular locus. Finally $\mathrm{rank} \mathrm{Sym}^m \mathcal{A} = \binom{e+m}{e} < d$ if $m < h-1$; there are no maximal minors. For $A = \langle 1, z_1, \dots, z_e \rangle$ the stable map is surjective, so this threshold cannot be lowered. $\square$

For $e = 2$ this is a genuine three-plane theorem for every $\mathbb{C}[x, y]/(x, y)^h$: the contact length is $\binom{h+1}{2}$ and the failure multiplicity is $\binom{h+1}{3}$. Already $h = 3$ gives a length-six, embedding-dimension-two algebra whose failure divisor has multiplicity four. These are schemes on a full generating Grassmannian, not selected transverse slices. They are distinct from the curvilinear hyperplane schemes and the reduced pair-diagonal ideal.

21.1. A global conductor theorem in arbitrary projective dimension. The preceding algebra is not the algebra of a divisor on a curve. To apply its scheme calculation to global sections, we require a separate conductor theorem. Its range depends on the order h of the fat point, not on its length d .

Theorem 21.2 (Fat-point conductor transport). *Let $e \geq 1$, $h \geq 2$, let $p \in \mathbb{P}^e$, and let Z be the fat point with ideal $\mathcal{I}_p^h$. Suppose $n \geq 2h - 1$. For any generating k -plane $A \subset H^0(Z, \mathcal{O}_Z(n))$, let*

$$W = H^0(\mathbb{P}^e, \mathcal{I}_p^h(n)), \quad U = \mathrm{res}_Z^{-1}(A).$$

For every $m \geq 2$ one has

$$WU^{m-1} = H^0(\mathbb{P}^e, \mathcal{I}_p^h(mn)). \quad (21.4)$$

Restriction induces an isomorphism of coherent cokernels on the entire relative generating Grassmannian, also with p varying,

$$\mathrm{coker}(\mathrm{Sym}^m U \rightarrow H^0(\mathcal{O}(mn)) \otimes \mathcal{O}) \simeq \mathrm{coker}(\mathrm{Sym}^m \mathcal{A} \rightarrow \pi_* \mathcal{O}_Z(mn)). \quad (21.5)$$

It and all its Fitting ideal equalities commute with arbitrary base change. The bound is sharp uniformly over all generating dimensions k and all $m \geq 2$; no rank-specific sharpness is asserted.

Proof. First fix $p = [1 : 0 : \cdots : 0]$ and work on its affine chart with coordinates $z_1, \dots, z_e$. A section of $\mathcal{O}(n)$ is a polynomial of degree at most n , and W has basis the monomials of total degree between h and n . Choose $u \in U$ with constant coefficient u_0 a unit. This is possible over a local parameter base because the augmentation of A and restriction onto A are surjective.

For every integer t with $h \leq t \leq mn$, one can choose $1 \leq j \leq m$ with $jh \leq t \leq jn$. Indeed the integral intervals $[jh, jn]$ have no gaps: $jn + 1 \geq (j + 1)h$ follows from $n \geq 2h - 1$. Every monomial z^α of total degree t can be factored into j monomials of degrees between h and n . Their product, multiplied by u^{m-j} , belongs to WU^{m-1} and has the form

$$u_0^{m-j} z^\alpha + \text{monomials of total degree strictly greater than } t.$$

Its degree is at most mn . Choosing this column for every monomial of degree between h and mn gives a square triangular matrix with units on the diagonal. It is invertible over the coefficient ring. These columns are thus a basis of the whole space on the right of (21.4). The reverse containment is immediate.

Restriction in degrees n and mn is surjective, since those degrees are at least $h - 1$. Its kernels and targets are vector bundles, with monomial bases on the current chart. In the locally split sequence $0 \rightarrow W \rightarrow U \rightarrow A \rightarrow 0$, the kernel on symmetric powers is the image of $W \otimes \text{Sym}^{m-1} U$. Equation (21.4) identifies its multiplication image with the kernel of target restriction. Taking the quotient of this restriction square proves (21.5). The triangular proof works over every coefficient ring once u_0 is a unit. For moving p , choose projective coordinates smoothly locally; restriction and the cokernel isomorphism are intrinsic, so they descend. All the split restriction sequences and surjective maps remain so after arbitrary pullback.

At the boundary $n = 2h - 2$, take $A = \langle 1 \rangle$. Then $U = \langle 1 \rangle + W$, and its quadratic monomials have degrees in $\{0\} \cup [h, 2h - 2] \cup [2h, 4h - 4]$. They omit z_1^{2h-1} , although it lies in the conductor kernel in degree $2n$. Its nonzero global cokernel class restricts to zero on Z . Thus the cokernel comparison already fails in degree two. $\square$

Corollary 21.3 (Global noncurvilinear primary schemes). *For $e \geq 2$, $h \geq 3$, and $n \geq 2h - 1$, consider all inverse-image series U of generating $(e + 1)$ -planes on $Z = (\mathcal{I}_p^h)$. For every $m \geq h - 1$, their global multiplication failure scheme has the exact primary ideal $\mathcal{I}_\Delta^{\binom{e+h-1}{e+1}}$. Every power is primary and there are no embedded associated points on this fixed- p generating parameter space. In degrees $2 \leq m < h - 1$ the failure ideal is zero, so the whole parameter space fails. The same primary assertions hold on the relative parameter space with $p \in \mathbb{P}^e$ varying; the coherent cokernel comparison itself holds after every base change.*

Proof. Apply Theorems 21.1 and 21.2. For moving p there is the following canonical construction. On $P \times P$, $P = \mathbb{P}^e$, let $\mathcal{D}_h$ be the subscheme defined by the h th power of the diagonal ideal, and let p_1, p_2 be the projections. Set

$$\mathcal{B}_h = (p_1)_* \mathcal{O}_{\mathcal{D}_h}, \quad \mathcal{L}_n = (p_1)_*(\mathcal{O}_{\mathcal{D}_h} \otimes p_2^* \mathcal{O}_P(n)).$$

The first is the rank- d jet algebra; the second is an invertible $\mathcal{B}_h$ -module, locally free of rank d over P . Let $\mathcal{M}_n$ be the kernel of $\mathcal{L}_n \rightarrow \mathcal{O}_P(n)$. Its first normal quotient is $\Omega_P^1 \otimes \mathcal{O}_P(n)$. On the relative generating open of $\text{Gr}_P(e + 1, \mathcal{L}_n)$, the universal augmentation kernel K therefore has the determinant line map

$$\det K \longrightarrow \det(\Omega_P^1 \otimes \mathcal{O}_P(n)).$$

Its zero divisor is canonical. The map $A \mapsto K$ to $\text{Gr}_P(e, \mathcal{M}_n)$ is an affine-bundle torsor, locally the space of lifts of $\mathcal{O}_P(n)$ in $\mathcal{L}_n/K$. The divisor is the pullback of the relative Schubert divisor

for $\mathcal{M}_n \rightarrow \Omega_P^1 \otimes \mathcal{O}_P(n)$. Local splittings of these vector-bundle sequences exhibit the latter as the product with the standard integral Schubert divisor. Since P is integral and the fibres are nonempty, this divisor and its affine-bundle pullback are integral and Cartier. The total generating parameter space is smooth, so the same primary-power argument applies. No primary-decomposition assertion is inferred for arbitrary nonreduced specializations of this parameter space. $\square$

22. PRIMARY DESCENT OVER MULTIPLICITY STRATA

The following descent statement concerns ideals, not just the permutation action on their generic points.

Lemma 22.1 (Etale descent of weighted prime divisors). *Let X be a regular integral Noetherian $\mathbb{C}$ -scheme and $p : \tilde{X} \rightarrow X$ a finite etale Galois cover, with group G . Let $\{D_\alpha\}$ be a finite G -stable collection of distinct integral effective Cartier divisors on $\tilde{X}$. Work componentwise if the cover is disconnected. Suppose the positive integers b_α are constant on G -orbits. For an orbit O , there is a unique integral effective Cartier divisor D_O on X whose ideal satisfies*

$$p^* \mathcal{I}_{D_O} = \prod_{\alpha \in O} \mathcal{I}_{D_\alpha} = \bigcap_{\alpha \in O} \mathcal{I}_{D_\alpha}.$$

If a descended ideal $\mathcal{J}$ pulls back to $\prod_\alpha \mathcal{I}_{D_\alpha}^{b_\alpha}$, then its complete primary decomposition is

$$\mathcal{J} = \prod_O \mathcal{I}_{D_O}^{b_O} = \bigcap_O \mathcal{I}_{D_O}^{b_O}. \quad (22.1)$$

There are no embedded associated points, and the formula holds at intersections as well as generically. The descended prime divisor need not be smooth or geometrically unbranched.

Proof. In each regular local ring upstairs, equations for distinct prime Cartier divisors are distinct prime factors. Unique factorization gives product equal to intersection, with or without positive powers. The reduced orbit union has an invariant invertible ideal subsheaf of $\mathcal{O}_{\tilde{X}}$ with its canonical descent data. Faithfully flat descent gives an invertible ideal subsheaf of $\mathcal{O}_X$. It defines an effective Cartier divisor. Its support is the image of any one member of the orbit, hence is irreducible, since the map is finite. Reducedness descends faithfully flat, so this divisor is integral.

The same factorization upstairs, and flatness for finite intersections, identify the pullbacks of all three ideals in (22.1). Faithfulness makes them equal downstairs. Locally downstairs each $\mathcal{I}_{D_O}$ is generated by a prime element of a regular local ring; its b_O th power is primary. Their intersection is irredundant after discarding components absent from the chart, so it is the complete primary decomposition and has no embedded primes. No normal-crossing hypothesis was used. An orbit can have several members through a point upstairs, giving several etale branches of the integral descended divisor. This explains the final distinction. $\square$

Apply the lemma to the relative generating pencil Grassmannian over a multiplicity stratum. Ordering the distinct support points gives a finite etale cover with permutations among equal multiplicities. On it the divisors T_i, E_{ij} are the relative rank-one Schubert divisors. Their ideals are prime Cartier, and their weights are $\binom{d_i}{2}, d_i d_j$. Formula (22.1) supplies the whole descended primary decomposition. In particular it remains valid at a braid intersection where several values agree. The fixed labelled divisors are smooth; the descended orbit divisors are asserted integral, not automatically smooth or unbranched.

This argument applies to the earlier pencil calculation. For the contact-hyperplane family, Theorem 14.2 gives a scheme isomorphism with the quotient incidence and does not require a conjectural primary descent rule.

23. CONTACT PENCILS: EQUATIONS, PRIMARY COMPONENTS, AND INFINITESIMAL LAYERS

Fix $d \geq 4$ and $n \geq 2d - 2$. In this section the divisor D is fixed, although it need not be reduced. Set $G_D^\circ = \text{Gr}(2, E_{n,D})^\circ$. It is a smooth irreducible variety of dimension $2d - 4$. The conductor construction embeds it as the open generating locus in the Grassmannian of subseries containing W_D . Let $\mathcal{J}_m$ be the zeroth Fitting ideal of $\text{coker}(\text{Sym}^m \mathcal{U} \rightarrow V_{mn})$ on this variety.

Trivialize $\mathcal{O}_D(n)$, and put $R = H^0(\mathcal{O}_D)$. There is a smooth surjective frame map

$$\mathcal{F}_D = \{(a, v) : a \in R^\times, v \in R \setminus \mathbb{C}\} \longrightarrow G_D^\circ, \quad (a, v) \longmapsto \langle a, av \rangle. \quad (23.1)$$

It is the open part of the frame bundle where the first basis vector is a unit. Its relative dimension is four. All calculations on $\mathcal{F}_D$ below are therefore exact faithfully flat local calculations for G_D° , not transverse slices inferred from pointwise ranks.

Lemma 23.1 (The power-basis presentation). *On $\mathcal{F}_D$ the multiplication cokernel is presented by*

$$K_m(v) = [1, v, v^2, \dots, v^m] : \mathcal{O}^{m+1} \longrightarrow R \otimes \mathcal{O}. \quad (23.2)$$

For $m \geq d - 1$, every later column is an $\mathcal{O}$ -linear combination of the first d columns, and

$$\mathcal{J}_m \mathcal{O}_{\mathcal{F}_D} = (\Delta_D(v)), \quad \Delta_D(v) = \det[1, v, \dots, v^{d-1}]. \quad (23.3)$$

For $2 \leq m < d - 1$, $\mathcal{J}_m = 0$.

Proof. By Theorem 11.2, it suffices to multiply sections in the pencil $\langle a, av \rangle$ on D . Its image is generated by $a^m v^j$ for $0 \leq j \leq m$. Multiplication by a^{-m} is an invertible target operation and gives (23.2). Multiplication by v is an endomorphism of the rank- d free module $R \otimes \mathcal{O}$. Its monic characteristic polynomial annihilates it over the coefficient ring, by the adjugate proof of Cayley–Hamilton. Thus v^d and every higher power lie in the span of the first d powers, with regular coefficients. Column operations reduce the presentation to $[K_{d-1}(v), 0]$. This proves (23.3), as an identity of ideals and not merely of radicals. If $m + 1 < d$, all d -minors are zero. Faithful flatness descends this assertion. $\square$

Write $D = \sum_{i=1}^s d_i p_i$, with distinct p_i . Choose local parameters z_i and the Chinese-remainder identification

$$R = \prod_{i=1}^s \mathbb{C}[z_i]/(z_i^{d_i}), \quad v_i = \lambda_i + \sum_{r=1}^{d_i-1} c_{ir} z_i^r.$$

All λ_i, c_{ir} are independent affine coordinates before removing the scalar locus. For $d_i \geq 2$ define T_i by failure of A to separate $2p_i$; for $i < j$ define E_{ij} by failure to separate $p_i + p_j$. On the frame space they have equations $c_{i1} = 0$ and $\lambda_j - \lambda_i = 0$, respectively.

Theorem 23.2 (Full fixed-contact classification). *For every $m \geq d - 1$, in the ordered jet basis $1, z_i, \dots, z_i^{d_i-1}$,*

$$\Delta_D(v) = \prod_{i:d_i \geq 2} c_{i1}^{\binom{d_i}{2}} \prod_{i < j} (\lambda_j - \lambda_i)^{d_i d_j}. \quad (23.4)$$

The complete primary decomposition on G_D° is

$$\mathcal{J}_m = \bigcap_{i:d_i \geq 2} \mathcal{I}_{T_i}^{\binom{d_i}{2}} \cap \bigcap_{i < j} \mathcal{I}_{E_{ij}}^{d_i d_j}. \quad (23.5)$$

All displayed divisors are nonempty, smooth, irreducible, and distinct. They are precisely the associated subvarieties; no embedded associated points occur. The failure divisor has dimension $2d - 5$ and divisor class $\binom{d}{2}$ times the restricted Pluecker class.

Every multiplication corank, for every $m \geq 2$, is determined as follows. Put $e_i = \text{ord}_{z_i}(v_i - \lambda_i)$, with $e_i = d_i$ for a constant v_i , and put $\ell_i = \lceil d_i/e_i \rceil$. For each distinct value λ among the λ_i , set $L_\lambda = \max_{i:\lambda_i=\lambda} \ell_i$, and let $s(v) = \sum_\lambda L_\lambda$. Then

$$\text{corank}(\text{Sym}^m U_A \longrightarrow V_{mn}) = d - \min\{m + 1, s(v)\}. \quad (23.6)$$

On the frame space every higher Fitting ideal is, exactly, the corresponding minor ideal of (23.2).

Proof. For a polynomial P of degree at most $d - 1$, Taylor expansion gives

$$P(v_i(z_i)) = \sum_{r=0}^{d_i-1} \frac{P^{(r)}(\lambda_i)}{r!} (v_i(z_i) - \lambda_i)^r \pmod{z_i^{d_i}}.$$

The linear change from the divided derivatives of P at λ_i to the coefficients of $P(v_i(z_i))$ is triangular, with diagonal $1, c_{i1}, \dots, c_{i1}^{d_i-1}$. Its determinant is $c_{i1}^{\binom{d_i}{2}}$, including when this coefficient is zero. The determinant of the divided-derivative evaluation map

$$P \longmapsto \left(\frac{P^{(r)}(\lambda_i)}{r!} \right)_{i, 0 \leq r < d_i}$$

in the basis $1, T, \dots, T^{d-1}$ is the confluent Vandermonde $\prod_{i < j} (\lambda_j - \lambda_i)^{d_i d_j}$. Here is an explicit normalization of that identity. Start with d distinct variables, cluster d_i of them near λ_i , and in each cluster replace successive evaluation rows by divided-difference rows. The ordinary Vandermonde determinant is divided by the product of the differences *within* each cluster. In the cluster limit the rows are the divided derivatives above, while the remaining cross-cluster factors tend to the displayed product. This proves the identity on a dense open set and hence as a polynomial identity, including all collisions. Combining the two determinants proves (23.4) without inverting any of its factors.

For any length-two $Z \leq D$, restriction $E_{n,D} \rightarrow E_{n,Z}$ is surjective. The locus in $\text{Gr}(2, E_{n,D})$ where its restriction to A has rank at most one is the corresponding irreducible Schubert divisor. Its rank-zero locus is excluded by generation on D . Its rank-one locus is smooth: choose a basis of A with one nonzero image, and vary the second basis vector in a complementary target direction; the determinant then has a nonzero linear derivative. For irreducibility one may also mark the image line in $E_{n,Z}$; the resulting incidence is a Grassmannian bundle over $\mathbb{P}(E_{n,Z})$, and projects birationally onto the rank-one locus. The generating open is nonempty for each divisor when $d \geq 4$. On frames this follows directly by choosing $a = 1$ and a nonscalar v with the single prescribed linear equality. The same coordinates show that the divisors are distinct.

On each affine frame chart the coordinate ring is a localization of a polynomial ring. The distinct linear factors in (23.4) generate distinct prime ideals there. Unique factorization gives $(\prod f_\alpha^{b_\alpha}) = \bigcap (f_\alpha^{b_\alpha})$; each ideal $(f_\alpha^{b_\alpha})$ is primary. These identities descend by faithful flatness to (23.5). Equivalently, at each point of the smooth base the local ring is regular, each displayed divisor is smooth, and its powers are primary. This also proves absence of embedded primes. The dimension follows since this is a nonzero Cartier divisor. Each component is a Pluecker hyperplane section, and

$$\sum_i \binom{d_i}{2} + \sum_{i < j} d_i d_j = \binom{d}{2},$$

which proves the class formula.

Finally, in $\mathbb{C}[z_i]/(z_i^{d_i})$ the element $v_i - \lambda_i$ has nilpotency index ℓ_i . Its minimal polynomial is $(T - \lambda_i)^{\ell_i}$. In the product algebra the minimal polynomial is therefore

$$P_v(T) = \prod_\lambda (T - \lambda)^{L_\lambda}. \quad (23.7)$$

No polynomial of smaller degree can annihilate v , and division by this monic polynomial shows that $1, v, \dots, v^{s(v)-1}$ form a basis of $\mathbb{C}[v]$. Consequently the rank of $K_m(v)$ is $\min\{m+1, s(v)\}$. The cokernel isomorphism of Theorem 11.2 gives (23.6) and the assertion about Fitting ideals. $\square$

Theorem 23.3 (Exact infinitesimal layers and singularities). *Let $m \geq d-1$. At a point of a smooth frame chart discard the nonvanishing factors of (23.4). Write the remaining distinct linear factors as f_α , with multiplicities b_α , and put*

$$F = \prod_{\alpha} f_{\alpha}^{b_{\alpha}}, \quad P = \prod_{\alpha} f_{\alpha}.$$

In the regular local ring B of that chart the failure ring is $B/(F)$ and its nilradical is $\mathcal{N} = (P)/(F)$. Its exact nilpotency index is $\max_{\alpha} b_{\alpha}$, with index one for a reduced local ring. For every $q \geq 1$, there is an isomorphism of B -modules, with indicated generator P^q ,

$$\mathcal{N}^q / \mathcal{N}^{q+1} \simeq B / \left(P, \prod_{\alpha} f_{\alpha}^{(b_{\alpha}-q)+} \right). \quad (23.8)$$

The completed local equation is the same weighted linear arrangement, with unused frame and higher-jet coordinates as smooth parameters. Its tangent cone is defined by the homogeneous product of the active linear factors with these same multiplicities. Its singular scheme on this chart has ideal $(F, \partial F)$, and its singular support is the union of the components with $b_{\alpha} \geq 2$ and the pairwise intersections of distinct components. These descriptions descend smoothly to G_D° .

Proof. The local ring in question is a localization of a polynomial ring in the unit coordinates of a and the independent jet coordinates of v . The factors f_{α} are prime, distinct, and vanish at the selected point. Unique factorization shows that $P^q \in (F)$ if and only if $q \geq b_{\alpha}$ for every α . This proves the nilpotency assertion, including at multiple intersections.

The homomorphism $B \rightarrow \mathcal{N}^q / \mathcal{N}^{q+1}$ sending 1 to the class of P^q is onto. Its kernel is

$$((P^{q+1}, F) : P^q) = (P) + (F : P^q) = \left(P, \prod_{\alpha} f_{\alpha}^{(b_{\alpha}-q)+} \right).$$

For the middle equality, if $rP^q = sP^{q+1} + tF$, subtract sP from r ; cancellation gives membership in $(F : P^q)$. The last colon is computed by prime-factor exponents. This proves (23.8). Globally the chosen generator carries the corresponding line-bundle trivialization; the displayed isomorphism is a local one and makes no claim of a preferred global generator.

At the chosen point every active factor is linear in centered coordinates. Every discarded factor is a unit, so the completed ideal is exactly the displayed product, not a truncation of it. The initial ideal is generated by the same homogeneous product, which proves the tangent-cone statement. The hypersurface cotangent presentation gives $(F, \partial F)$. At a point on exactly one component of multiplicity one the differential is a unit times $df_{\alpha} \neq 0$. At every point on a multiple component or on two distinct components it is zero. This proves the singular support assertion. Smooth faithfully flat pullback preserves these local statements; in particular the four frame parameters do not alter the nilpotency index or the singular-support criterion. $\square$

For illustration, length four already gives subseries of codimension two, in every V_n with $n \geq 6$. For every $m \geq 3$ their fixed-contact primary equations are as follows. Put $\delta_{ij} = \lambda_j - \lambda_i$; only first-jet coefficients of multiple points are displayed.

Multiplicities of D	Exact equation on frames	Largest local nilpotency index
$(1, 1, 1, 1)$	$\prod_{i < j} \delta_{ij}$	1
$(2, 1, 1)$	$c_{11} \delta_{12}^2 \delta_{13}^2 \delta_{23}$	2
$(2, 2)$	$c_{11} c_{21} \delta_{12}^4$	4
$(3, 1)$	$c_{11}^3 \delta_{12}^3$	3
(4)	c_{11}^6	6

The reduced arrangement need not have normal crossings: three coincident values give three braid hyperplanes in a two-dimensional normal space. The equation, rather than a normal-crossing label, records that singularity. Formula (23.6) also distinguishes points with the same support of failure but different higher-jet vanishing orders.

24. MOVING CONTACTS AND THE NORMALIZATION OF THE FAILURE DIVISOR

Let $X_d^\circ = X_{d,2}^\circ$ be the relative generating Grassmannian in Theorem 11.2. It is smooth and irreducible of dimension $3d - 4$. The fixed-divisor schemes of the preceding section are its scheme-theoretic fibres, not its irreducible components. For $m \geq d - 1$ denote its failure scheme by Y_m .

24.1. Components over every multiplicity stratum. For a partition $\mathbf{d} = (d_1, \dots, d_s)$ of d , let $B_{\mathbf{d}} \subset \text{Sym}^d \mathbb{P}^1$ be the locally closed stratum of divisors with exactly this multiplicity partition, and restrict X_d° to it. Let M be the set of distinct multiplicities, and let $r_e = \#\{i : d_i = e\}$ for $e \in M$.

Corollary 24.1 (Complete component descent on multiplicity strata). *For $m \geq d - 1$, the failure scheme over $B_{\mathbf{d}}$ has*

$$\#\{e \in M : e \geq 2\} + \binom{\#M}{2} + \#\{e \in M : r_e \geq 2\} \quad (24.1)$$

irreducible components, all of dimension $s + 2d - 5$, and no embedded associated points. A ramification component is indexed by each $e \geq 2$ in M and has primary multiplicity $\binom{e}{2}$. An identification component is indexed by each unordered pair of multiplicities e, f that occurs at distinct support points and has primary multiplicity ef . The ideal of the scheme is the intersection of these prime-divisor ideals raised to the stated powers. On the labelled etale cover its full local equations and nilradical layers are exactly those of Theorem 23.3.

Proof. The ordered configuration space of s distinct points of $\mathbb{P}^1$ is an integral finite etale cover of $B_{\mathbf{d}}$, with deck group $\prod_{e \in M} \mathfrak{S}_{r_e}$. The relative first-jet and two-point restriction maps are surjective over this cover. Consequently each labelled T_i and E_{ij} is an irreducible relative rank-one Schubert divisor. The local factorization (23.4) holds with the support coordinates free, so there are no further components. The deck group permutes T_i exactly according to the value d_i , and permutes E_{ij} exactly according to the unordered pair $\{d_i, d_j\}$. The images of the orbit unions are precisely the irreducible components downstairs: each is the image of an integral labelled component, and two such images agree exactly for components in the same orbit. This gives (24.1).

Lemma 22.1 applies to this regular ambient space and its invariant weighted divisors, including all branch intersections. It proves that each component downstairs is a prime Cartier divisor; irreducibility here does not assert smoothness or geometric unbranchedness. Pulling its reduced ideal back to the labelled cover gives the reduced product of its labelled branch equations. Their multiplicity is constant on an orbit, so the primary ideal identity in (23.5) descends faithfully flatly. This also proves absence of embedded points and gives the stated local layers. The base stratum has dimension s , giving the dimension formula. A descended component can have intersecting

local branches; smoothness of each labelled divisor is not a claim that every descended component is smooth. $\square$

Theorem 24.2 (An integral failure divisor with smooth normalization). *For $d \geq 4$, $n \geq 2d - 2$, and $m \geq d - 1$, the schemes Y_m are all the same integral Cartier divisor Y of dimension $3d - 5$. In particular Y has no embedded associated points. Its normalization is*

$$\tilde{Y} = \{(D, A, Z) : (D, A) \in X_d^\circ, Z \leq D, \deg Z = 2, \text{rank}(A \rightarrow H^0(\mathcal{O}_Z(n))) = 1\}. \quad (24.2)$$

The forgetful map $\tilde{Y} \rightarrow Y$ is finite and birational, and $\tilde{Y}$ is smooth and irreducible.

More explicitly, on an affine support chart write $D = (g(t))$ with g monic of degree d , and on a frame chart put $v = b/a \in \mathbb{C}[t]/(g)$, represented by a polynomial of degree less than d . The failure equation is

$$\Delta(g, v) = \det[1, v, \dots, v^{d-1}]_{\mathbb{C}[t]/(g)} = 0. \quad (24.3)$$

The normalization on this chart is parametrized by

$$g = hq, \quad v = \lambda + hw, \quad (24.4)$$

where h, q are monic of degrees $2, d - 2$, respectively, $\lambda \in \mathbb{C}$, and $\deg w \leq d - 3$. The scalar-pencil locus is omitted; the unit a is a free smooth parameter.

Proof. Theorem 11.2 applies on the whole relative space. The proof of Lemma 23.1 applies over the polynomial coefficient ring of a varying monic g , not only on its fibres. Thus every Y_m has the identical ideal (24.3) on a smooth covering by frame charts. This determinant is not identically zero: $v = t$ makes its matrix the identity for every g .

The space of pairs $Z \leq D$ with lengths $2, d$ is $\text{Sym}^2 \mathbb{P}^1 \times \text{Sym}^{d-2} \mathbb{P}^1$; the map sends (Z, Q) to $D = Z + Q$. It is finite. Indeed it is proper, and a fixed effective divisor has only finitely many degree-two subdivisors, obtained by choosing multiplicities at its finitely many support points. Restriction $E_{n,D} \rightarrow E_{n,Z}$ is a surjection of vector bundles. The rank-at-most-one condition on a pencil is a relative Schubert divisor; its rank-zero part is excluded by the generating hypothesis. This makes (24.2) a closed subscheme of the finite base change to X_d° , so its projection is finite.

Smoothness and irreducibility can be checked explicitly. A divisor with finite support is contained in an affine coordinate chart. On that chart Z is defined by a unique monic quadratic h , and $Z \leq D$ means $g = hq$. Since a is a unit, rank one on Z is equivalent to $v \bmod h$ being a scalar λ . Division by the monic h gives uniquely $v = \lambda + hw$, with the stated degree bound. Thus the framed incidence is an open part of affine space with coordinates the coefficients of h, q, w , the scalar λ , and the unit a . This proves smoothness. Globally irreducibility also follows from the relative rank-one Schubert description over the integral pair-divisor base; each fibre is irreducible, and its generating open is nonempty. These charts describe the same incidence, so no extra component appears at infinity.

An element v fails to generate the length- d algebra $R = H^0(\mathcal{O}_D)$ precisely when it either takes the same value at two distinct support points or has zero first derivative at a multiple support point. This follows at once from the minimal polynomial (23.7): its degree is d if and only if the support values are distinct and each local $\ell_i = d_i$. These two failure mechanisms are precisely the existence of a length-two $Z \leq D$ on which v is constant. Therefore the image of the finite incidence is the whole support of (24.3). That support is irreducible.

To determine its multiplicity, work over divisors with distinct support, label their roots after an étale base change, and use evaluation coordinates for v . The fixed-fibre formula becomes a product of the distinct differences $\lambda_j - \lambda_i$. At a point with exactly one equal pair the determinant has a nonzero linear derivative. Hence its multiplicity along its unique irreducible support is one. Since the ambient space is smooth, a Cartier hypersurface has no embedded associated points; a generically reduced such hypersurface is reduced. One may see the last assertions locally from

unique factorization in the regular local ring, or from the Cohen–Macaulay and $R_0 + S_1$ criterion [24, Tag 031R]. Thus Y is integral.

Over the dense open subset with reduced D and precisely one equal pair there is exactly one possible Z . The finite projection is therefore birational in characteristic zero. Its source is smooth and hence normal, so it is the normalization. The affine parametrization is exactly (24.4), as already proved. The dimension of the divisor is $3d - 5$. $\square$

Proposition 24.3 (Exact singular-support test across collisions). *At a closed point $(D, A) \in Y$, call a length-two $Z \leq D$ scalar if A has rank one on Z . The point is smooth on Y if and only if there is precisely one scalar Z and the following linear system has no nonzero solution. In the associated factorization (24.4), the unknown is $r(t)$ of degree less than two, and the system is*

$$h \mid qr, \quad wr \bmod h \text{ is constant.} \quad (24.5)$$

In particular, this criterion is effective at nonreduced D and is not merely the distinct-support pair-count criterion. The full singular scheme on a frame chart has ideal

$$(\Delta, \partial\Delta/\partial g_0, \dots, \partial\Delta/\partial g_{d-1}, \partial\Delta/\partial v_0, \dots, \partial\Delta/\partial v_{d-1}). \quad (24.6)$$

Proof. If the normalization has more than one point above (D, A) , the local ring of Y cannot be regular. Suppose there is just one. We first justify the differential criterion for a finite normalization $\nu : N \rightarrow Y$ with N smooth. A regular local ring is normal, so a smooth point of Y has an isomorphic normalization near it. Conversely let $A_0 \subset B_0$ be the local finite extension at a point with a unique preimage and common residue field $\mathbb{C}$. Injectivity of $d\nu$ means

$$\mathfrak{m}_{B_0}/(\mathfrak{m}_{B_0}^2 + \mathfrak{m}_{A_0}B_0) = 0.$$

Nakayama’s lemma first gives $\mathfrak{m}_{B_0} = \mathfrak{m}_{A_0}B_0$. It then gives, for the finite A_0 -module B_0 , that 1 generates B_0 , since the quotient by $\mathfrak{m}_{A_0}$ is $\mathbb{C}$. The injective map $A_0 \rightarrow B_0$ is consequently surjective. Thus $A_0 = B_0$ is regular. Local finiteness here follows by localizing the finite morphism; uniqueness of the preimage makes the resulting semilocal ring local.

In the affine normalization coordinates, a tangent vector with $\dot{g} = \dot{v} = 0$ satisfies

$$0 = q\dot{h} + h\dot{q}, \quad 0 = \dot{\lambda} + w\dot{h} + h\dot{w}.$$

Set $r = \dot{h}$, so $\deg r < 2$ because h is monic. The first equation is soluble exactly when $h \mid qr$, and then determines $\dot{q} = -qr/h$. The second is soluble exactly when the remainder of wr modulo h is constant; it determines $\dot{\lambda}$ as the negative of that constant and determines $\dot{w}$ by division. The resulting degrees satisfy the required bounds. If $r = 0$ all four variations are zero. Thus the differential has nonzero kernel exactly when (24.5) has a nonzero solution. The unit frame parameter has zero variation in a kernel and creates no additional one. Smooth frame descent proves the criterion on Y . Finally (24.6) is the hypersurface Jacobian presentation in its smooth affine ambient space. $\square$

Remark 24.4 (A smooth total point with a triple primary fibre). Take $d = 4$, $g = t^3(t - 1)$, and $A = \langle 1, t^2 \rangle$ on $D = (g)$. The only scalar length-two subdivisor is $h = t^2$. In (24.4) one has $q = t(t - 1)$ and $w = 1$. For $r = r_0 + r_1t$, divisibility $h \mid qr$ forces $r_0 = 0$, while constancy of $wr \bmod h$ forces $r_1 = 0$. Hence the total divisor is smooth here. On the fixed-divisor fibre, the two support values are distinct and (23.4) is a unit times c_{11}^3 . Its exact local nilpotency index is three. Thus even a smooth point of the integral total divisor can have a nonreduced primary fibre.

Proposition 24.5 (Discriminants and specialization). *Let $\chi_v(T)$ be the characteristic polynomial of multiplication by v in $\mathbb{C}[t]/(g)$. With the usual monic polynomial normalization of discriminants,*

$$\text{disc}(\chi_v) = \text{disc}(g) \Delta(g, v)^2. \quad (24.7)$$

This is a polynomial identity on the entire coefficient space. The equation $\Delta = 0$, not $\text{disc}(\chi_v) = 0$, defines the failure divisor, including when g has multiple roots. Its fibre over each fixed D is exactly the primary scheme in (23.5).

Proof. When g has distinct roots $t_1, \dots, t_d$, the evaluation matrix changes the basis $1, t, \dots, t^{d-1}$ to point values. Therefore

$$\Delta(g, v) = \frac{\prod_{i < j} (v(t_j) - v(t_i))}{\prod_{i < j} (t_j - t_i)}.$$

Squaring gives (24.7), because $\chi_v(T) = \prod_i (T - v(t_i))$ on this open set. Both sides are polynomial in the coefficients, so the identity holds everywhere. For $v = t$ the determinant is one even when g is nonreduced; this also shows why the factor $\text{disc}(g)$ must not be included in the failure equation. The scheme-theoretic fibre statement follows from Fitting base change in Theorem 11.2, or by specializing the polynomial determinant and applying (23.4). $\square$

Proof of Theorem 10.1. The fixed-contact assertions are Theorems 23.2 and 23.3. The moving-contact assertions are Theorem 24.2 and Proposition 24.3. All four concern the original multiplication map, by the coherent cokernel isomorphism of Theorem 11.2. $\square$

25. PENCIL COLLISIONS AND COMPARISON OF LOCAL TYPES

The normalization criterion above separates phenomena which a set-theoretic discriminant calculation conflates. In the affine normalization write $g = hq$ and $v = \lambda + hw$, with h monic of degree two. The kernel test is

$$h \mid qr, \quad wr \bmod h \text{ constant}, \quad \deg r < 2.$$

The following three cases give concrete local distinctions.

If g has distinct roots, the normalization is unramified; a unique equal value pair gives a smooth point, and two disjoint equal pairs give an ordinary double crossing. Three equal values give the three braid factors $\delta_{12}\delta_{13}(\delta_{13} - \delta_{12})$, not a normal-crossing divisor in a two-dimensional normal space. These are exact equations because root labelling is étale here.

For $g = t^3(t - 1)$ and $v = t^2$, the scalar subdivisor $h = t^2$ is unique. One has $q = t(t - 1)$ and $w = 1$. Divisibility forces the constant term of r to vanish and the second condition forces its linear term to vanish. The total pencil divisor is smooth, although its fixed- D fibre has primary equation a unit times c_{11}^3 .

For $g = t^4$ and $v = t^2$, again $h = t^2$ is the unique scalar subdivisor. Now $q = t^2$ and $w = 1$. The constant $r = 1$ satisfies the kernel test. Thus the total pencil divisor is singular and its normalization is ramified at the unique preimage. This last example is asserted to have a unique normalization point, not an unproved analytic cusp normal form. In contrast, the moving contact-hyperplane scheme is smooth at all these divisor collisions by Theorem 14.4. Its fixed fourfold-contact transverse algebra is G_4 , with length six and nilpotency index five, rather than the pencil's sixth power of a prime divisor.

26. RELATION TO EARLIER CONSTRUCTIONS

The determinant $\det[1, v, \dots, v^{d-1}]$ is a classical local index form. Arpin, Bozlee, Herr and Smith construct schemes of monogenerators and polygenerators for finite locally free algebras [29]. Their Proposition 3.6 and Definition 3.12 identify the index-form open and its closed complement; Proposition 3.14 gives the polygenerator open by minors. Thus neither the generator functor nor its index determinant is claimed as new here. Our use concerns the coherent cokernel of a global section-multiplication map and, for generating hyperplanes, its exact closed incidence with a rank-two quotient. Lemma 14.1 and Theorem 14.2 prove that closed-scheme identification on nonreduced bases, rather than deriving it from agreement of generator opens.

The algebra G_r is the classical two-row Grassmannian cohomology ring; [31, Theorem 2.7] provides the rectangular Schur basis in a more general quotient. Its role here is an exact transverse algebra of multiplication failure, determined by a rank-two quotient of $\mathbb{C}[z]/(z^r)$. The explicit residual equations (14.7) verify this identification without a tangent-cone approximation. The diagonal-ideal and all-powers identities in the three-plane result are precisely [30, Corollary 3.8.3]; the argument here supplies the finite total-degree presentation and conductor transport, not a new proof of that theorem.

Normal generation in sufficiently positive degree on a curve is classical [14]. The curve conductor theorem uses it only for $L(-D)$, and then proves the precise quotient comparison by filling the double-vanishing space and its first normal layer. On $\mathbb{P}^1$ the evaluation-kernel argument gives the sharper rank-dependent bound and its boundary example.

For pencils, confluent Vandermonde factorization and the discriminant identity are classical algebra. The moving normalization is the explicit scalar-pair incidence of that index-form divisor. Its finite map and differential test are proved in Section 24; the normalization is not offered as a construction independent of index-form geometry. The contact-hyperplane incidence is different: it is the failure scheme itself, not just its normalization, and its total space is smooth even at collisions.

Subalgebra varieties, quotient flags, and the additional comparisons. Iovanov and Sistko [32, Theorem 0.1] classify maximal subalgebras of finite-dimensional algebras over fields under their Schur hypothesis. In the commutative algebraically closed case, the alternatives include identifying two residue factors and imposing a first-order condition at one factor. These closed-point mechanisms are classical and are not claimed here as new. Sistko [33, Definition 3.1, Theorems 1.1 and 4.1] constructs subalgebra varieties and computes the maximal-subalgebra vanishing ideal for basic algebras. The distinction needed here is between such a variety and the full relative quadratic-equation functor: for $\mathbb{C}[z]/(z^r)$ the latter already has the transverse algebra G_r , of length $\binom{r}{2}$. Agreement on the reduced variety does not identify this thickening.

The Hilbert functor used in Section 14 is the standard functor of locally free quotient algebras; its representability is not a contribution of this article. Likewise, nested effective divisors on a curve are classically expressed by a divisor and its residual divisor. What is established here is the identification of these functors with specified multiplication Fitting schemes, the canonical cokernel line (14.2), and the global-section transports. These are separate assertions from the classical classification of geometric hyperplanes.

The faithful-action viewpoint is also classical; compare [33, Proposition 3.4] for subalgebras of matrix algebras. Our conductor flags record the *specified inclusion* into B , the action on B/E , and its exact-rank scheme structure. A conductor rank bound alone is not the flag equivalence, nor does either imply unrestricted base change across an action-rank jump. The one-variable codimension-two classification is already given in [34, Theorem 39]. We use it to locate the fibres of Example 15.5, not as a new classification of polynomial subalgebras. The quotient-tower statement concerns arbitrary finite B and its relative exact-rank functors.

In [29, Example 4.16], the two-generator jet calculation has curvilinear target $k[\varepsilon]/\varepsilon^{m+1}$. Our target in (21.1) has embedding dimension e . The symmetric-power determinant identity in the proof is classical linear algebra. Its role is to determine the full primary power of the tangent determinant, uniformly in e, h , and then identify that power with global section multiplication on $\mathbb{P}^e$ using Theorem 21.2. Merely knowing the polygenerator open would not determine this exponent or the global degree range. These comparisons identify the inputs and additional claims; they are not a certificate of exhaustive priority against all literature.

Codimension, embedded incidences, and regularity. The elementary dimension bound for an increasing sequence of subspaces is not a scheme-theoretic stabilization theorem. The

proof of Theorem 13.1 retains the compressed commutator relation over the coefficient ring before using polarized Cayley–Hamilton. The polygenerator minors of [29] remain the classical ambient construction. The assertions added here are the codimension-sensitive degree bound, the full codimension-two matrix, and its conductor incidence on nonreduced bases. We do not claim an exhaustive priority comparison for every possible length bound.

The binary cubic for a rank-three algebra is the explicit square condition for a line to lift to a subalgebra. No independent novelty claim is made for that classical cubic-algebra calculation. Its role in Proposition 16.2 is to describe an exceptional fibre of the same incidence which transfers the embedded local rings in Theorem 18.1. The contact algebra G_h and its Schur basis are classical inputs; the mixed equations bv, b^2u , their displayed primary intersection, and their realization as local rings of full multiplication failure are proved separately.

Sidman’s Theorems 1.3 and 1.8 in [35] supply the relationship between regularity and saturation, and the ordinary-square estimate $\text{reg}(I^2) \leq 2 \text{reg } I$ for a finite projective scheme. Sidman also records the earlier power bounds of Chandler and of Geramita–Gimigliano–Pitteloud. None of these regularity inequalities is claimed as new. Their use here is the normal-layer identity (19.8) and the resulting isomorphism of actual multiplication cokernels, including relative families. This separation is particularly important: bounding the regularity of an ideal is not by itself a theorem identifying the Fitting scheme of a prescribed incomplete linear series.

Ballico’s failure-locus work [4, 5] is a necessary nearest-source comparison. The accessible statements of [5, Theorem 0.2, Proposition 2.2, Remark 2.3] concern failure on finite linear sections of a fixed completely embedded variety. The complete theorem pages of [4] have not been obtained in this revision, so their exact relation to the present scheme statements remains unverified. We make no inference of non-anticipation from that access limitation. The results here are submitted with complete proofs and explicit classical inputs; priority of the resulting package is a separate question for source comparison and editorial assessment.

26.1. Generators, algebra length, and the boundary classification. The scheme-theoretic study of generators is an established input. Arpin–Bozlee–Herr–Smith [29] construct the index-form non-monomer scheme (Definition 3.12) and describe the polygenerator open by maximal minors (Proposition 3.14); the numbering here refers to the identified preprint version. Thus neither representability, index forms, nor the use of maximal minors is a novelty claim of this paper. In their construction Cayley–Hamilton bounds powers of each individual generator. The additional tasks here are the codimension-sensitive equality of image modules, descent to the generating Grassmannian, the conductor incidence, and the complete primary decompositions of the full boundary in the stated classes.

The field-level length inequality is separated from the relative assertion in Section 13.1. We do not use a special numerical estimate from the algebra-length literature as an unproved premise. The elementary dimension argument and the sharpness family are proved in the text. The coefficient-ring proof supplies the Fitting-ideal consequence needed when the base is nonreduced.

Theorems 17.1–17.3 identify the present boundary contribution precisely: a full multiplication ideal, all its associated strata for Hilbert function $(1, 2, 2)$, their primary representatives, and their conductor position. They do not assert a primary classification for every finite algebra. The higher-contact and complete contact-hyperplane results remain separate theorems with their original scope. The discussion of [4] in Section 26 remains subject to the same source-access limitation; no exhaustive priority comparison with that article is asserted on the basis of its title or bibliographic record.

26.2. Logical dependencies. The finite-algebra core is self-contained. It does not take a theorem from a manuscript labelled A1 as an unstated premise. The evaluation and conductor comparisons

proved here are the interfaces through which a global linear-series problem becomes a finite-algebra problem. The outputs available to further work are canonical multiplication cokernels and their Fitting ideals, exact-action conductor flags, and the primary boundary classification. A downstream application must verify the stated generating and regularity hypotheses and retain the scheme structure; the determinant support alone loses the embedded strata. The historical polar, residual and statistical sections are not prerequisites for this core. This logical dependency statement does not certify that any particular version of a later manuscript already invokes these results.

26.3. Determinantal inputs and the elliptic family. The primality and normality of a generic minor ideal are classical; we use [39, Theorem 3.5(c)]. We also use the elementary rank-stratum codimension count and prove the needed symbolic-order inclusion by a Schur complement. The weighted ideal in Theorem 6.3 is not offered as a new theory of generic determinantal varieties. Its contribution here is the explicit primary intersection for the full multiplication ideal, including the extra linear determinant factor, and its use for the universal restriction flag. The incidence resolution is the usual image-subspace construction, applied to the specific embedded supports and with its fibres proved in Theorem 7.2.

The geometry of the Hesse pencil is also classical. We use the Weierstrass coefficients in [40, Section 2] to normalize the j -invariant. We do not claim a new classification of plane cubics. The additional statements are the explicit finite algebra (8.1), the full failure ideal on U , the square-zero line module (8.6), and the realization of the varying elliptic curves as the bases of its embedded supports. Those statements are derived from multiplication in the algebra and not from a resemblance between reduced equations.

The distinction from generator representability and the elementary field-level algebra-length bound is explained in Section 26.1. In particular a field-level dimension bound does not assert equality of image submodules over a nonreduced coefficient ring or identify their Fitting ideals there; the coefficient-ring argument and sharpness construction are provided in full in this article.

The full text of [4] has not been obtained for a theorem-by-theorem comparison. Its bibliographic record and publisher reference listing are not sufficient evidence to affirm or deny anticipation of any theorem here. No exhaustive priority conclusion with respect to that article is asserted. This documentary question is separate from the proofs above and remains explicitly recorded in the accompanying response and literature audit.

27. THE RAMIFICATION COMPARISON AND ITS SCOPE

The Jacobian description of ramification, the hypersurface adjunction formula, and the cohomology of projective bundles are classical. The K3 facts used here are the quartic example and the characteristic-zero non-uniruledness argument in [41, Chapter 1, Example 1.3(i); Chapter 9, proof of Corollary 4.4]; the needed birational cancellation is proved in Lemma 4.1. The Hesse invariant remains attributed to [40]. None of these ingredients is presented as a new theorem of this article.

The result proved here links them to multiplication failure by the cokernel identification (2.2), the primary identity (3.3), and the intrinsic annihilator (3.9). Its conclusion is a reconstruction statement for the nonreduced failure scheme, not a claim to have introduced ramification hypersurfaces or K3 surfaces. The general determinantal input and the retained finite-contact comparisons are discussed in Sections 26.3 and the preceding literature sections.

The nearest-source comparison with Ballico's higher-order failure-locus article [4] is not completed. The bibliographic record and the inherited record of its first-page inspection do not supply its full theorem statements. The present revision again did not obtain the complete article by the accessible publisher or Library routes. No statement that it anticipates, or fails to anticipate,

the results above is inferred from that access limitation. The accompanying literature audit specifies the comparison axes and the evidence available for each. Proof of the displayed results, an assessment of their priority, and a journal decision are distinct questions.

REFERENCES

- [1] R. Balan, P. Casazza, and D. Edidin, On signal reconstruction without phase, *Appl. Comput. Harmon. Anal.* **20** (2006), 345–356.
- [2] E. Ballico and Ph. Ellia, The maximal rank conjecture for nonspecial curves in $\mathbb{P}^3$, *Invent. Math.* **79** (1985), 541–555.
- [3] E. Ballico and Ph. Ellia, The maximal rank conjecture for non-special curves in $\mathbb{P}^n$, *Math. Z.* **196** (1987), 355–367.
- [4] E. Ballico, On the Failure Locus of Higher Order Properties of Embeddings in Projective Spaces, *Math. Nachr.* **163** (1993), 5–13, doi:10.1002/mana.19931630102.
- [5] E. Ballico, On the failure cycles for the quadratic normality of a projective variety, *Pacific J. Math.* **172** (1996), 307–313.
- [6] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, second ed., Springer, 2006.
- [7] A. Conca, M. Mostafazadehfard, A. K. Singh, and M. Varbaro, Hankel determinantal rings have rational singularities, *Adv. Math.* **335** (2018), 111–129; arXiv:1709.05685, Sections 1–2.
- [8] J. A. Eagon and D. G. Northcott, Ideals defined by matrices and a certain complex associated with them, *Proc. Roy. Soc. London Ser. A* **269** (1962), 188–204.
- [9] D. Eisenbud, C. Huneke, and B. Ulrich, Heights of ideals of minors, *Amer. J. Math.* **126** (2004), 417–438; arXiv:math/0209184.
- [10] A. Fröhbis-Krüger and M. Zach, Determinantal singularities, arXiv:2106.04855v1, 2021; Lemma 1.3, Remark 1.18, Section 1.3.2, and Proposition 2.12.
- [11] W. Fulton, *Intersection Theory*, second ed., Springer, 1998, Theorem 14.4.
- [12] I. M. Gelfand, M. M. Kapranov, and A. V. Zelevinsky, *Discriminants, Resultants, and Multidimensional Determinants*, Birkhäuser, 1994.
- [13] C. J. Geyer, On the asymptotics of constrained M -estimation, *Ann. Statist.* **22** (1994), 1993–2010.
- [14] M. Green and R. Lazarsfeld, On the projective normality of complete linear series on an algebraic curve, *Invent. Math.* **83** (1986), 73–90.
- [15] A. B. Hempel, P. J. Goulart, and J. Lygeros, Inverse parametric quadratic programming and an application to hybrid control, *IFAC Proceedings Volumes* **45**(17) (2012), 68–73.
- [16] G. Kempf and D. Laksov, The determinantal formula of Schubert calculus, *Acta Math.* **132** (1974), 153–162.
- [17] J. Kohonen, A meet-in-the-middle algorithm for finding extremal restricted additive 2-bases, *J. Integer Seq.* **17** (2014), Article 14.6.8.
- [18] J. Kohonen, An improved lower bound for finite additive 2-bases, *J. Number Theory* **174** (2017), 518–524.
- [19] A. R. Kustin, Canonical complexes associated to a matrix, arXiv:1509.04367, 2015, discussion following (1.0.2) and Section 2.9.
- [20] E. Larson, The maximal rank conjecture, arXiv:1711.04906, Theorem 1.2.
- [21] B. Mourrain and V. Y. Pan, Multivariate polynomials, duality, and structured matrices, *J. Complexity* **16** (2000), 110–180.
- [22] N. A. Nguyen, S. Olaru, P. Rodriguez-Ayerbe, M. Hovd, and I. Necoara, Constructive solution of inverse parametric linear/quadratic programming problems, *J. Optim. Theory Appl.* **172** (2017), 623–648.
- [23] C. Scheiderer, Extreme points of Gram spectrahedra of binary forms, *Discrete Comput. Geom.* **67** (2022), 1174–1190.
- [24] The Stacks Project Authors, *The Stacks Project*, Tag 031R, Lemma 10.157.3, the $R_0 + S_1$ reducedness criterion; consulted September 22, 2026.
- [25] The Stacks Project Authors, *The Stacks Project*, Section 15.8, Tags 07Z8 and 07ZA, presentation independence and base change of Fitting ideals; consulted September 22, 2026.
- [26] A. W. van der Vaart, *Asymptotic Statistics*, Cambridge University Press, 1998.
- [27] G. Yu, Upper bounds for finite additive 2-bases, *Proc. Amer. Math. Soc.* **137** (2009), 11–18.
- [28] The Stacks Project Authors, *The Stacks Project*, Section 10.63, Tag 00L9; Lemmas 10.63.3 and 10.63.5 and Proposition 10.63.6, associated primes; consulted September 22, 2026.
- [29] S. Arpin, S. Bozlee, L. Herr, and H. Smith, The scheme of monogenic generators I: representability, *Res. Number Theory* **9** (2023), article 14, doi:10.1007/s40993-022-00419-5. Proposition and definition numbering used here follows arXiv:2108.07185v2 (2022), especially Proposition 3.6, Definition 3.12, and Proposition 3.14.

- [30] M. Haiman, *Hilbert schemes, polygraphs, and the Macdonald positivity conjecture*, J. Amer. Math. Soc. **14** (2001), 941–1006; arXiv:math/0010246v2, Corollary 3.8.3.
- [31] D. Grinberg, *A basis for a quotient of symmetric polynomials*, arXiv:1910.00207v2 (2021), Theorem 2.7 and the subsequent discussion.
- [32] M. C. Iovanov and A. Sistko, *Maximal subalgebras of finite-dimensional algebras*, arXiv:1705.00762v2 (2017), Theorem 0.1.
- [33] A. Sistko, *Automorphism groups of finite-dimensional algebras acting on subalgebra varieties*, arXiv:1809.09760v2 (2019), Definition 3.1, Proposition 3.4, and Theorems 1.1 and 4.1.
- [34] R. Grönkvist, E. Leffler, A. Torstensson, and V. Ufnarovski, *Describing subalgebras of $K[x]$ using derivatives*, arXiv:2107.11916v1 (2021), Theorem 39, pp. 94–95.
- [35] J. Sidman, *On the Castelnuovo–Mumford regularity of products of ideal sheaves*, arXiv:math/0110184v2 (2001), Theorems 1.3 and 1.8.
- [36] C. J. Pappacena, An upper bound for the length of a finite-dimensional algebra, *J. Algebra* **197** (1997), 535–545.
- [37] O. V. Markova, Upper bound for the length of commutative algebras, *Sb. Math.* **200** (2009), no. 12, 1767–1787.
- [38] M. A. Khrystik, *Combinatorics on finite words and the length of a finite-dimensional associative algebra*, arXiv:2605.00141v1 (2026), Sections 1 and 6.
- [39] W. Bruns, T. Römer, and A. Wiebe, Initial algebras of determinantal rings, Cohen–Macaulay and Ulrich ideals, *Michigan Math. J.* **53** (2005), 71–81.
- [40] M. Artebani and I. Dolgachev, The Hesse pencil of plane cubic curves, *Enseign. Math.* (2) **55** (2009), 235–273.
- [41] D. Huybrechts, *Lectures on K3 surfaces*, Cambridge University Press, 2016. The cited internal locations refer to the author-hosted prepublication draft; Chapter 1, Example 1.3(i), and Chapter 9, proof of Corollary 4.4.